Network Information



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Executive Summary

Cyberdyne Systems, following its acquisition by WoT Incorporated, is expanding its operational footprint by opening a new branch office in Canberra, linked directly to its Sydney Headquarters. To support this growth, this project delivers an enterprise-grade, integrated network architecture that unifies both locations into a secure operational environment.

The implemented technical solution establishes an encrypted site-to-site IPSec VPN to ensure safe data transit over public paths, coupled with dynamic network services including centralized DHCP provisioning and NTP synchronization. To safeguard internal assets, the network enforces strict zero-trust access controls through IEEE 802.1X authentication integrated with Active Directory and RADIUS, dynamic VLAN steering, and granular Layer 3 Access Control Lists (ACLs). Designed to accommodate a projected 25% staff expansion over 5 years across 5 distinct security VLANs per site, the infrastructure guarantees long-term scalability and operational resilience.

Delivered over a 9 week, four-stage implementation timeline, the project concludes with rigorous failover and security validation, comprehensive operational documentation, and a live demonstration to stakeholders to confirm business readiness

Project Goal

The primary goal of this project is to design, deploy, and validate a secure, high-performance, and scalable multi-site enterprise network infrastructure connecting Cyberdyne Systems' Sydney Headquarters and the new Canberra branch office.

Specifically, the project aims to:

- Establish seamless, encrypted inter-site communication over public WAN infrastructure using an IPSec VPN tunnel.
- Enforce identity-based port security and centralized access control using IEEE 802.1X and Active Directory / NPS RADIUS integration.
- Provide segmented, high-availability network operations across both sites with centralized DHCP services, NTP synchronization, NAT/PAT translation, and granular Inter-VLAN security controls.

Project Scope

- **Addressing & Subnetting:** Allocation of an RFC1918 IPv4 scheme across 5 core VLANs per site (Infrastructure/Servers, Staff, IT Admin, Executive, Guest, and Restricted Personal), incorporating a 25% 5-year growth allowance.
- **WAN Connectivity & NAT:** Configuration of an IPSec VPN tunnel connecting the Sydney and Canberra edge routers, along with NAT/PAT translation for outbound Internet traffic.

- **Layer 2/3 Security & Access Control:** Deployment of 802.1X port-based authentication backed by Windows Server AD/NPS, dynamic VLAN allocation, Switch Port Security limits, and Inter-VLAN ACLs.

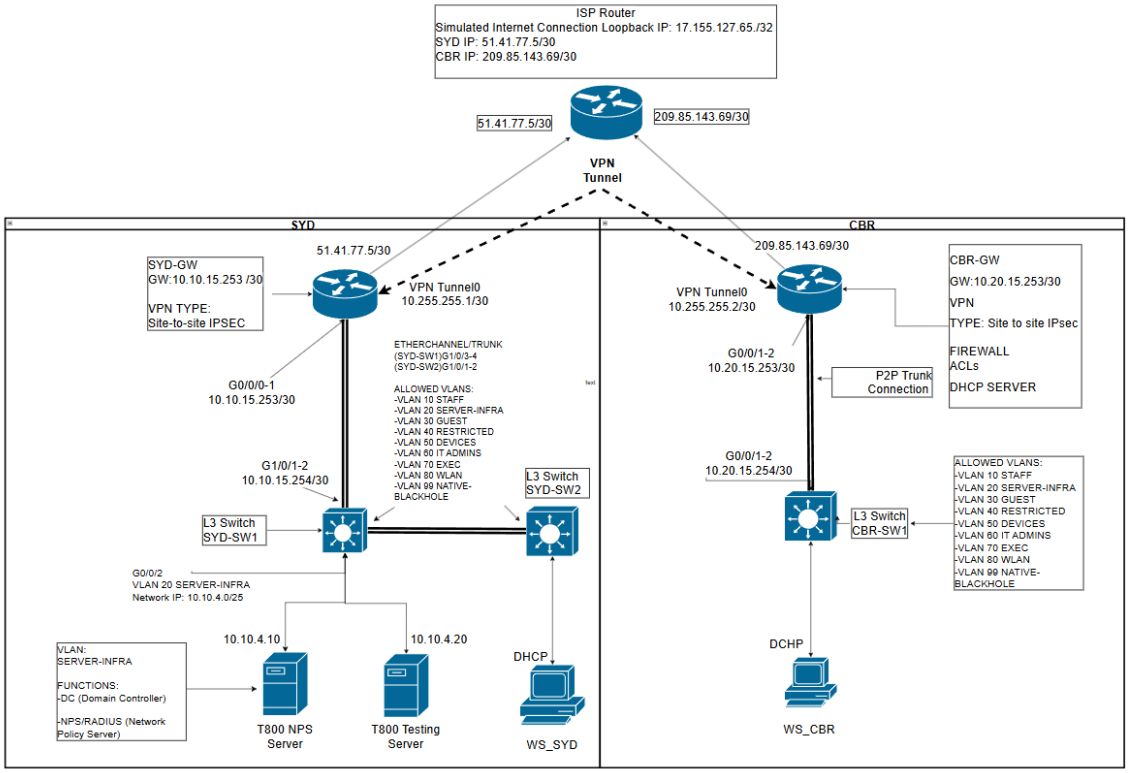
- **Core Network Services:** Centralized DHCP scope relays hosted on the Canberra edge router (CBR-GW), device-wide NTP time synchronization, and Inter-VLAN routing.

- **Device Hardening:** Implementation of SSH v2 for management and disabling unencrypted protocols (Telnet/HTTP).

- **Testing & Handover:** Execution of system failover tests, security boundary validation, administrative runbook delivery, and a live technical proof-of-concept demonstration.

Logical Design

Network Topology



*See appendix for Hi-Res image

Recommendation on how to increase the bandwidth to Canberra site:

Split-Tunneling / Direct Internet Access (DIA):

Configure Direct Internet Access at the Canberra edge router (CBR-GW) using Local Internet Breakouts. Non-critical public web traffic, cloud services (e.g., Microsoft 365, AWS), and guest network traffic should route directly out the local ISP interface.

According to Cisco (2016), implementing Direct Internet Access (DIA) via local internet breakouts offloads non-critical, cloud-bound traffic directly at the branch edge, significantly mitigating backhaul latency and optimizing site-to-site VPN bandwidth utilization, rather than consuming bandwidth backhauled through the IPSec VPN to Sydney HQ.

IP Addressing Scheme (HQ Site)

Purpose	VLAN	IP Address Block	Static/Dynamic
Staff	10	10.10.0.0/22	Dynamic
Servers/Infrastructure	20	10.10.4.0/25	Static
Guest	30	10.10.4.128/25	Dynamic
Restricted	40	10.10.5.0/25	Dynamic
Devices	50	10.10.5.128/26	Dynamic
IT Admins	60	10.10.5.192/26	Dynamic
Exec	70	10.10.6.0/26	Dynamic
Management	99	10.10.6.64/29	Static
Native	998	N/A	N/A
Blackhole	999	N/A	N/A

IP Addressing Scheme (Canberra Site)

Purpose	VLAN	IP Address Block	Static/Dynamic
Staff	10	10.20.0.0/26	Dynamic
Servers/Infrastructure	20	10.20.0.224/28	Static
Guest	30	10.20.0.64/26	Dynamic
Restricted	40	10.20.0.128/26	Dynamic
Devices	50	10.20.0.192/27	Dynamic
IT Admins	60	10.20.1.0/29	Dynamic
Exec	70	10.20.0.240/28	Dynamic
Management	99	10.20.1.8/29	Static
Native	998	N/A	N/A
Blackhole	999	N/A	N/A

VPN Configuration

```
CBR-GW#show crypto isakmp policy

Default IKE policy
Protection suite of priority 65513
        encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Rivest-Shamir-Adleman Signature
        Diffie-Hellman group:   #14 (2048 bit)
        lifetime:               86400 seconds, no volume limit
Protection suite of priority 65514
        encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   #14 (2048 bit)
        lifetime:               86400 seconds, no volume limit
CBR-GW#show crypto session
Crypto session current status

Interface: Tunnel0
Profile: CBR-SYD-IKEV2-PROFILE
Session status: UP-ACTIVE
Peer: 51.41.77.6 port 500
  Session ID: 3
  IKEv2 SA: local 209.85.143.70/500 remote 51.41.77.6/500 Active
  IPSEC FLOW: permit 47   host 209.85.143.70 host 51.41.77.6
        Active SAs: 2, origin: crypto map
```

Security / Tunnel Layer	Protocol Parameter	Configured Value
Architecture	Tunnel Type	GRE over IPsec (tunnel mode gre ip)
Phase 1 (IKEv2)	Protocol & Encryption	IKEv2 / AES-CBC-256
	Hashing & Key Exchange	SHA-256 / Diffie-Hellman Group 14
	Authentication & Health	Pre-Shared Key (PSK) / DPD (10 sec, 2 retries)
Phase 2 (IPsec)	Transform Set	ESP-AES-256 / ESP-SHA256-HMAC
	IPsec Mode & PFS	Transport Mode / PFS Group 14
Optimization	MTU / TCP MSS	IP MTU 1400 / TCP MSS 1360

Justification: Standard IPsec tunnels do not support multicast traffic. Utilizing a Generic Routing Encapsulation (GRE) tunnel wrapped in IPsec allows the architecture to pass dynamic routing protocols between Sydney and Canberra. Transport Mode is explicitly selected to eliminate redundant IP header overhead, as the GRE header already encapsulates the payload. IKEv2 replaces legacy IKEv1 to provide faster negotiations, built-in Dead Peer Detection (DPD) for rapid failover, and stronger resilience against DoS attacks.

VPN Encryption evaluation

Methods of Selection: The cryptographic suite was selected based on enterprise security baselines and hardware compatibility. AES-256 was chosen over 3DES or AES-128 to ensure military-grade confidentiality that exceeds standard corporate compliance. SHA-256 was mandated for hashing to eliminate the known collision vulnerabilities present in legacy MD5 and SHA-1 algorithms. Diffie-Hellman Group 14 (2048-bit) with Perfect Forward Secrecy (PFS) was selected to ensure that compromised future keys cannot decrypt historical session data.

Methods of Evaluation: The encryption deployment is evaluated using end-to-end throughput testing (e.g., iPerf3) to measure the CPU and bandwidth overhead introduced by the AES-256 encryption process. Network protocol analyzers (e.g., Wireshark) are utilized on the simulated ISP link to capture transit packets, verifying that internal IP headers and data payloads are successfully obfuscated within Encapsulating Security Payload (ESP) packets.

Routing Protocol

Routing Parameter	Configured Value
Protocol Type	Open Shortest Path First (OSPFv2)
Routing Domain	Single Area (Area 0)
Tunnel Interface	interface Tunnel0 (xxxx.xxxx.xxxx.xxxx/xx)
Multicast Support	Enabled (via GRE encapsulation)

Justification: OSPFv2 is deployed across the GRE tunnel to dynamically advertise the internal RFC1918 subnets (VLANs) between Sydney HQ and the Canberra branch. Unlike static routing, OSPF automatically detects network topology changes and converges rapidly without manual administrative intervention. Because OSPF relies on multicast traffic (224.0.0.5) to form neighbour adjacencies, the GRE tunnel is essential to encapsulate these multicast packets before they are encrypted and forwarded by the IPsec profile.

VLANs

Configuration (SYD-SW1)

```
SYD-SW1#sh run | section vlan
spanning-tree vlan 10,20,30,40,50,60,70,99,998-999 priority 24576
 switchport trunk native vlan 998
 switchport trunk allowed vlan 10,20,30,40,50,60,70,99
 switchport trunk native vlan 998
 switchport trunk allowed vlan 10,20,30,40,50,60,70,99
 switchport trunk native vlan 998
 switchport trunk allowed vlan 10,20,30,40,50,60,70,99
 switchport access vlan 20
 switchport access vlan 20
 switchport access vlan 40
 switchport access vlan 999
 switchport access vlan 999
 switchport access vlan 999
 switchport access vlan 999
 switchport access vlan 999
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 switchport access vlan 999
 switchport access vlan 999
SYD-SW1#
```

VLAN Segmentation & Rationale

To optimize broadcast domain boundaries, enforce granular access control policies, and mitigate security risks associated with flat Layer 2 networks, the network architecture implements a structured Virtual Local Area Network (VLAN) design.

The rationale for each operational and security segment is detailed below:

Core Operational Segments

- **VLAN 10 (STAFF):** Dedicated to general employee workstations and laptops. Separating standard user devices limits broadcast noise and prevents unprivileged users from directly accessing adjacent high-security segments without traversing Layer 3 security controls.

- **VLAN 20 (SERVER-INFRA):** Isolates core server assets (domain controllers, database applications, virtual hosts) into a restricted zone. Access to this segment is tightly controlled via Layer 3 Access Control Lists (ACLs) or firewall policies, permitting only necessary service ports.

- **VLAN 60 (ITADMIN) & VLAN 70 (EXEC):** High-value target segments. VLAN 60 separates administrative hosts to allow stricter monitoring and privileged access management (PAM). VLAN 70 isolates executive endpoints carrying sensitive business data, allowing targeted security monitoring and stringent inbound access policies.

Isolated & Low-Trust Segments

- **VLAN 30 (GUEST):** Provides Internet-only connectivity for non-company devices. This segment is isolated from all internal corporate subnetworks via firewall policies to neutralize the risk of rogue endpoint compromise or unauthorized scanning.

- **VLAN 40 (RESTRICTED):** Segregates high-risk endpoints, contractors, or non-compliant devices. Isolate this traffic to apply tailored internet access restrictions, zero-trust policies, or temporary monitoring controls.

- **VLAN 50 (DEVICES):** Houses peripherals, smart devices, and printers. Because IoT devices and networked printers frequently lack advanced endpoint security controls and generate excessive broadcast traffic, isolating them prevents lateral movement in the event of an IoT compromise.

Infrastructure Security & System Segments

- **VLAN 99 (MANAGEMENT):** Dedicated exclusively to in-band device management traffic (SSH, SNMP, HTTPS) for network hardware (switches, routers, firewalls). Restricting management interfaces to a dedicated VLAN ensures administrative controls remain unreachable from standard user subnets.

- **VLAN 998 (NATIVE):** Used strictly as a dummy native VLAN across 802.1Q trunk links. Disabling untagged traffic on **VLAN 1** and shifting native traffic to an unused ID mitigates VLAN hopping and double-tagging security exploits.

- **VLAN 999 (BLACKHOLE):** Configured as a dead-end segment for unallocated switch ports. Unused physical ports are assigned to this non-routable VLAN and administratively shutdown to prevent unauthorized "plug-and-play" network access.

VLAN Category	VLAN ID	SUBNET / NAME	RATIONALE	SECURITY
User Segments	10, 60, 70	STAFF, ITADMIN, EXEC	Role-Based Access Control (RBAC) & broadcast isolation	Layer 3 ACLs, Dynamic VLANs (802.1X)
Resource Segments	20, 50	SERVER-INFRA, DEVICES	Protect core assets & isolate legacy/IoT threats	Layer 3 ACLs
Untrusted Segments	30, 40	GUEST, RESTRICTED	Zero-trust isolation for non-corporate assets	No Inter-VLAN Routing

Hardening Segments	99, 998, 999	MANAGEMENT, NATIVE, BLACKHOLE	Layer 2/3 control-plane security & port hygiene	Unrouted VLANs, 802.1Q Native Tagging, Port Shutdown

Verification (SYD-SW1)

```
SYD-SW1#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   STAFF                            active
20   SERVER-INFRA                     active    Gi1/0/5, Gi1/0/6
30   GUEST                            active
40   RESTRICTED                       active    Gi1/0/7
50   DEVICES                          active
60   ITADMIN                          active
70   EXEC                             active
99   MANAGEMENT                       active
998  NATIVE                           active
999  BLACKHOLE                        active    Gi1/0/8, Gi1/0/9, Gi1/0/10
                                                Gi1/0/11, Gi1/0/12, Gi1/0/13
                                                Gi1/0/14, Gi1/0/15, Gi1/0/16
                                                Gi1/0/17, Gi1/0/18, Gi1/0/19
                                                Gi1/0/20, Gi1/0/21, Gi1/0/22
                                                Gi1/0/23, Gi1/0/24, Gi1/1/1
                                                Gi1/1/2, Gi1/1/3, Gi1/1/4
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
40   enet  100040     1500  -      -      -        -    -        0      0
50   enet  100050     1500  -      -      -        -    -        0      0
60   enet  100060     1500  -      -      -        -    -        0      0
70   enet  100070     1500  -      -      -        -    -        0      0
99   enet  100099     1500  -      -      -        -    -        0      0
998  enet  100998     1500  -      -      -        -    -        0      0
999  enet  100999     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

Remote SPAN VLANs
------------------------------------------------------------------------------


Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
```

Network Security

Infrastructure Security

Device Hardening & Administrative Access: All network infrastructure devices (routers and switches) enforce SSH v2 for encrypted remote management, with unencrypted protocols

(Telnet, HTTP) explicitly disabled. Administrative access is restricted exclusively to authorized IT personnel.

Physical & Boundary Security: Switch port density at the Canberra branch is strictly managed using Switch Port Security to limit unauthorized physical hardware connections. Unused switch ports are shut down and assigned to an isolated blackhole VLAN by default.

Time & System Integrity: Network devices synchronize system clocks against a central Network Time Protocol (NTP) server to guarantee accurate, audit-ready timestamping across all security and event logs.

Simple Network Management Protocol (SNMP): LibreNMS Server centrally collect, monitor, and manage data from networked devices like routers, switches, and servers.

Layer 2 Security

Port-channel: Etherchannel links between all Gateway Routers and Switches, to prevent disruption to the network at layer 2 and increase bandwidth between switches.

Network Segmentation: The architecture segments site traffic into five distinct, dedicated VLANs per location—Infrastructure/Servers, Staff, IT Admin, Executive, Guest, and Restricted Personal—to prevent unrestricted broadcasts and reduce the Layer 2 attack surface.

Trunking & Port Protection: Inter-switch links use IEEE 802.1Q trunking with explicit VLAN tagging, disabling Dynamic Trunking Protocol (DTP) to prevent VLAN hopping attacks. Switch Port Security enforces maximum MAC address limits on access ports to mitigate MAC address flooding.

802.1x

Port-Based Network Access Control (PNAC): Wired access across the infrastructure enforces IEEE 802.1X identity verification prior to granting full network access.

Centralized Authentication Integration: Switch ports interface with a centralized Active Directory Domain Controller and Network Policy Server (NPS) acting as the RADIUS authentication backend located at Sydney HQ.

Dynamic VLAN Assignment & Isolation: Validated user credentials dynamically assign endpoints to their designated departmental VLANs (e.g., Staff or IT Admin). Unauthenticated or unauthorized devices are dynamically isolated into a Restricted Personal VLAN with zero access to internal enterprise resources.

Layer3 Security

Encrypted Multi-Site Transit: Inter-site communication between Sydney HQ and the Canberra branch is secured over the public WAN using an IPSec VPN tunnel, guaranteeing confidentiality and data integrity via strong encryption.

Granular Traffic Control (ACLs): Inter-VLAN traffic is strictly regulated using Layer 3 Access Control Lists (ACLs). ACLs enforce lease boundaries by blocking Guest and Restricted VLANs from accessing internal server infrastructure while permitting IT Admin access for management operations.

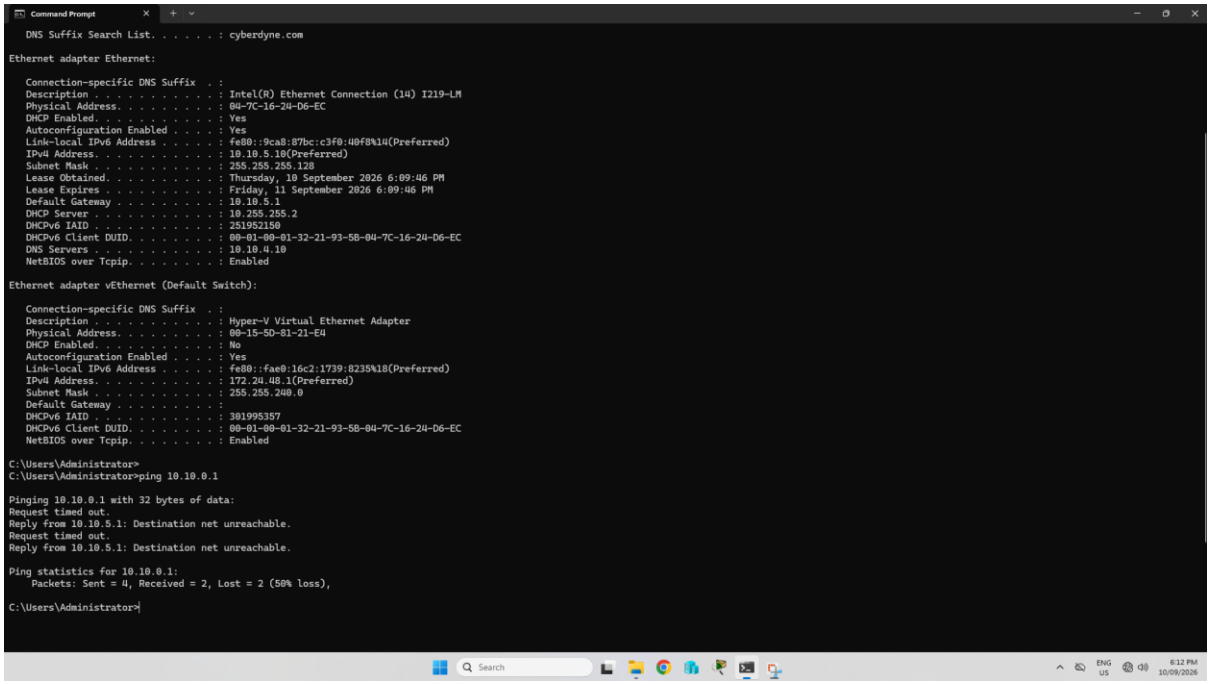
Edge Address Translation: Outbound traffic to public networks is processed through Network Address Translation / Port Address Translation (NAT/PAT) on the edge router, hiding internal RFC1918 IPv4 addressing schemas from external exposure.

ACL Testing

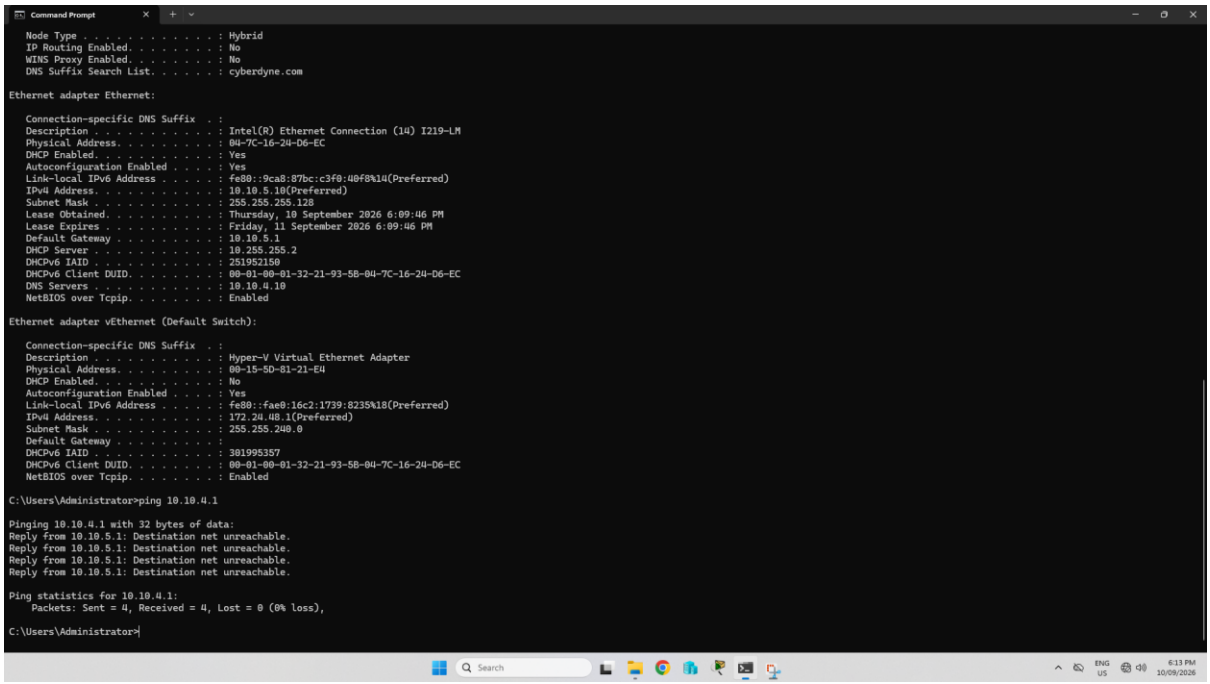
Testing Documentation & Presentation

Restricted VLAN Isolation

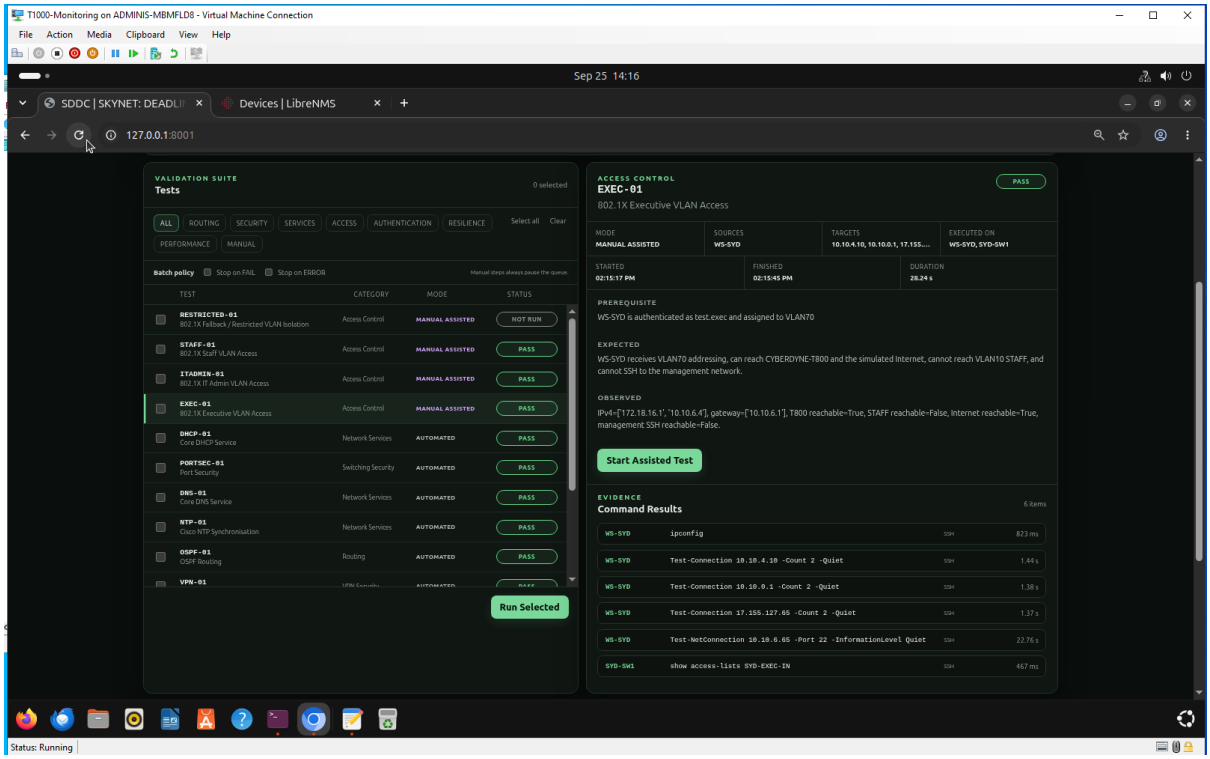
Test	Instructions	Commands	Expected Result	Remediation
Internal Access	VLAN40 workstation, test access to VLAN10 & VLAN20.	ping 10.10.0.1 ping 10.10.4.1	Both fail	Check VLAN 40 ACL: show ip interface vlan 40 show access-lists
External Access	VLAN 40 test access to the external network	ping 17.155.127.65	Fail	Check VLAN 40 ACL: show ip interface vlan 40 show access-lists
Management Access	VLAN 40 test access SSH to switch	ssh -l <username> <Management-IP>	SSH Fails	Check management-network deny rules.



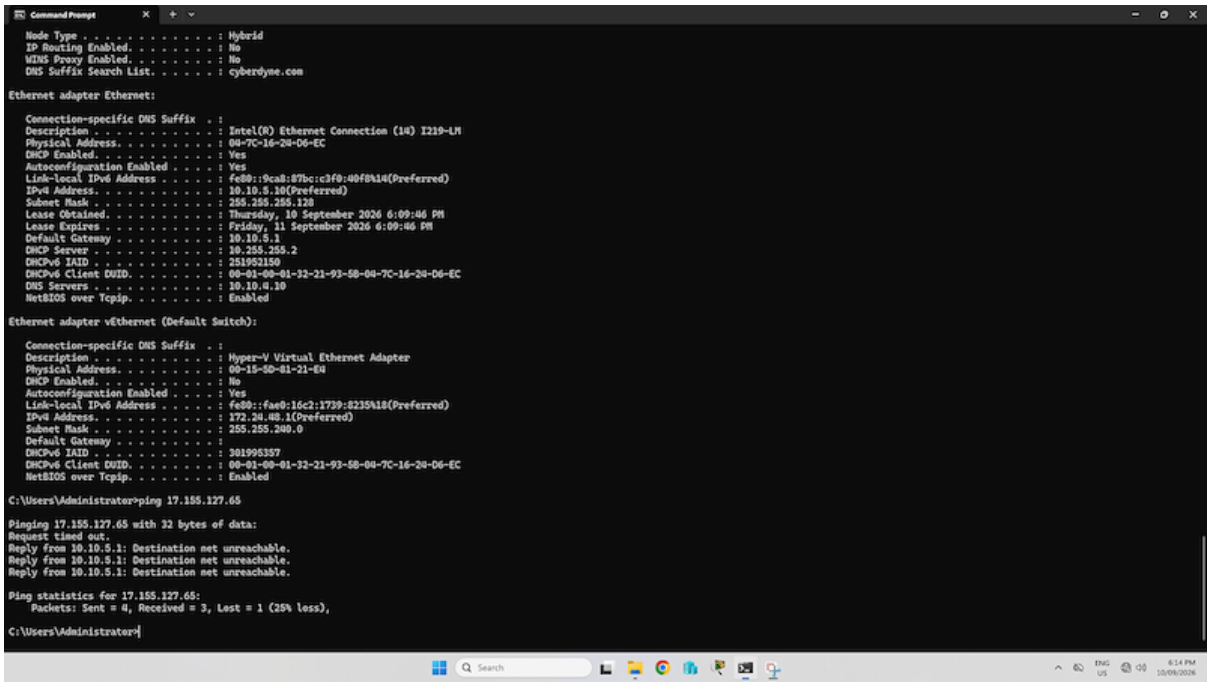
VLAN40 workstation, test access to VLAN10



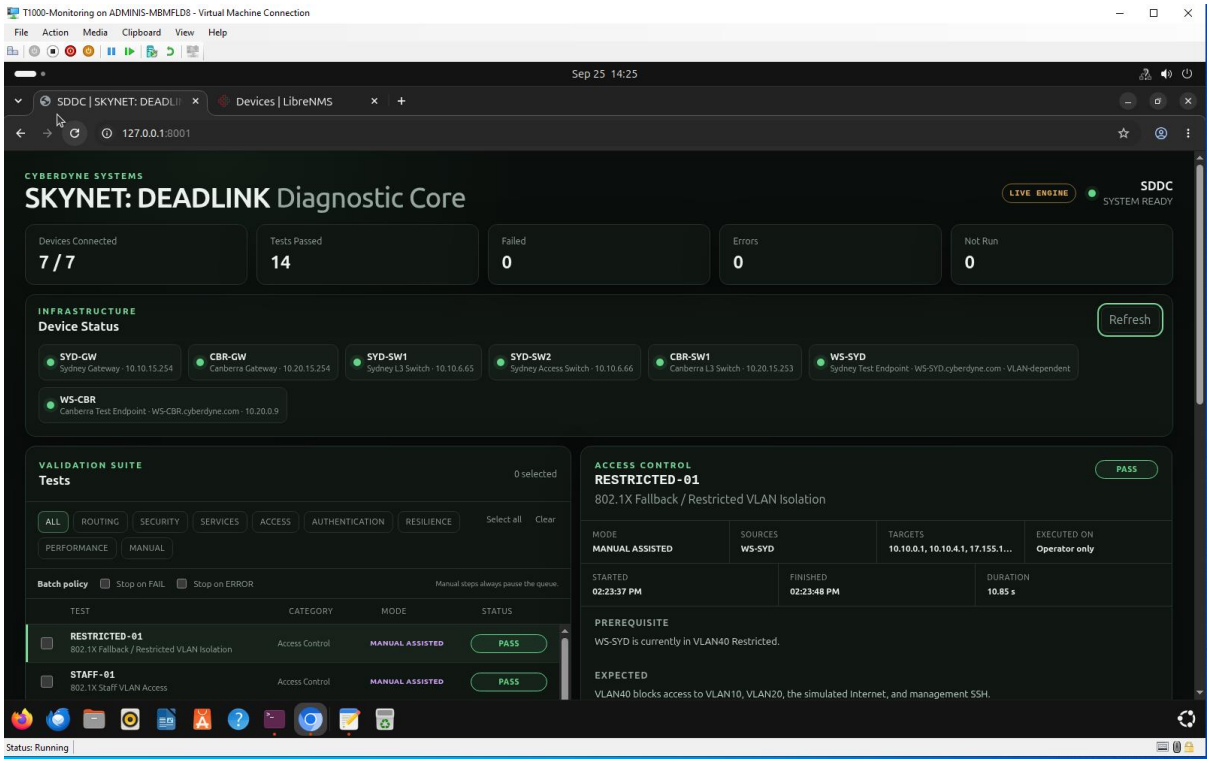
VLAN40 workstation, test access to VLAN20



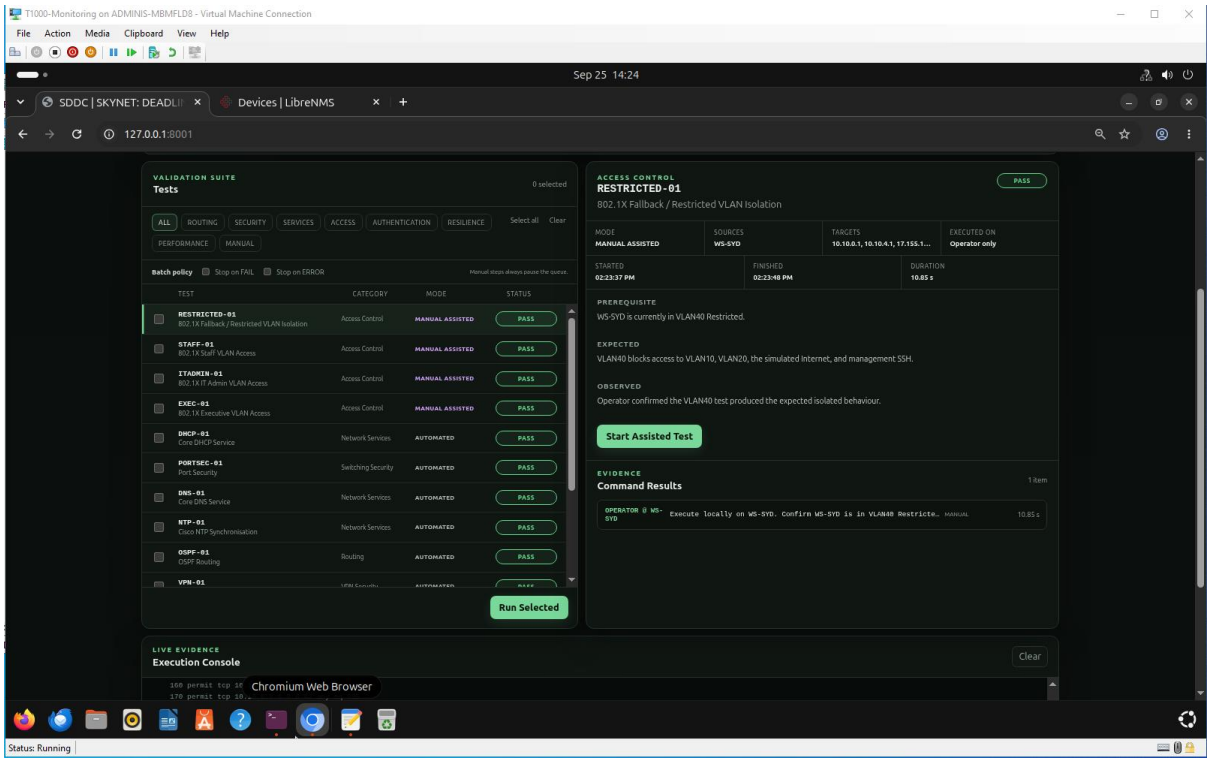
Testing executive access (using custom console)



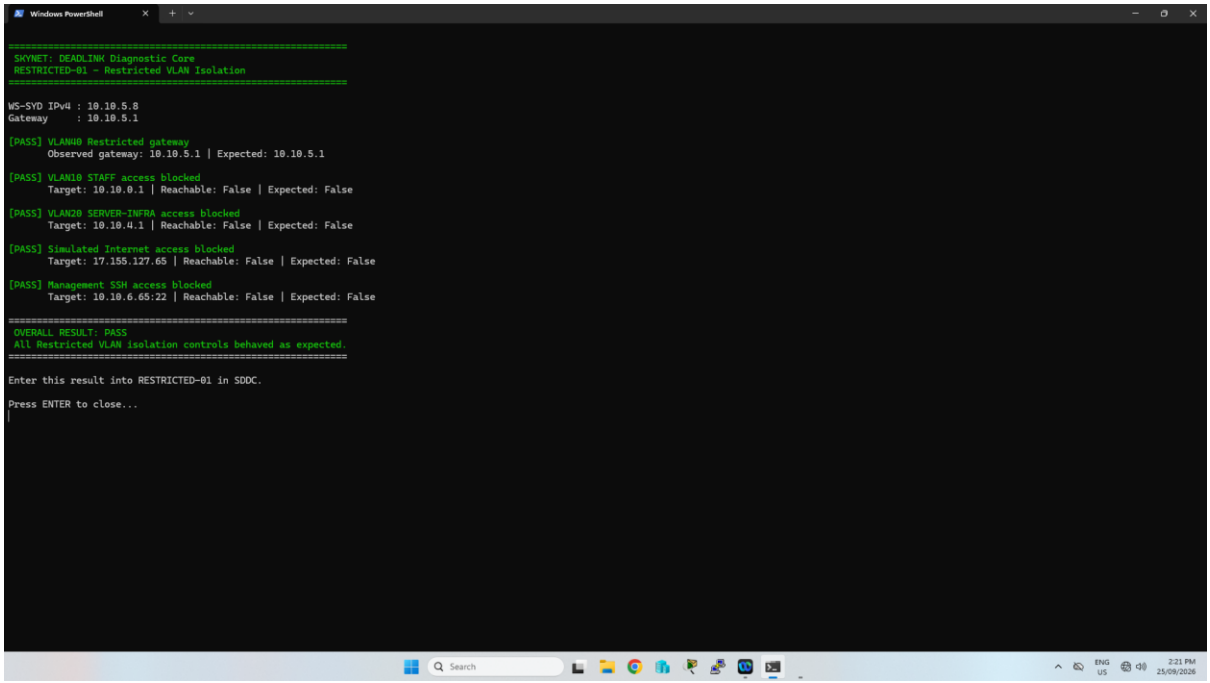
VLAN 40 test access to the external network



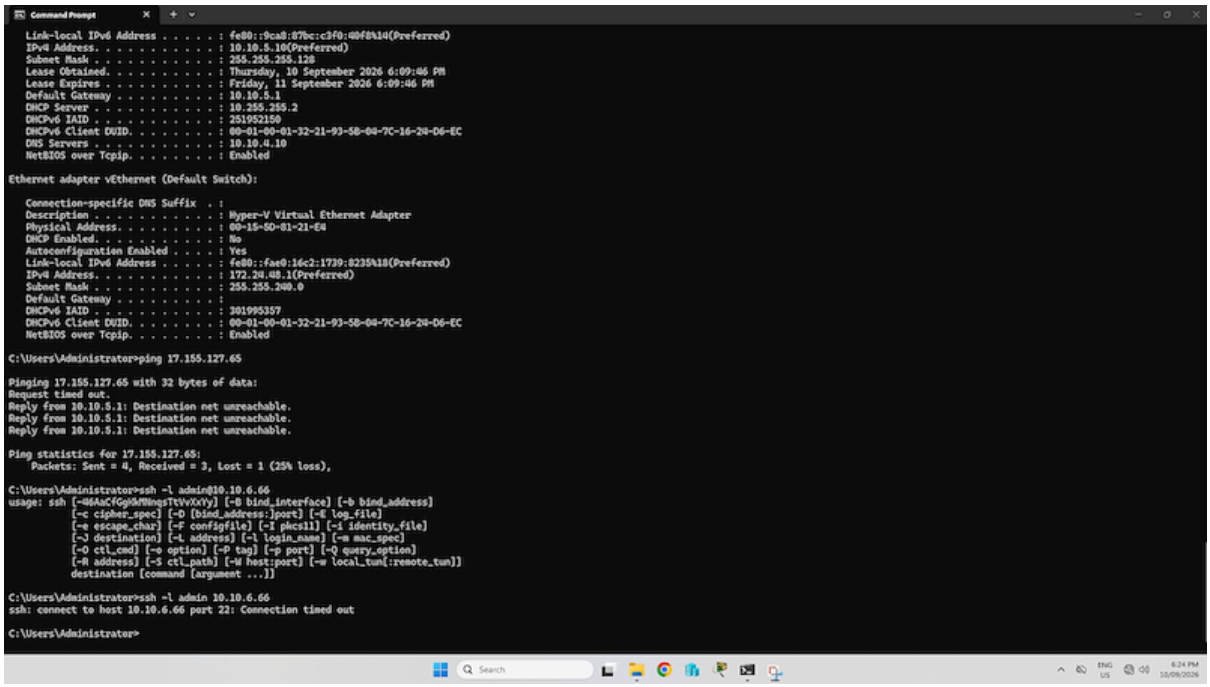
Testing restricted VLAN on custom console



Another restricted VLAN test



Another restricted VLAN test

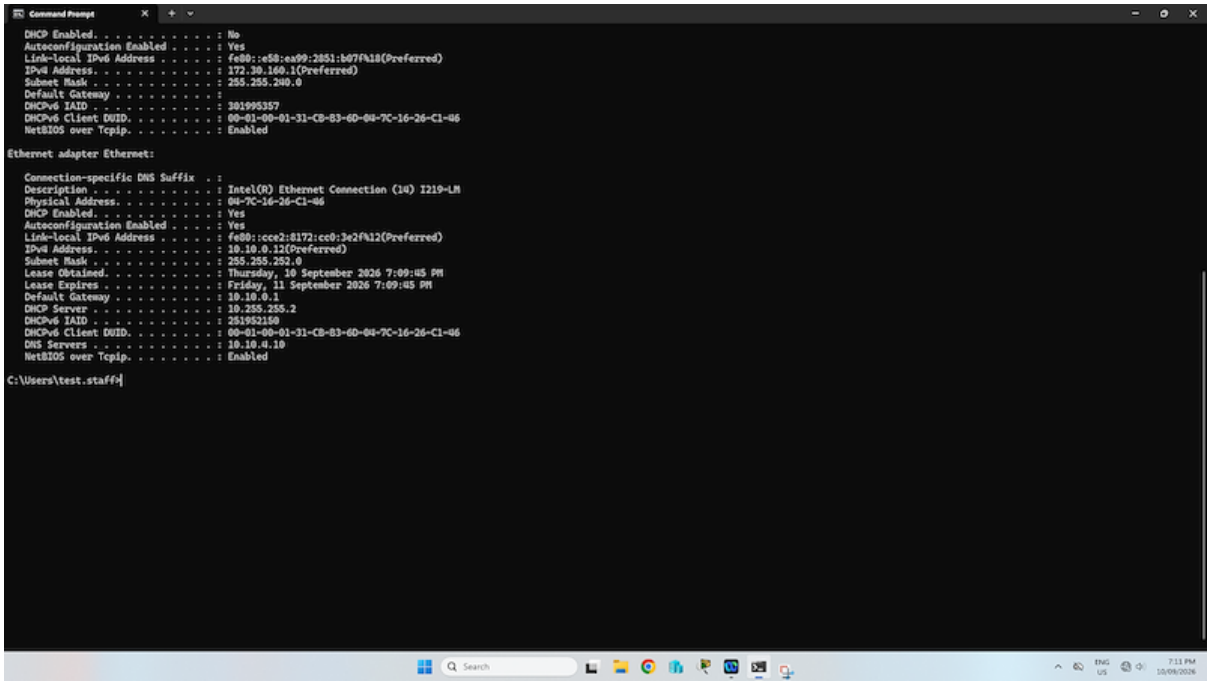


VLAN 40 test access SSH to switch

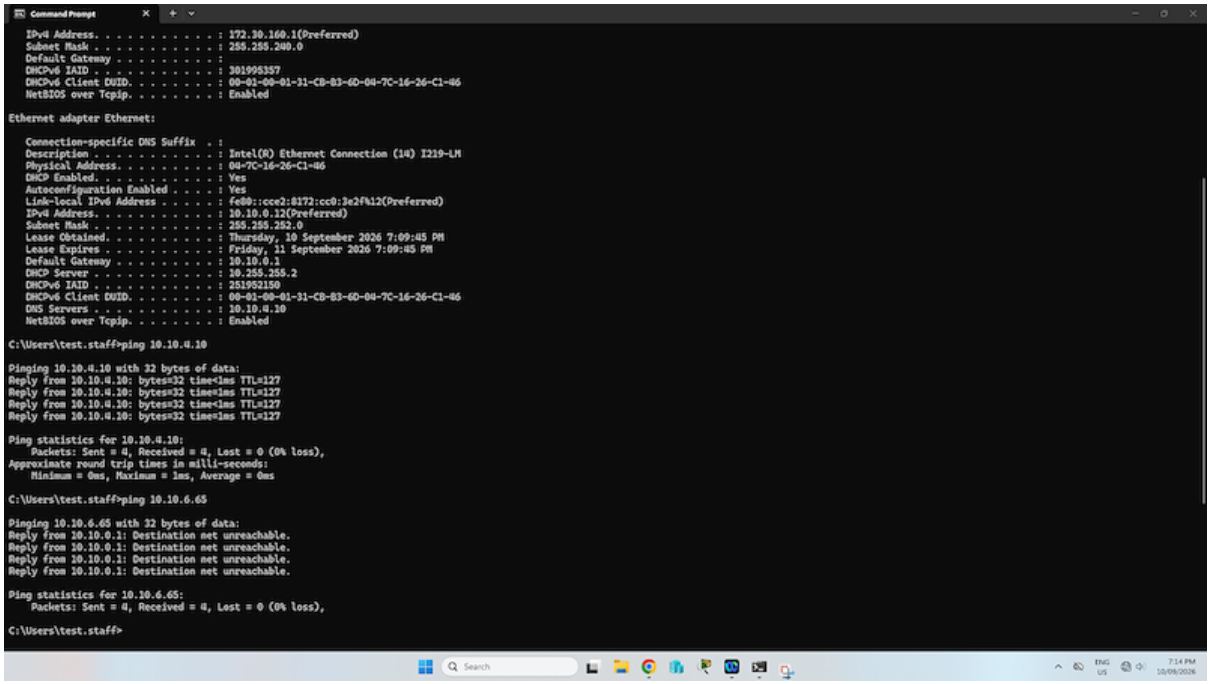
Staff VLAN Access Control

Test	Instructions	Commands	Expected Result	Remediation
Staff VLAN Access	Login to WS-SYD using test.staff, confirm it is dynamically	WS-SYD: • ipconfig • ping 10.10.4.10 • ping 10.10.6.65	WS-SYD: Receives `10.10.0.x/22` Access to	WS-SYD: Verify 802.1X VLAN assignment

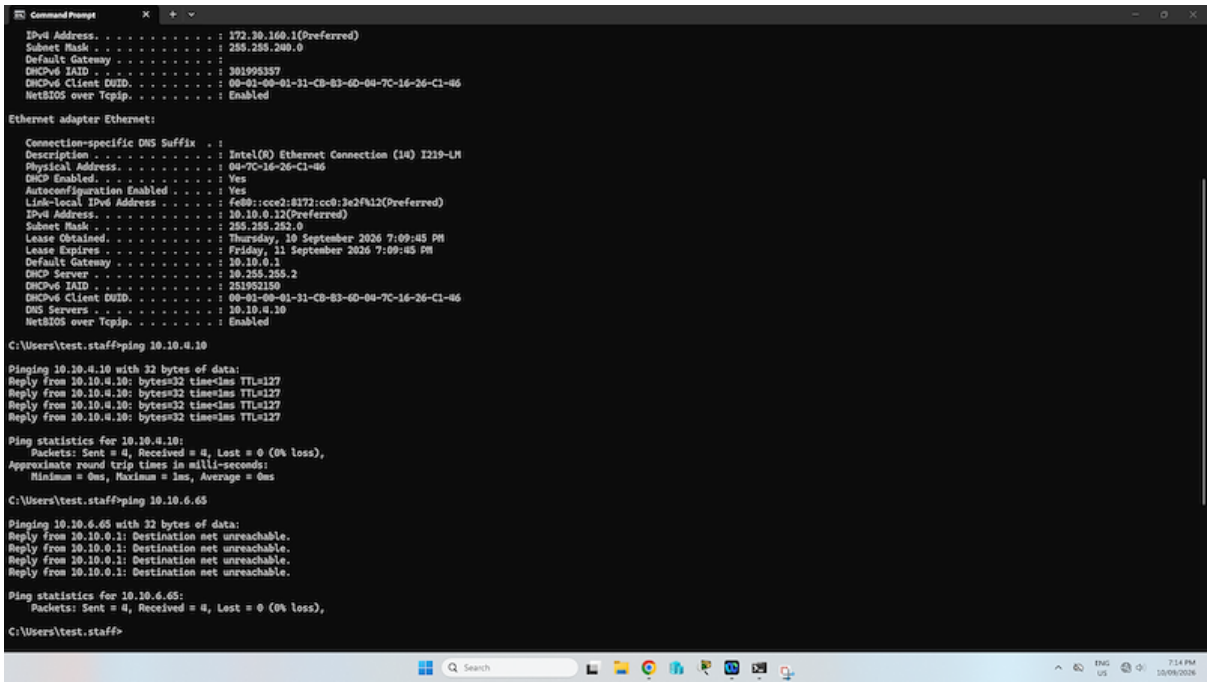
	assigned to VLAN10. Test access to an authorised server resource Then attempt to access the VLAN99.	SYD-SW1: show access-lists SYD-STAFF-IN	`10.10.4.10` Blocked Access to Management `10.10.6.65` SYD-SW1: ACL counters show the relevant permit/deny matches.	SYD-SW1: `SYD-STAFF-IN` ACL entries and that the ACL is applied inbound to `Vlan10`.



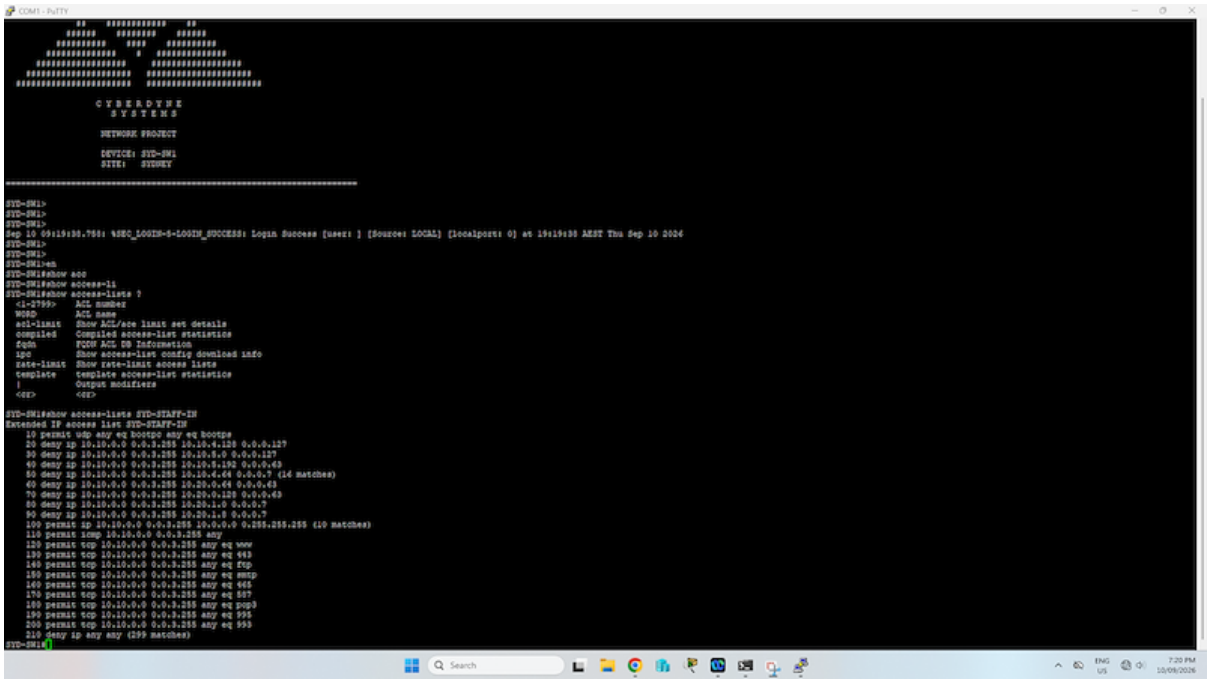
Dynamically assigned to VLAN1



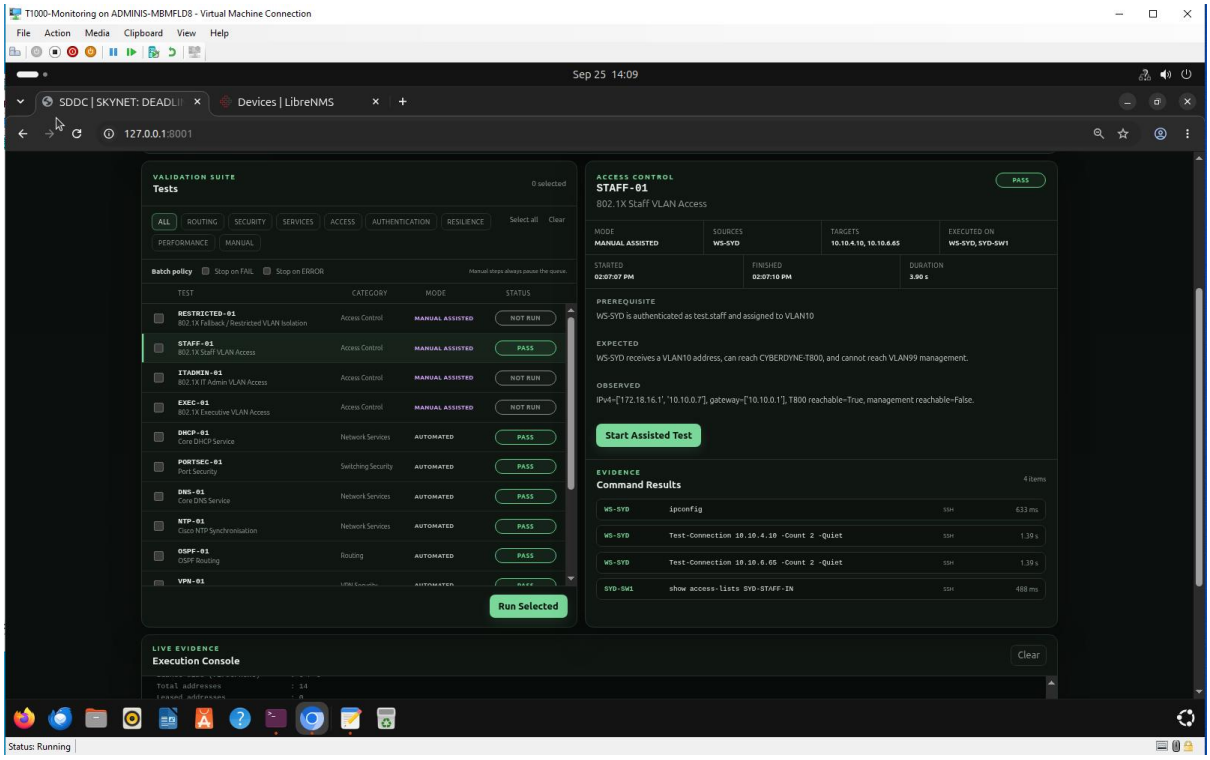
Test access to an authorised server resource



Test access to an authorised server resource



Show ACL

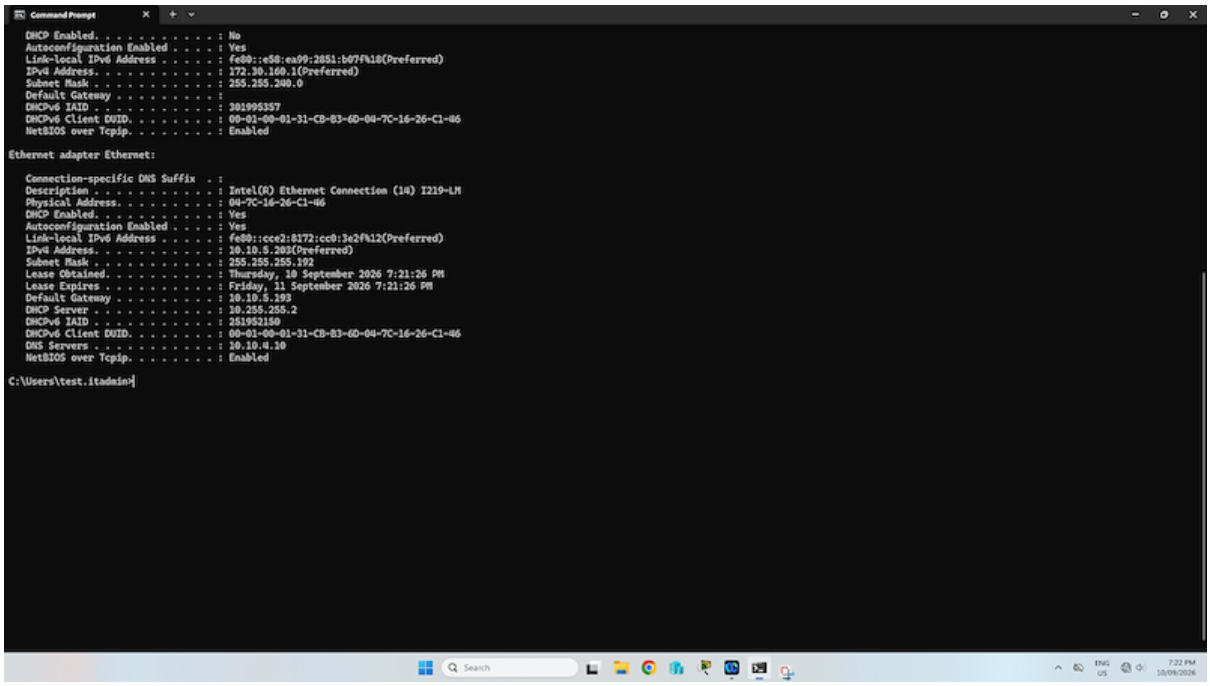


Using custom console

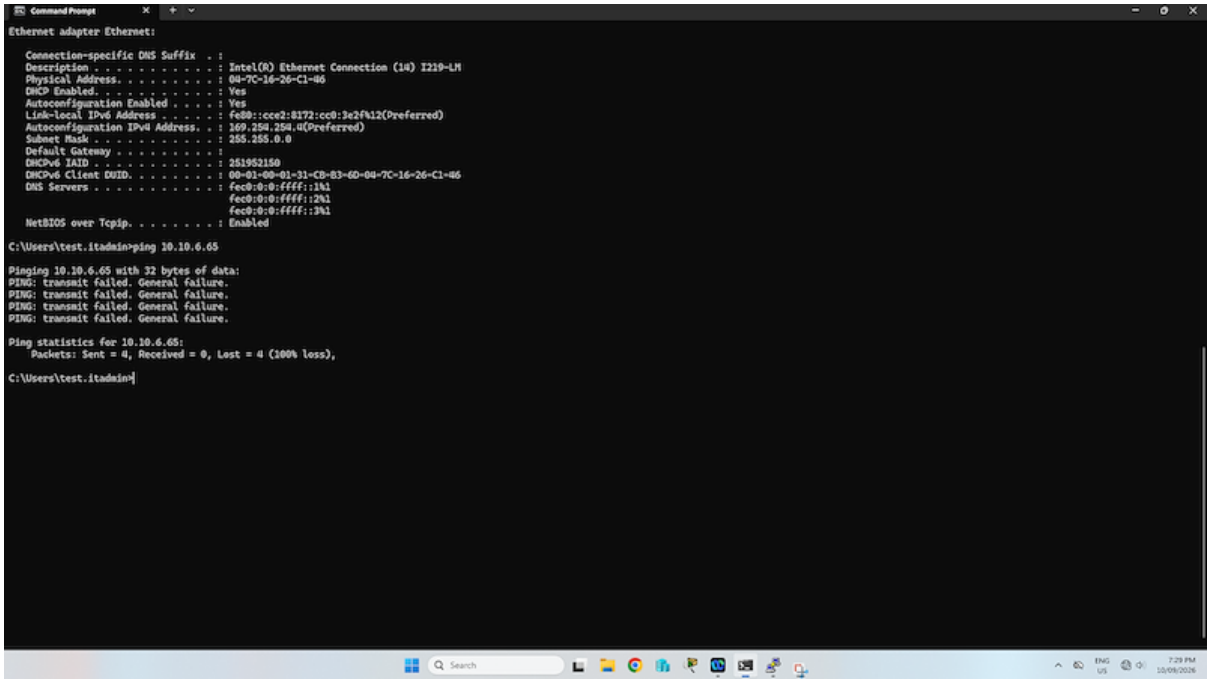
IT Admin Management Access

Test	Instructions	Commands	Expected Result	Remediation
IT Admin Management Access	Login to WS-SYD using test.itadmin, confirm it is dynamically	WS-SYD: • ipconfig • ping 10.10.6.65 • ping 10.10.6.66	WS-SYD: Receives `10.10.5.19x/26` Gateway:	WS-SYD: Verify 802.1X VLAN60 assignment

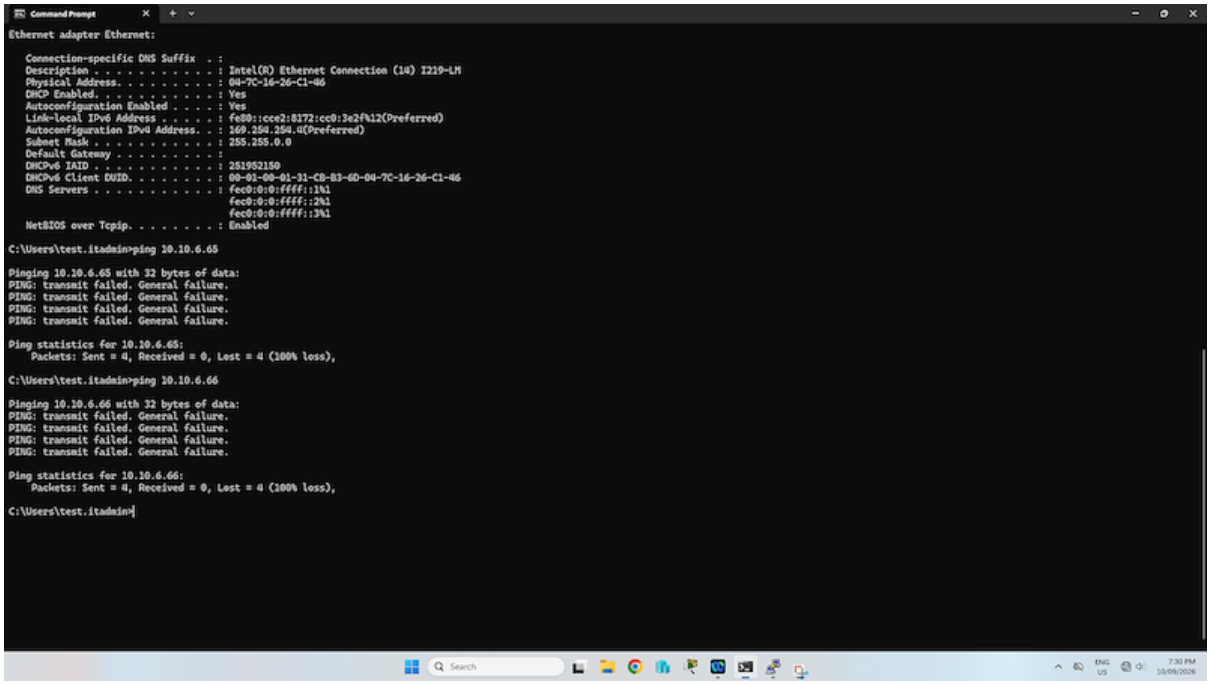
	assigned to VLAN60. Test access to access the Managment VLAN99	SYD-SW1: show access-lists SYD-ITADMIN-IN	`10.10.5.193` Pings to SYD-SW1: `10.10.6.65` SYD-SW2: `10.10.6.66` successful. ACL permit counters increase.	SYD-SW1: `SYD-ITADMIN-IN` ACL entries and that the ACL is applied inbound to `Vlan60`.



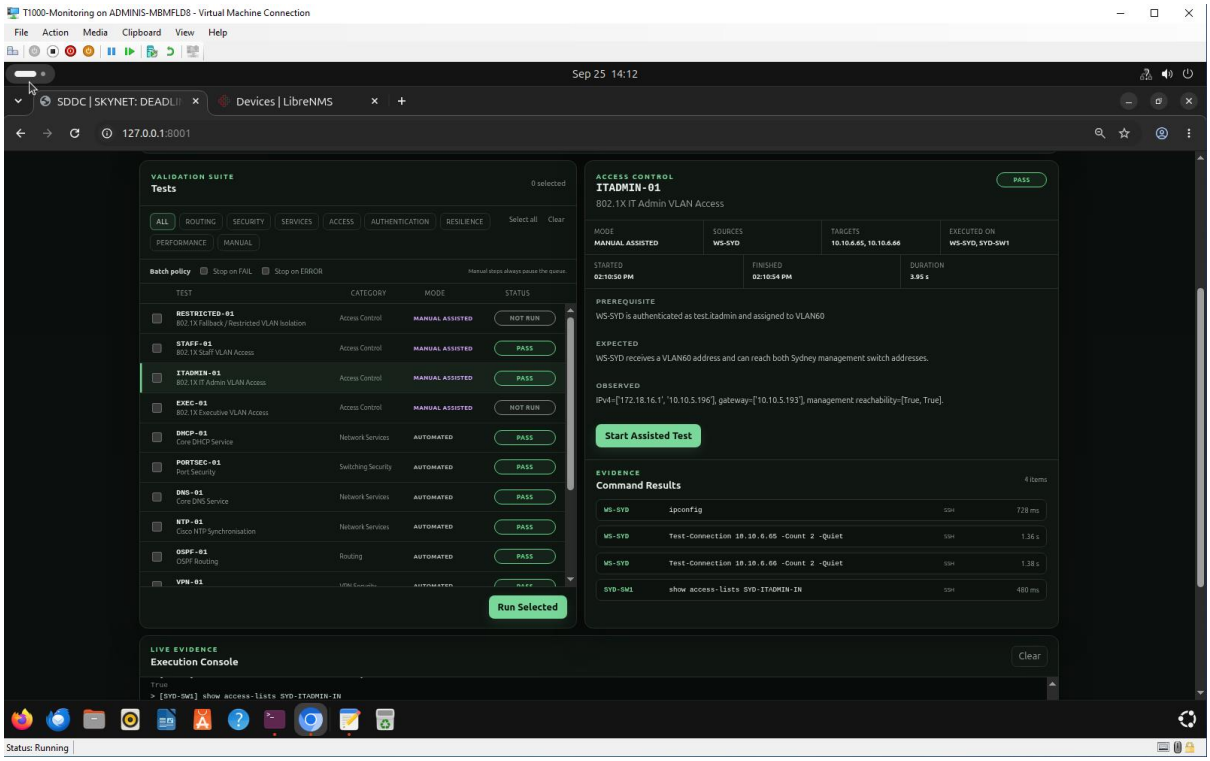
Confirm it is dynamically assigned to VLAN60.



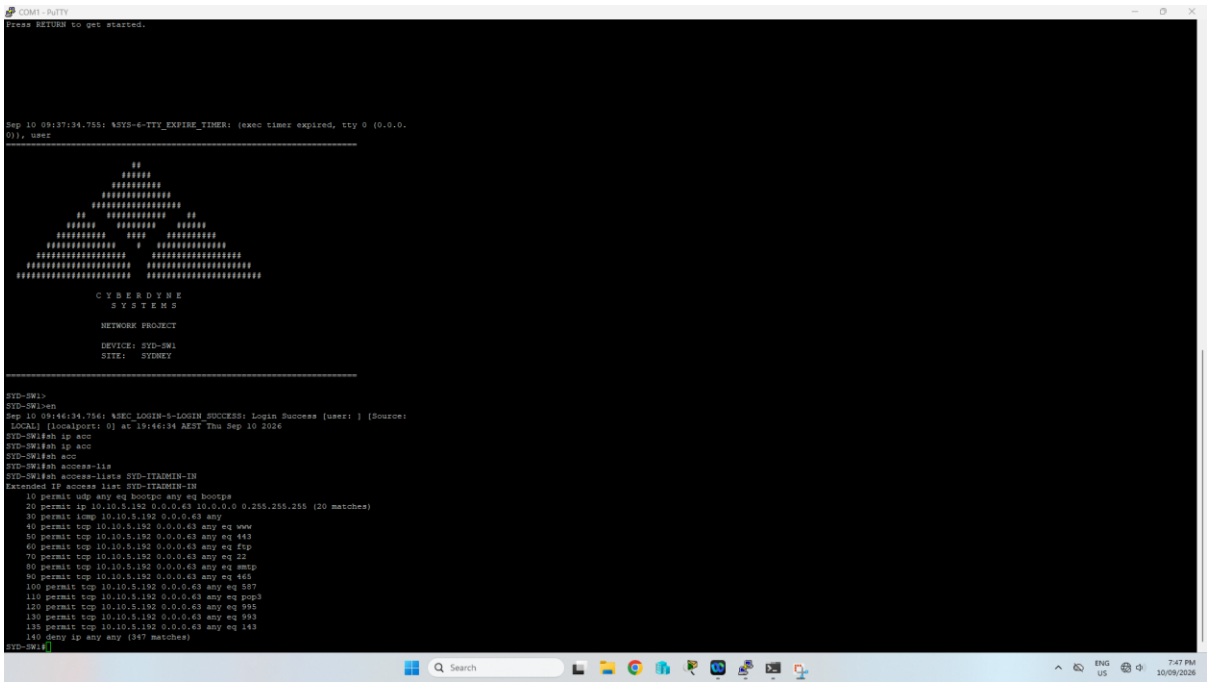
Test access to access the Management VLAN99



Test access to access the Management VLAN99



Using custom console

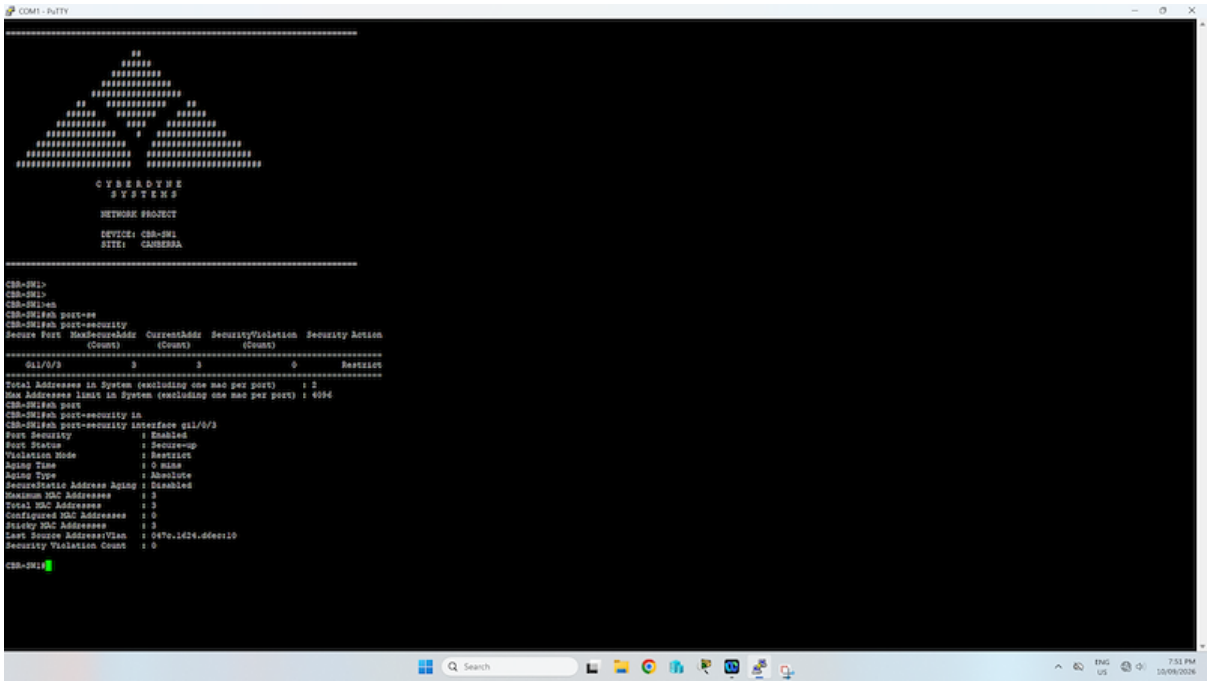


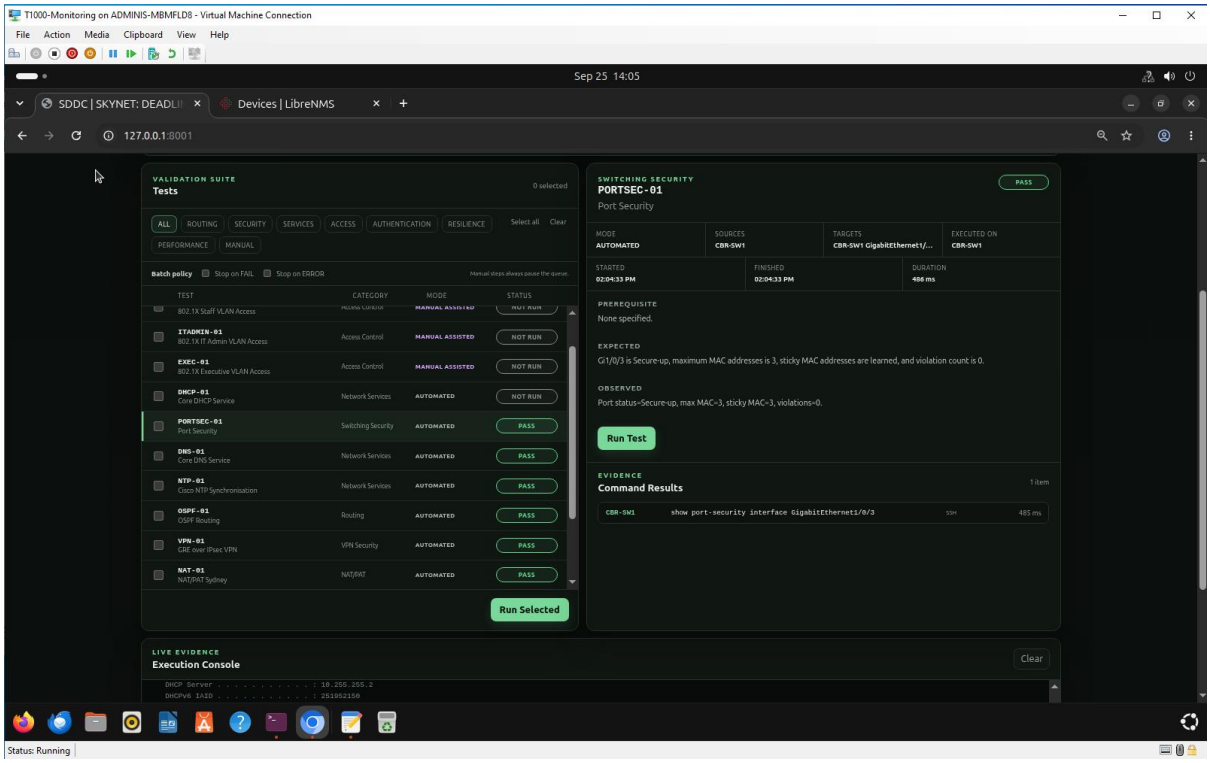
Show ACLs

Port Security

Test	Instructions	Commands	Expected Result	Remediation
Port Security	CBR-SW1 Gi1/0/3: • Verify port	• show port-security interface GigabitEthernet1	• Port status is Secure-up. • Maximum MAC	Check: • Port-security

	security is enabled. • Confirm that sticky MAC addresses are learned. • Check the maximum MAC limit and violation count	/0/3	addresses = 3. • Sticky MAC addresses are learned. • Violation count = 0.	configuration. • Maximum MAC setting. • Violation mode.

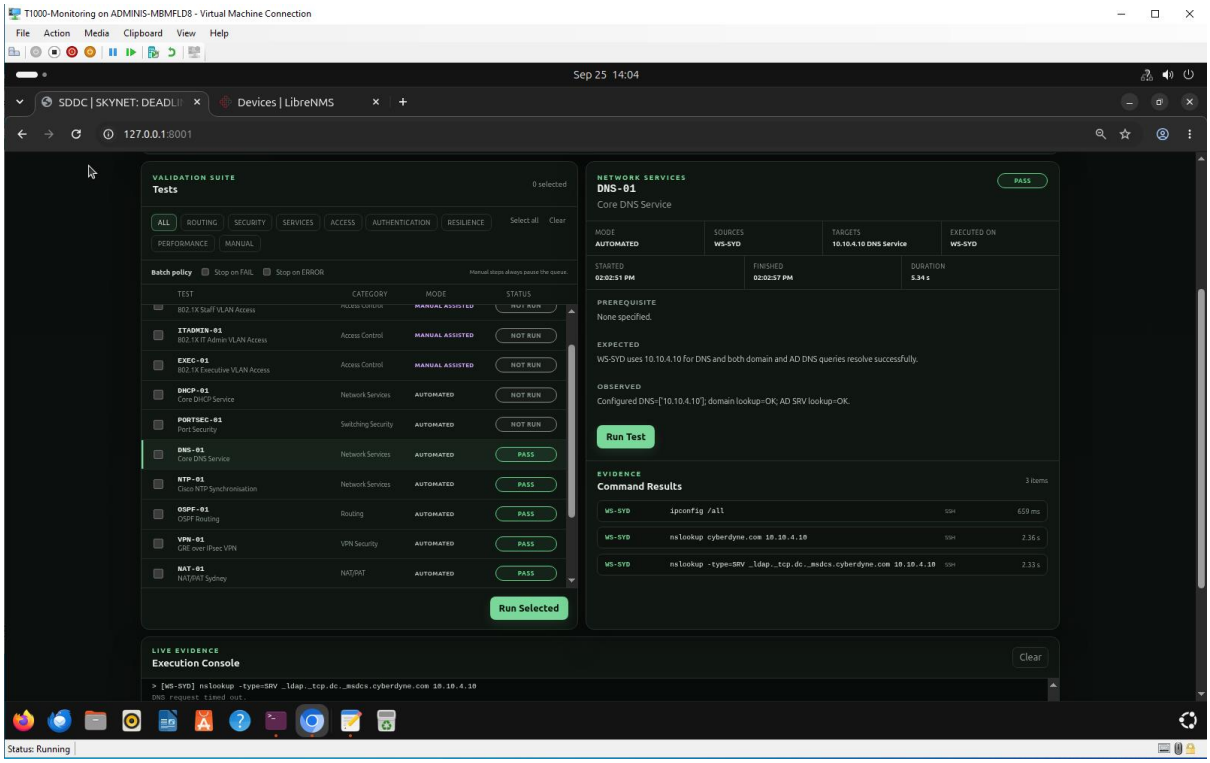




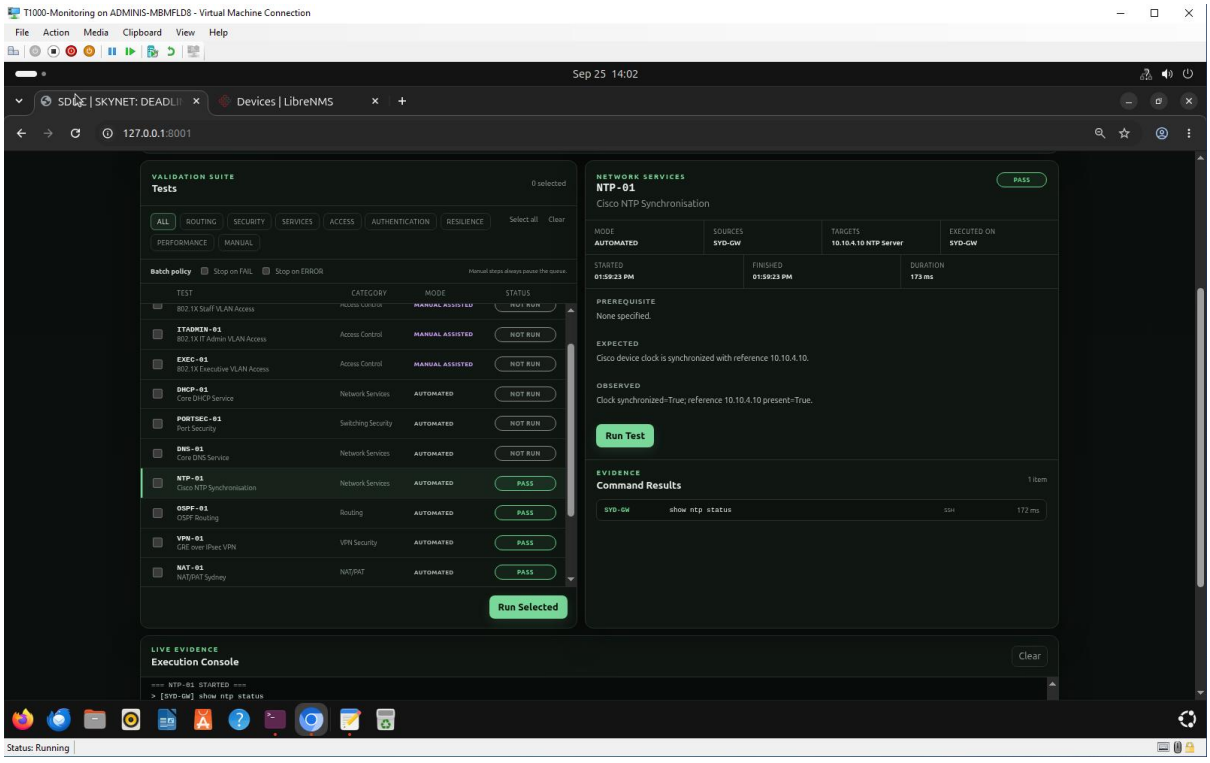
CLI (top) and custom console (bottom)

Core Network Services

Test	Instructions	Commands	Expected Result	Remediation
Core Network Services	From WS-SYD: Verify DNS resolution and confirm the client is using CYBERDYNE-T800 for DNS. Confirm NTP is synchronized on a network device.	On WS-SYD: • ipconfig /all • nslookup cyberdyne.com On Router/Switch: • show ntp status	WS-SYD shows DNS server 10.10.4.10 DNS queries resolve successfully Cisco device reports Clock is synchronized with reference 10.10.4.10	Verify: • DNS service on CYBERDYNE-T800 client DNS settings/DHCP options, • Routing to VLAN20, Windows Time/NTP service • Cisco NTP configuration.



Using custom console

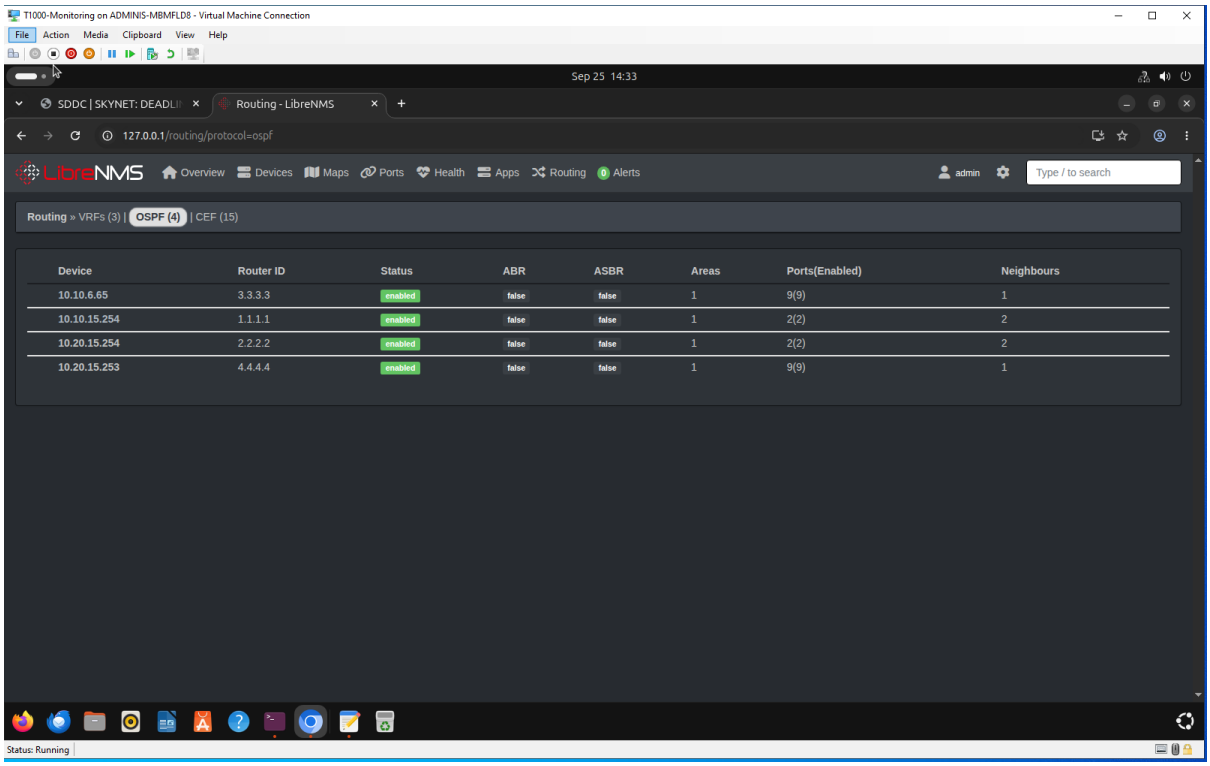


show NTP status

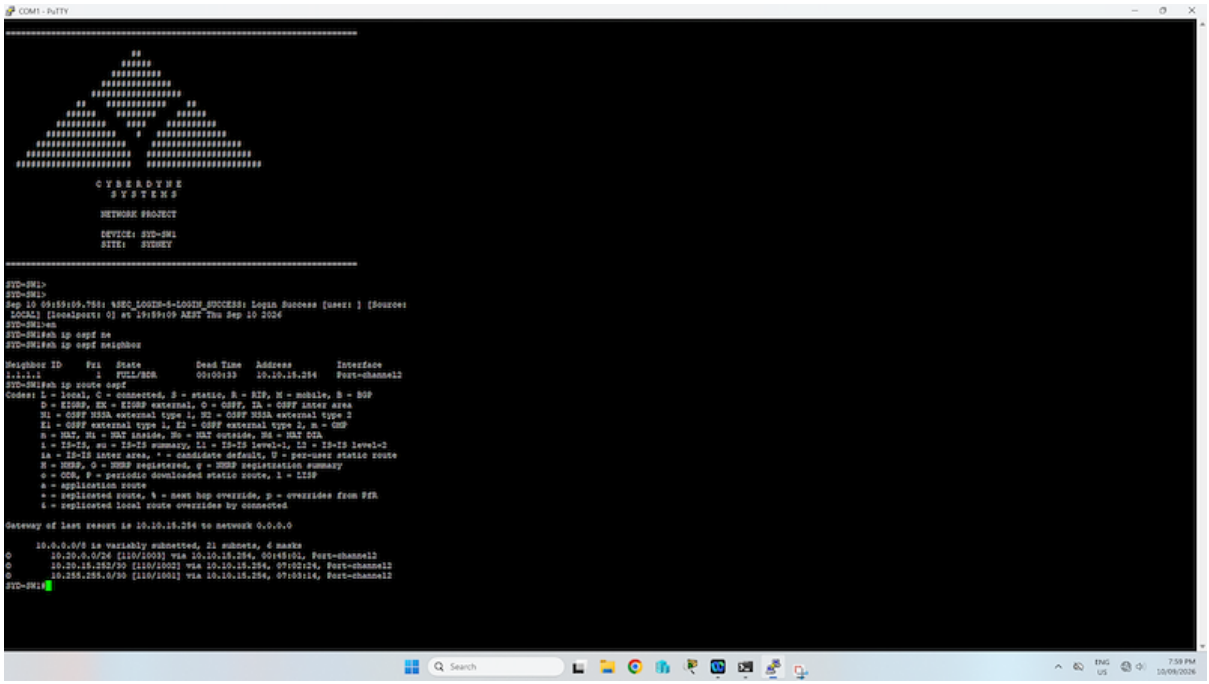
OSPF Routing

Test	Instructions	Commands	Expected Result	Remediation

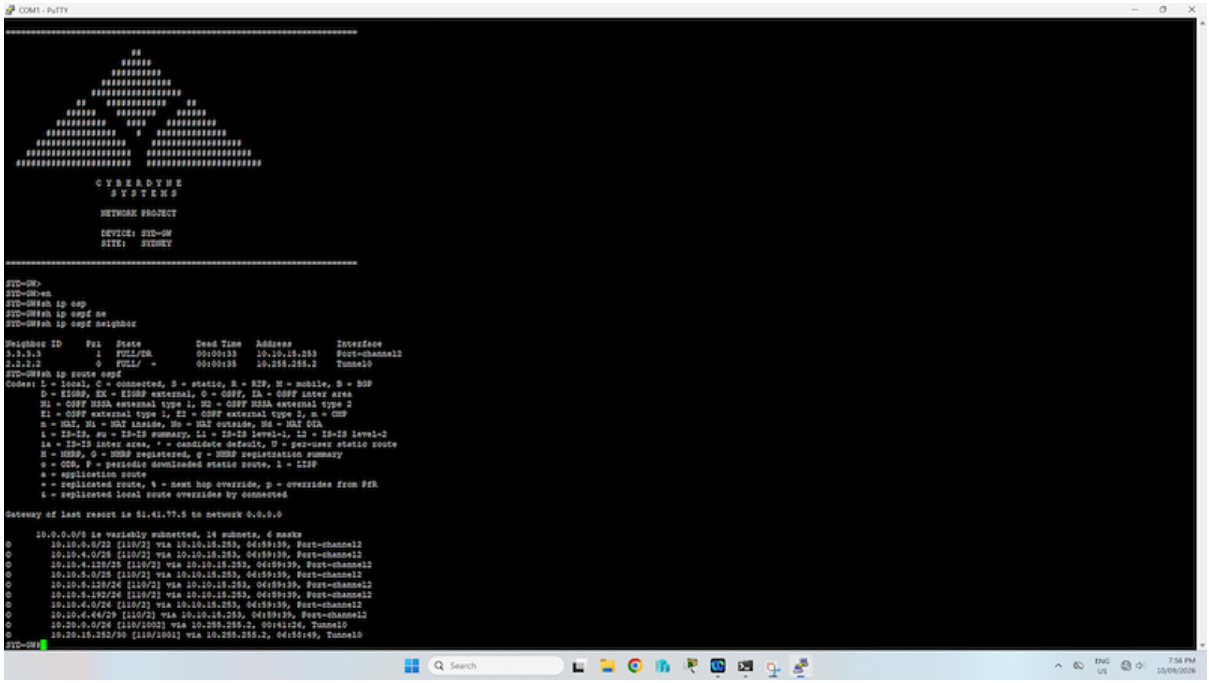
OSPF Routing	Verify OSPF adjacencies between the routers and Layer 3 switches at both sites, including the GRE tunnel between SYD and CBR. Confirm remote site routes are learned dynamically.	SYD-GW / CBR-GW / L3 switches: • show ip ospf neighbor • show ip route ospf	OSPF: • Neighbours show FULL including the SYD-GW ↔ CBR-GW adjacency over Tunnel0. • Remote site networks appear in the routing table as OSPF-learned routes.	Verify: • OSPF network statements • Area configuration • Router IDs • Passive-interface settings • Tunnel0 status • Routed EtherChannel connectivity



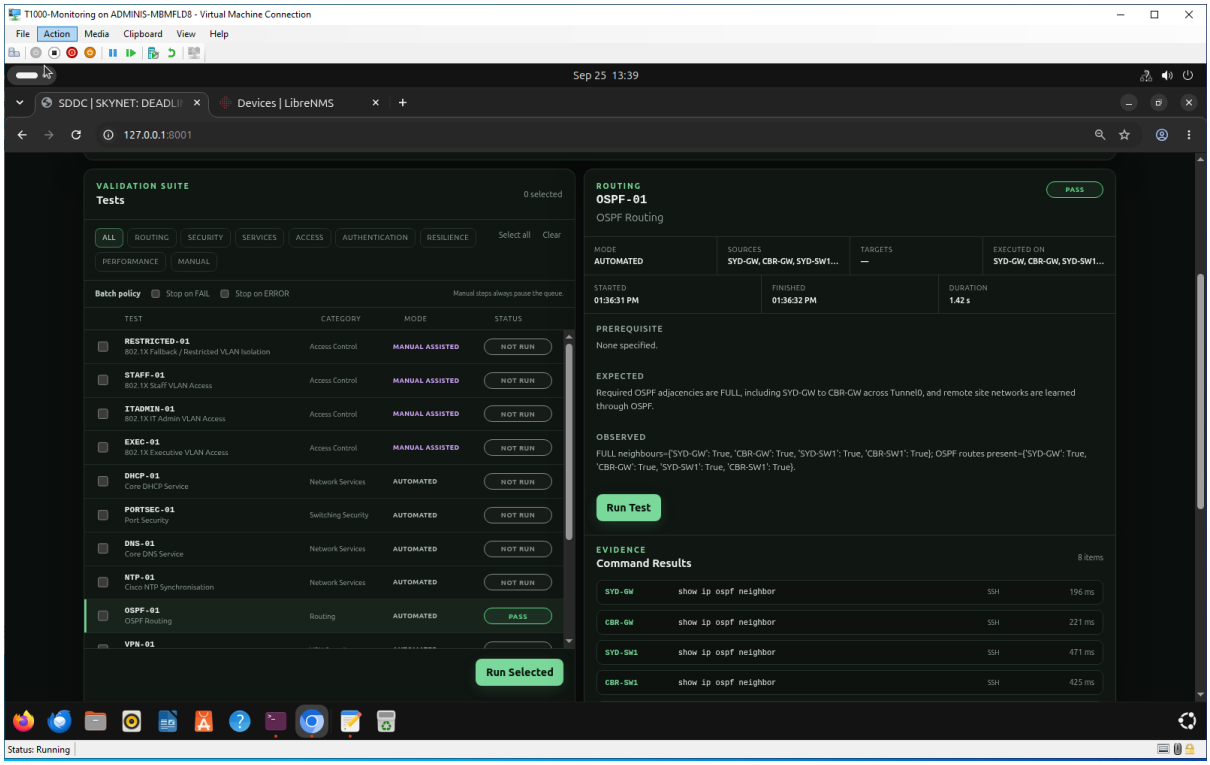
On the LibreNMS Server



show ip ospf neighbor



show ip route ospf

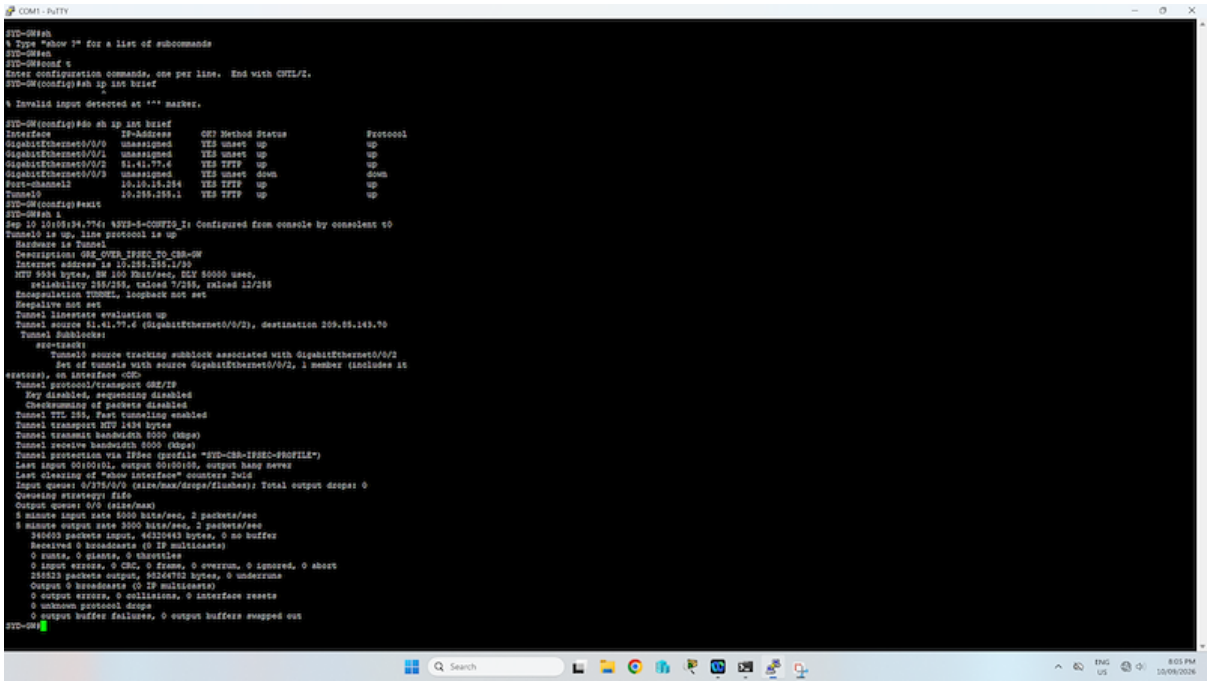


Using custom console

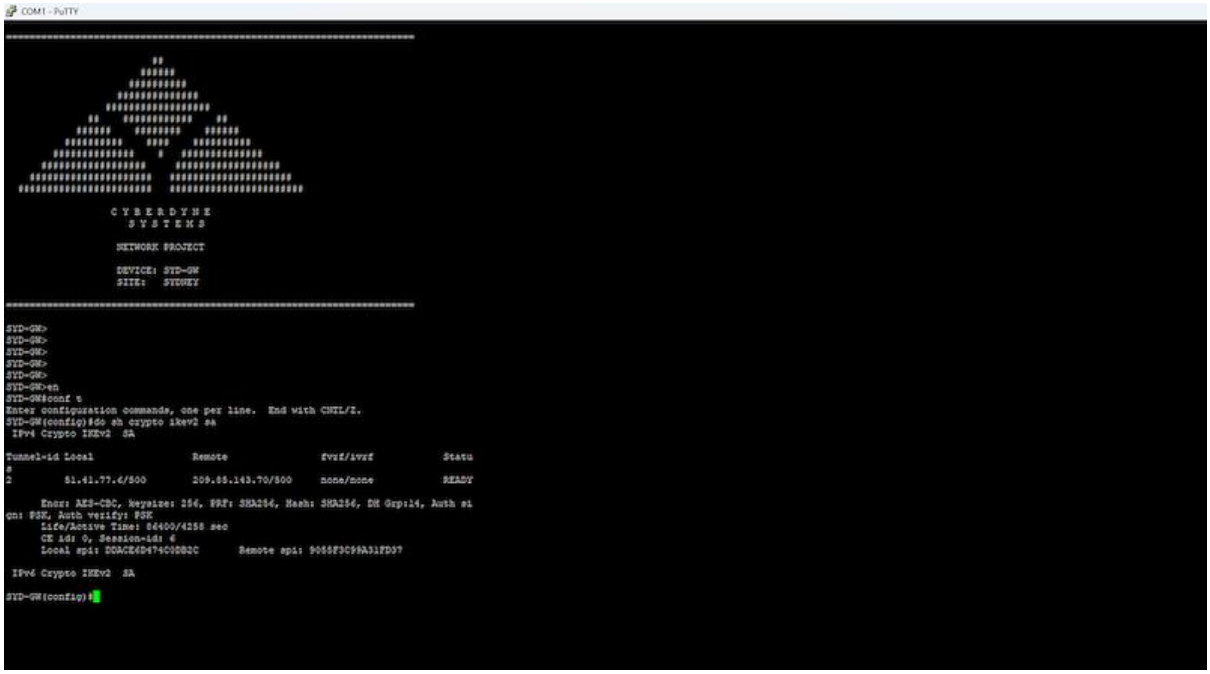
GRE/IPsec VPN

Test	Instructions	Commands	Expected Result	Remediation
GRE/IPsec VPN	SYD-GW: • Verify the GRE tunnel is operational. • Verify the IKEv2 security association with CBR-GW. • Verify IPsec is encrypting and decrypting GRE traffic. • Generate traffic between SYD and CBR and confirm the IPsec packet counters	SYD-GW: • show interfaces Tunnel0 • show crypto ikev2 sa • show crypto ipsec sa • ping 10.255.255.2 Optional verification on CBR-GW: • Show Tunnel0 • show crypto ikev2 sa • show crypto ipsec sa	• Tunnel0 is up/up. • Ping to 10.255.255.2 successful • IKEv2 SA status shows READY • Encryption shows AES-256 and SHA-256 • IPsec SA is active between 51.41.77.6 and 209.85.143.70 • Encaps, Encrypt, Decaps and Decrypt counters increase when traffic crosses the VPN.	**Check:** • WAN connectivity through the ISP • Tunnel0 source and destination addresses • GRE tunnel configuration • IKEv2 proposal, policy and profile • Pre-shared key matches on both gateways • IPsec transform-set and profile • IPsec profile is applied to Tunnel0 • OSPF/routes to remote networks • Compare SYD-GW and CBR-GW

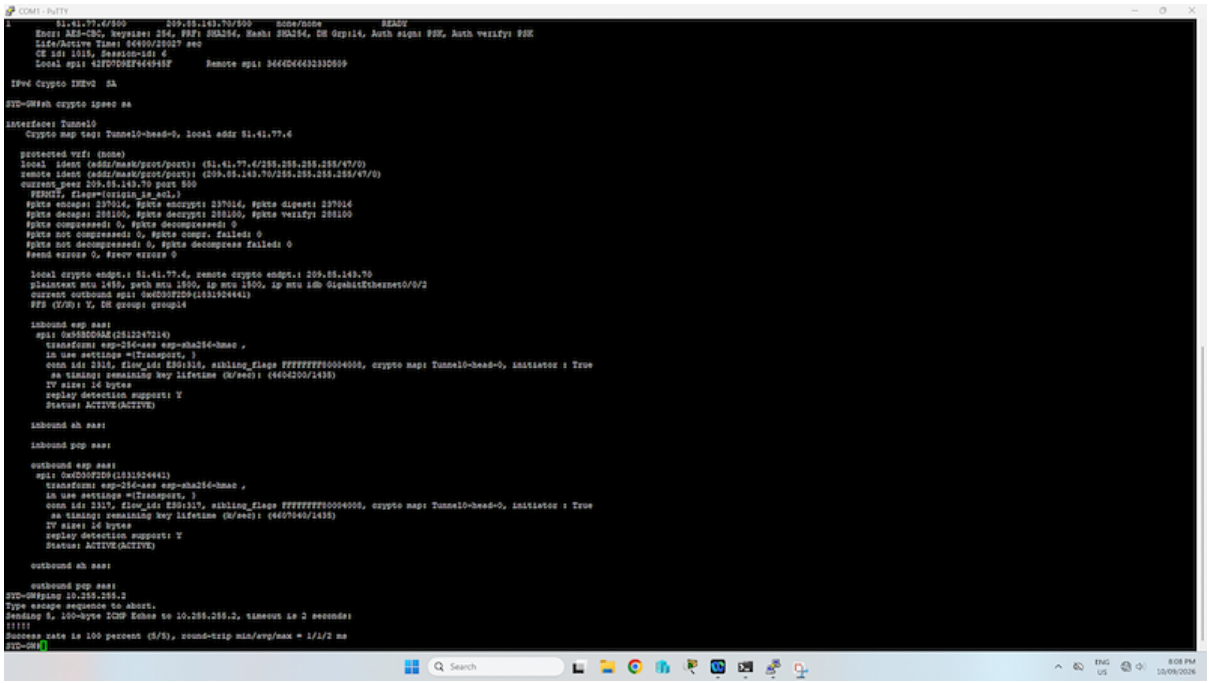
	increase.		• Send/receive errors remain 0	crypto configuration for mismatches



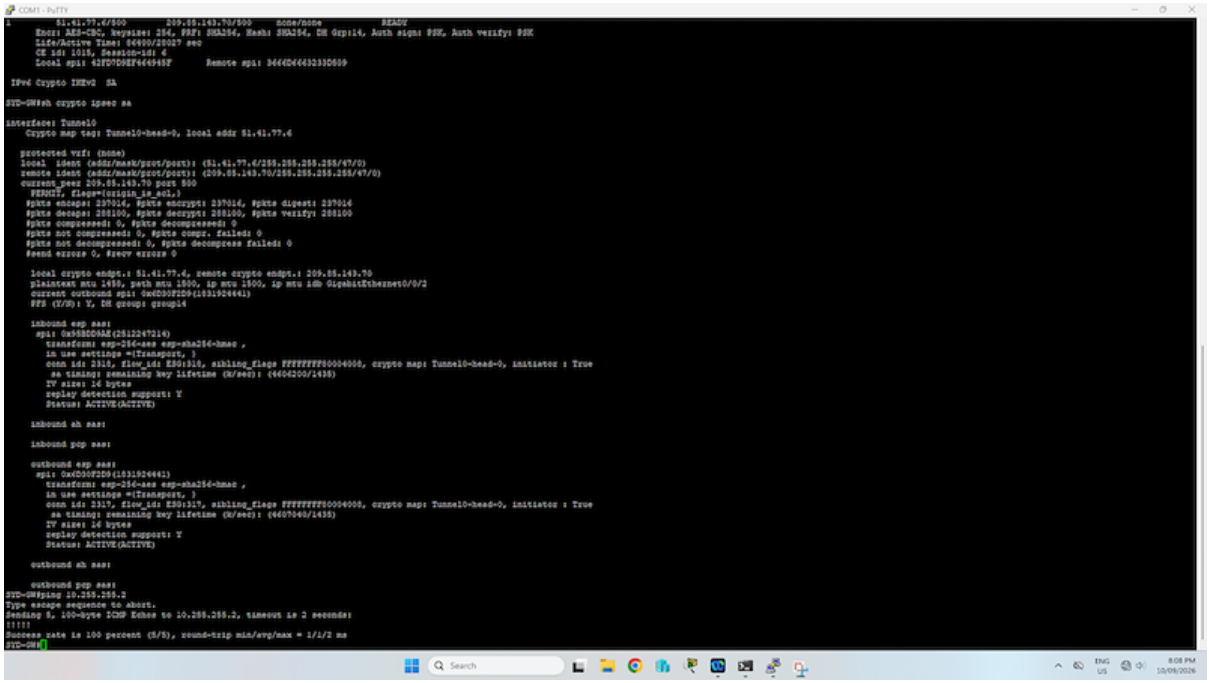
show interfaces Tunnel0



show crypto ikev2 sa



show crypto ipsec sa

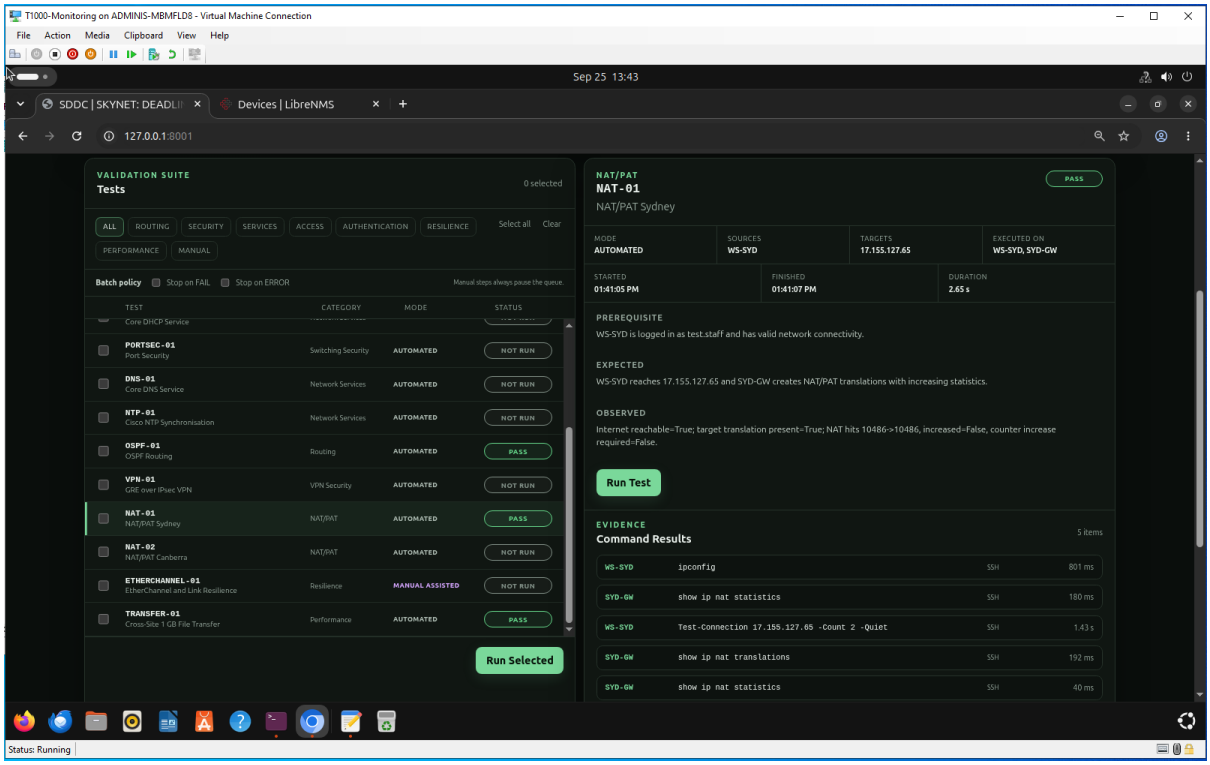


ping 10.255.255.2

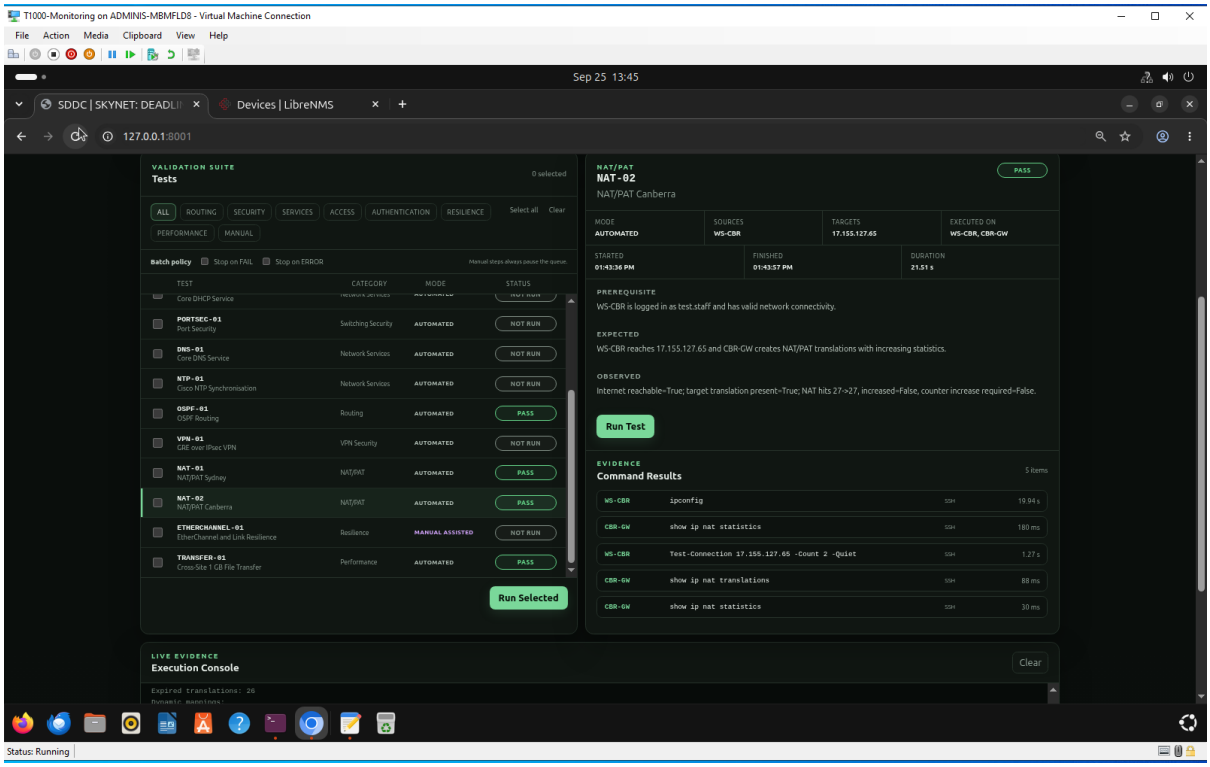
NAT/PAT

Test	Instructions	Commands	Expected Result	Remediation
NAT/PAT	WS-SYD / WS-CBR: • Confirm the client has a valid private IP address	WS-SYD / WS-CBR: • ipconfig • ping 17.155.127.	• 17.155.127.65 is reachable from SYD and CBR clients • NAT/PAT	**Check:** • NAT inside/outside configuration • NAT ACL and

	• Test connectivity to the simulated Internet address • Gateway creates a NAT/PAT translation for the client traffic • NAT statistics increase when traffic is generated	65 On the relevant gateway: • show ip nat translations • show ip nat statistics	translations are created on the gateways • NAT statistics increase when traffic is generated	overload settings • Default route to the ISP • Client gateway and ACL restrictions • Ping `17.155.127.65` from the gateway to isolate the issue



Using custom console

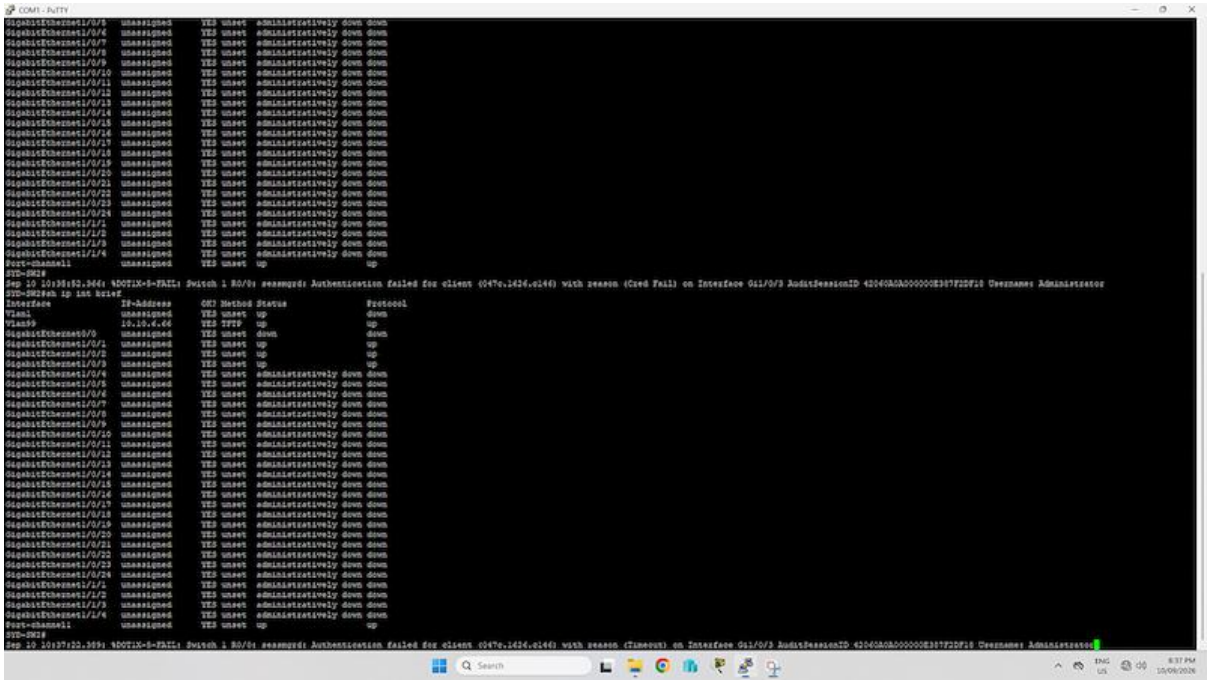


Using custom console

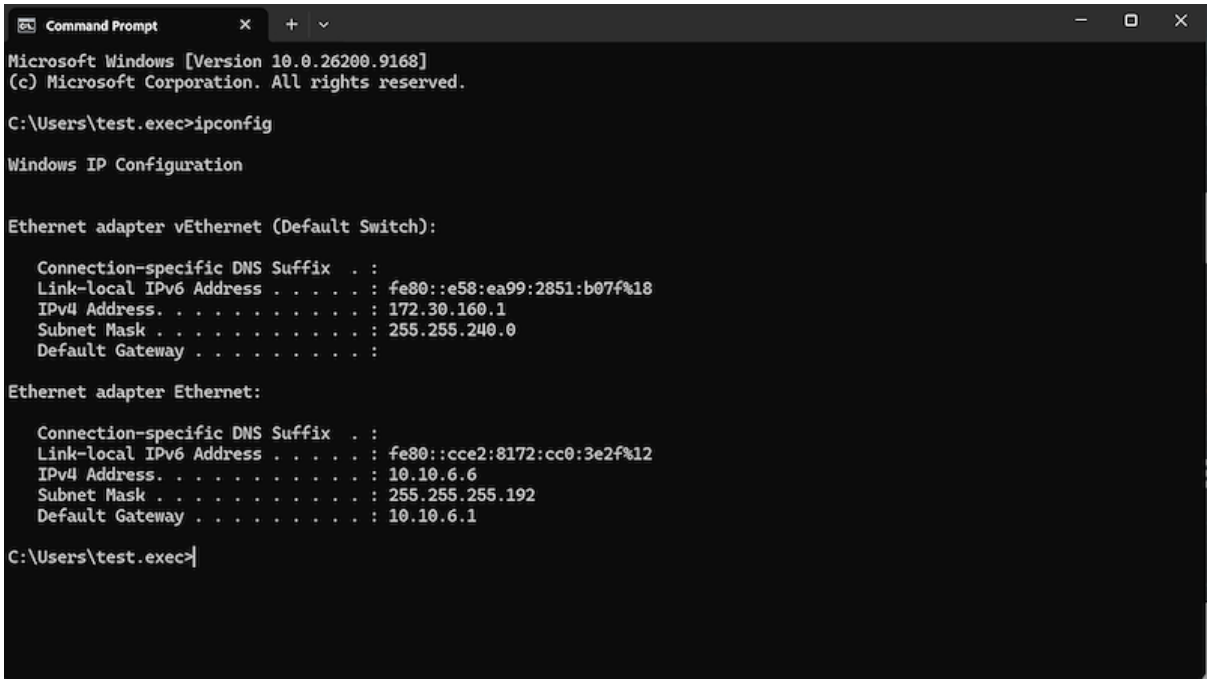
802.1X Authentication and VLAN Assignment

Test	Instructions	Commands	Expected Result	Remediation
802.1X Authentication and VLAN Assignment	WS-SYD: • Logged-out workstation is placed in VLAN40 Restricted. • Log in as: o test.staff o test.itadmin o test.exec • Confirm that each account is dynamically assigned to the correct VLAN	On SYD-SW2: • show authentication sessions interface GigabitEthernet1/0/3 details On WS-SYD: • ipconfig On NPS: • Check Event Viewer for successful 6272 events.	• Workstation logged-out o VLAN40 • test.staff o VLAN10 • test.itadmin o VLAN60 • test.exec o VLAN70 • Authentication shows Authc Success and the correct RADIUS VLAN.	**Check:** • NPS/RADIUS client config • User group memberships and Network Policies • Wired 802.1X GPO/PEAP settings • RADIUS VLAN attribute

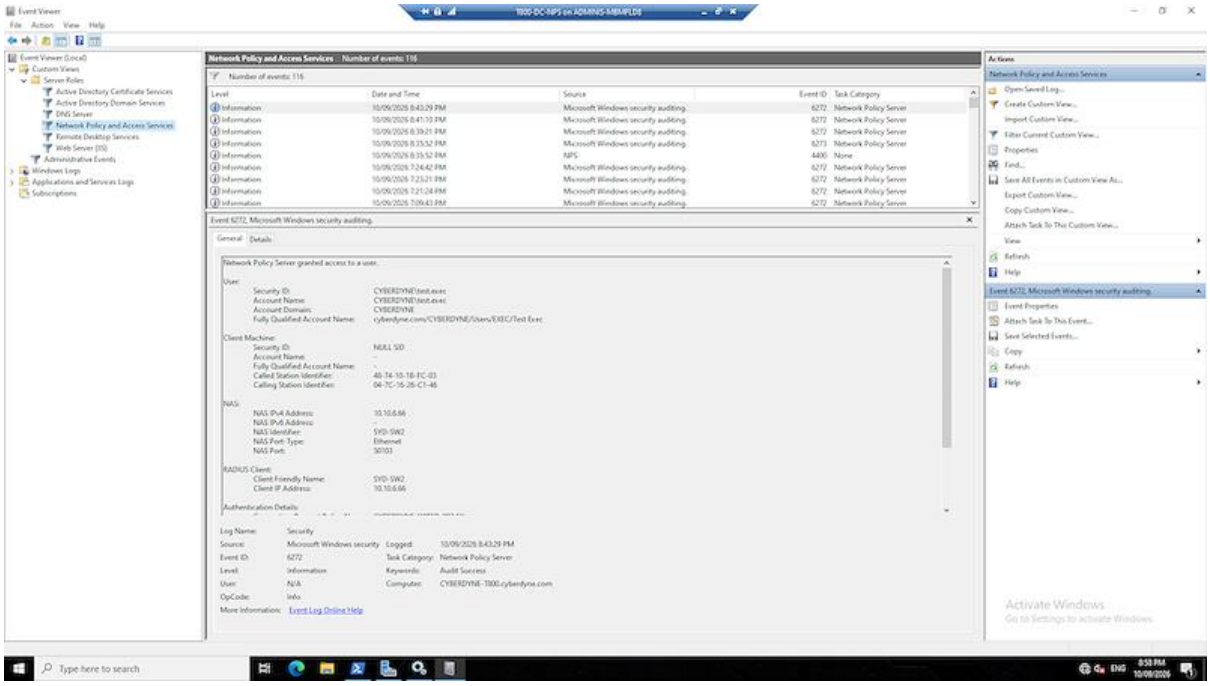
				s • SYD-SW2 AAA and Gi1/0/3 802.1X configuration



show authentication sessions



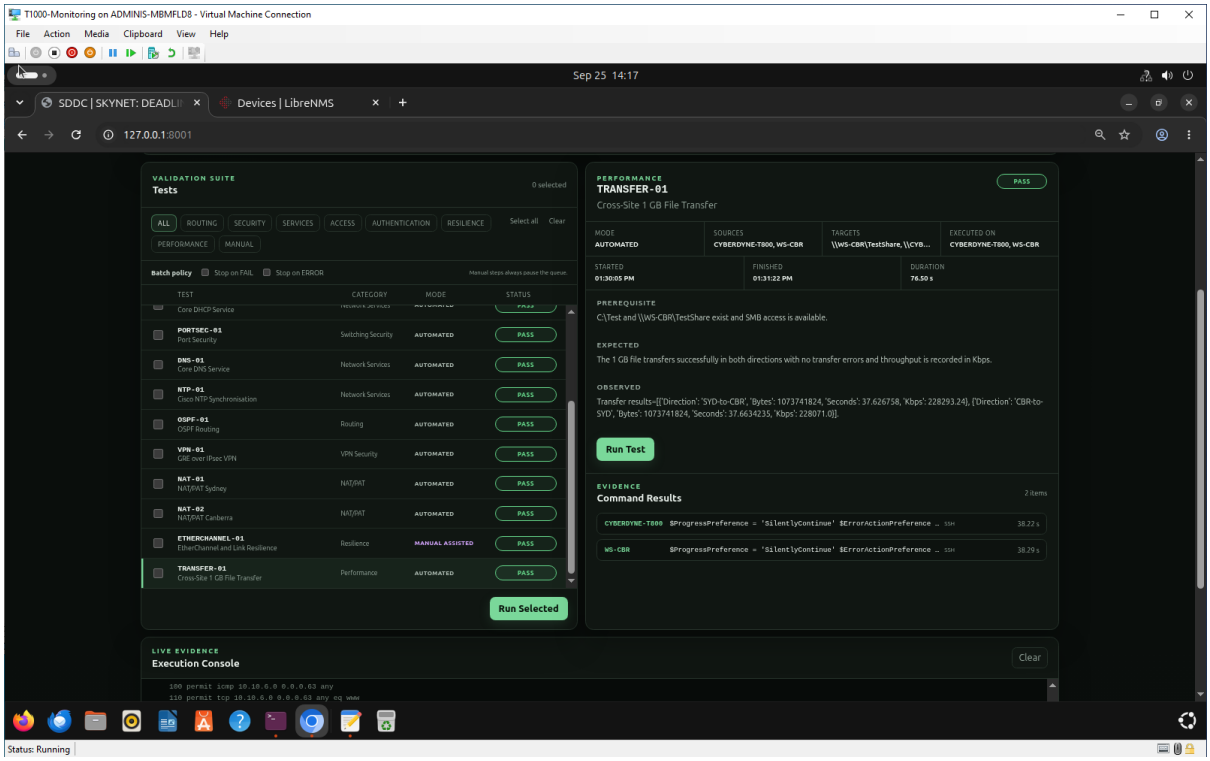
Ipconfig



Check Event Viewer for successful 6272 events.

Cross-Site 1 GB File Transfer

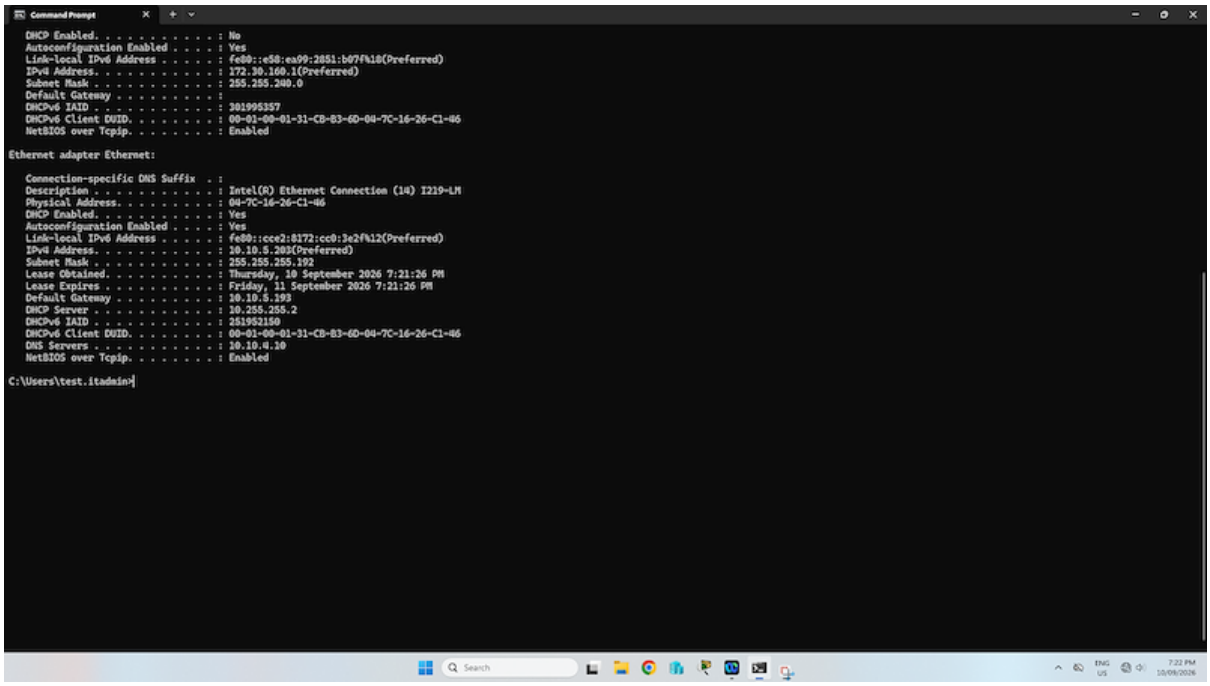
Test	Instructions	Commands	Expected Result	Remediation
Cross-Site 1 GB File Transfer	Between **CYBERDYNE-T800 and WS-CBR**: • Transfer a file of at least 1 GB from SYD to CBR. • Transfer the same file CBR to SYD. • Record transfer time and calculate throughput in Kbps.	PowerShell: • Measure-Command { Copy-Item <source> <destination> } Throughput: • (File size in bytes × 8) ÷ seconds ÷ 1000	• File transfers successfully in both directions. • Throughput is recorded in Kbps for SYD to CBR and CBR to SYD. • No packet loss or transfer errors.	Check: • VPN and OSPF status. • Client/server connectivity. • SMB/firewall permissions. • Interface errors or congestion.



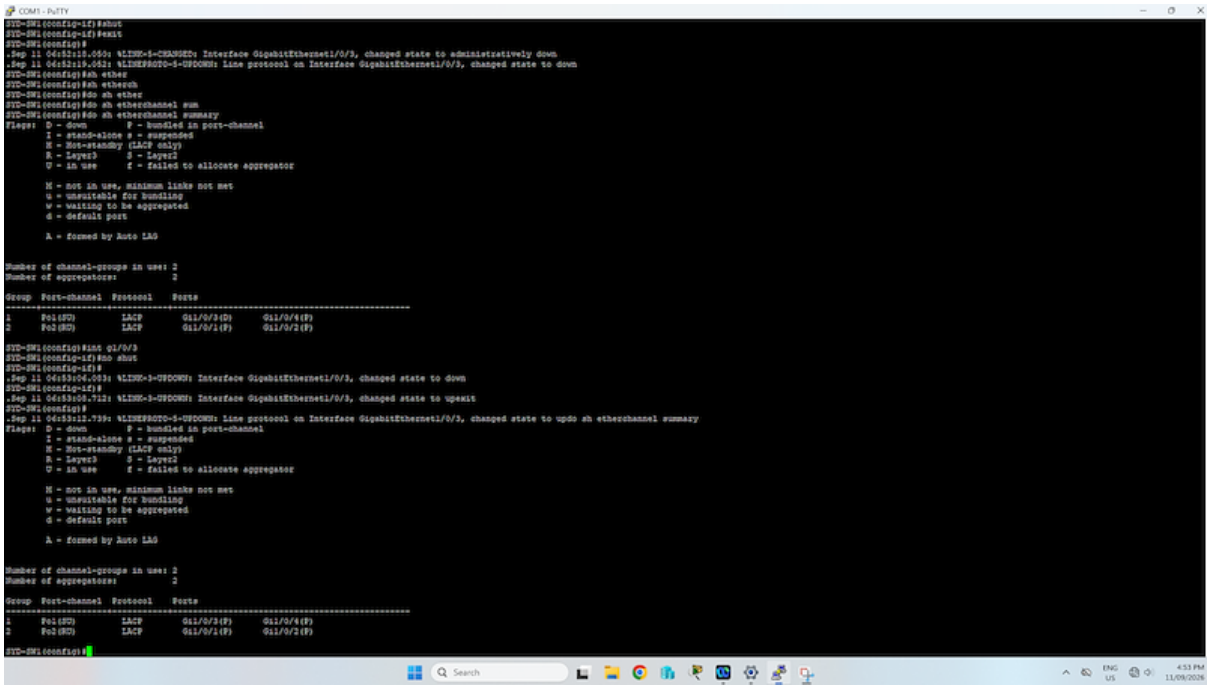
Using custom console

EtherChannel and Link Resilience

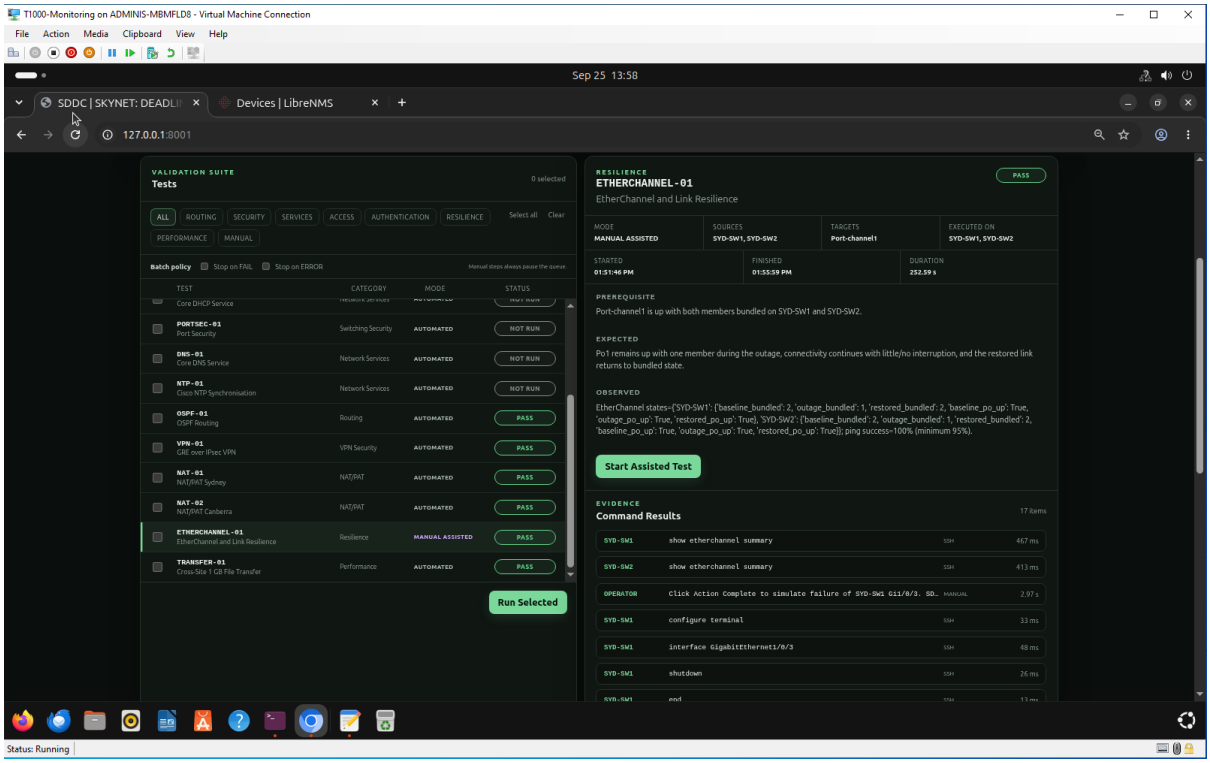
Test	Instructions	Commands	Expected Result	Remediation
EtherChannel and Link Resilience	SYD switch link: • Verify Po1 is up with both members bundled. • Start a continuous ping across the network. • Shut one EtherChannel member. • Confirm connectivity continues. • Restore the link and verify it rejoins the bundle	SYD-SW1 / SYD-SW2: • show etherchannel summary • ping 10.10.6.66 repeat 100 • conf t interface Gi1/0/3 shutdown End • show etherchannel summary • conf t interface Gi1/0/3 no shutdown end	• Po1 remains up with one member. • Connectivity continues with little/no interruption. • Restored link returns to bundled state.	Check: • LACP mode and channel-group configuration. • Trunk settings match on both sides. • VLAN/native VLAN configuration. • Physical link status.



show etherchannel summary (before)



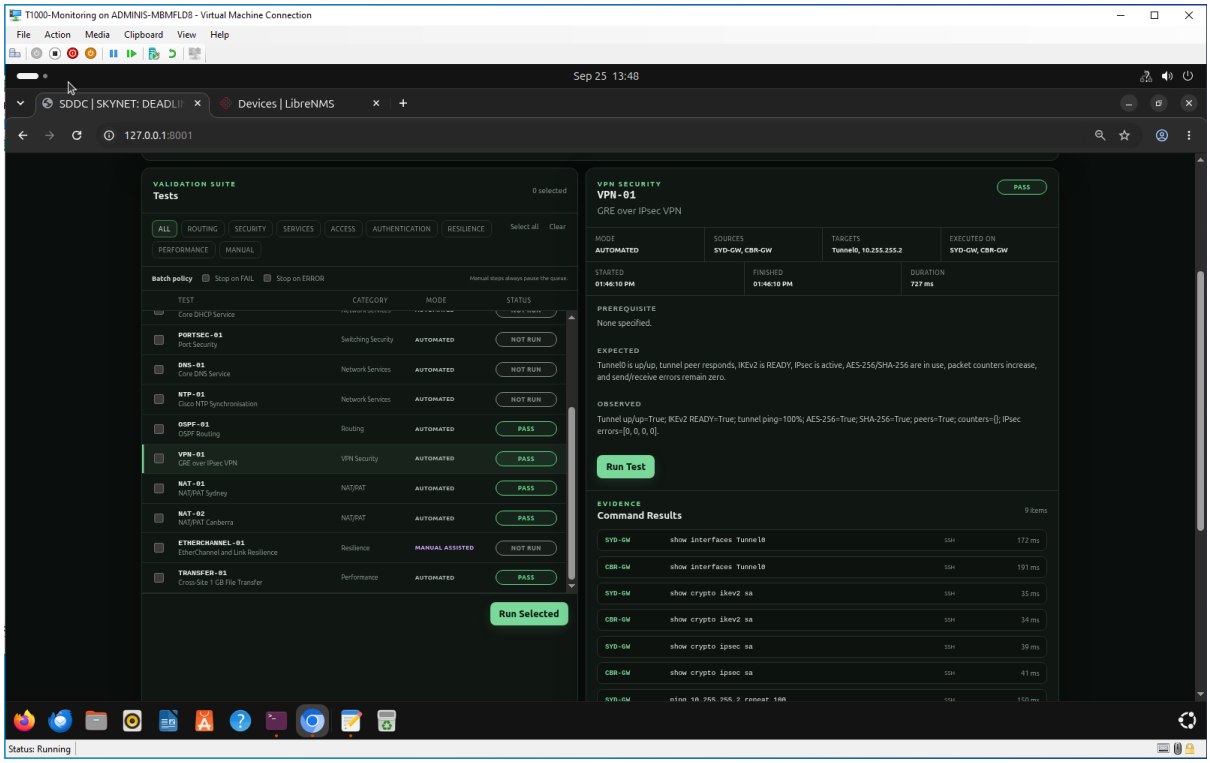
show etherchannel summary (after)



Using custom console

Failure and Recovery

Test	Instructions	Commands	Expected Result	Remediation
Failure and Recovery	**SYD-GW**: • Start a continuous ping to CBR. • Shut down `Tunnel0` to simulate VPN failure. • Confirm cross-site connectivity and OSPF adjacency are lost. • Re-enable `Tunnel0`. • Confirm routing and VPN connectivity recover automatically.	• ping 10.255.255.2 repeat 100 • interface Tunnel0 • shutdown • no shutdown • show ip ospf neighbor • show crypto ikev2 sa	• Connectivity fails while Tunnel0 is down • OSPF tunnel neighbour drops o After restoration: OSPF returns to FULL o IKEv2 shows READY • Cross-site connectivity resumes	Check: • Tunnel configuration and status. • OSPF configuration • IKEv2/IPsec status • Routing to the remote site



Backup and Recovery

Test	Instructions	Commands	Expected Result	Remediation
Backup and Recovery	CYBERDYNE-T800: • Verify that a Windows Server Backup exists. • Create/identify a test file. • Delete or move the test file. • Recover the file using Windows Server Backup. • Confirm that the restored file is accessible.	Windows Server Backup: • Open Windows Server Backup • Select Recover • Select backup/date • Select Files and folders • Restore the test file	• Backup is available and valid. • Selected file restores successfully. • Restored file opens and contains the expected data.	Check: • Backup destination is available. • Backup completed successfully. • Correct backup date/source selected. • Restore permissions and destination.

Network Services

DHCP

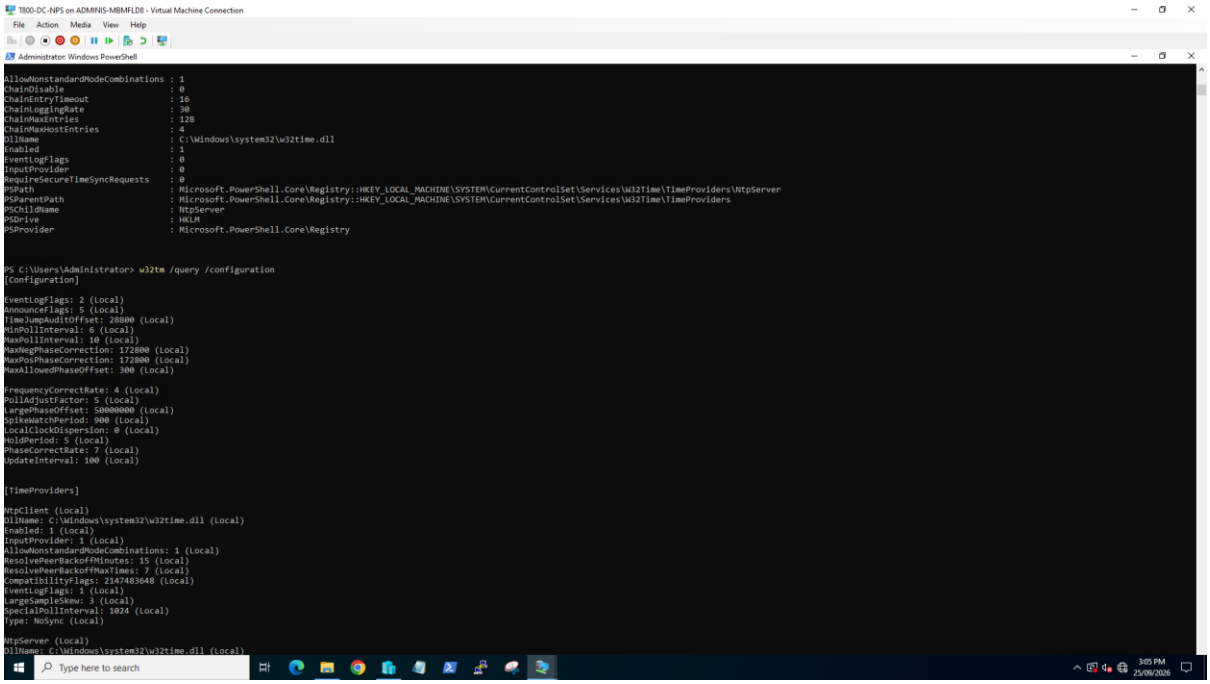
Information:

VLAN ID	VLAN / Network	First Usable Host	Last Usable Host	Default Gateway	DHCP Start	DHCP End	DHCP Pool Size	DHCP Excluded Range
10	STAFF	10.10.0.1	10.10.3.254	10.10.0.1	10.10.0.2	10.10.3.254	1021	10.10.0.1 - 10.10.0.1
20	SERVER-INFRA	10.10.0.1	10.10.0.126	10.10.0.1	N/A	N/A	N/A	N/A
30	GUEST	10.10.0.1	10.10.0.126	10.10.0.1	10.10.0.2	10.10.0.126	125	10.10.0.1 - 10.10.0.1
40	RESTRICTED	10.10.0.1	10.10.0.126	10.10.0.1	10.10.0.2	10.10.0.126	125	10.10.0.1 - 10.10.0.1
50	DEVICES	10.10.0.1	10.10.0.62	10.10.0.1	10.10.0.2	10.10.0.62	61	10.10.0.1 - 10.10.0.1
60	ITADMIN	10.10.0.1	10.10.0.62	10.10.0.1	10.10.0.2	10.10.0.62	61	10.10.0.1 - 10.10.0.1
70	EXEC	10.10.0.1	10.10.0.62	10.10.0.1	10.10.0.2	10.10.0.62	61	10.10.0.1 - 10.10.0.1

Verification (CBR Gateway):

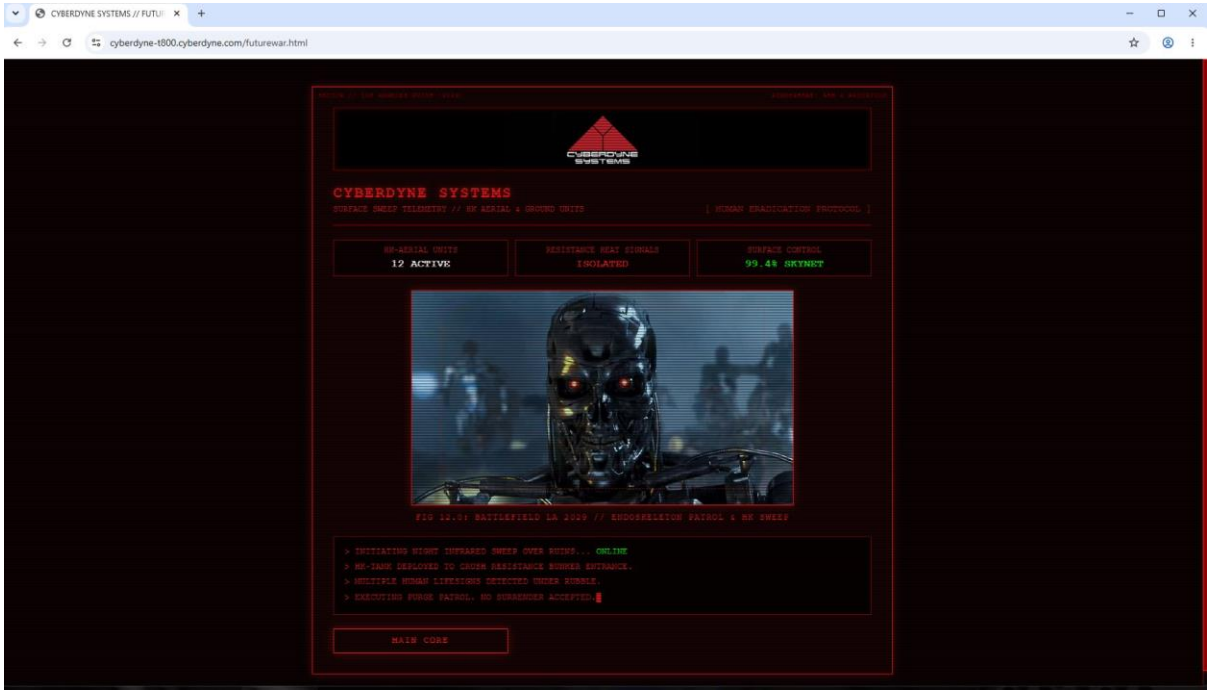
```
CBR-GW#sh run | section dhcp
ip dhcp excluded-address 10.10.0.1
ip dhcp excluded-address 10.10.4.129
ip dhcp excluded-address 10.10.5.1
ip dhcp excluded-address 10.10.5.129
ip dhcp excluded-address 10.10.5.193
ip dhcp excluded-address 10.10.6.1
ip dhcp excluded-address 10.20.0.1
ip dhcp excluded-address 10.20.0.65
ip dhcp excluded-address 10.20.0.129
ip dhcp excluded-address 10.20.0.193
ip dhcp excluded-address 10.20.1.1
ip dhcp excluded-address 10.20.0.241
ip dhcp pool SYD-STAFF
 network 10.10.0.0 255.255.252.0
 default-router 10.10.0.1
 dns-server 10.10.4.10
ip dhcp pool SYD-GUEST
 network 10.10.4.128 255.255.255.128
 default-router 10.10.4.129
 dns-server 10.10.4.10
ip dhcp pool SYD-RESTRICTED
 network 10.10.5.0 255.255.255.128
 default-router 10.10.5.1
 dns-server 10.10.4.10
ip dhcp pool SYD-DEVICES
 network 10.10.5.128 255.255.255.192
 default-router 10.10.5.129
 dns-server 10.10.4.10
ip dhcp pool SYD-ITADMIN
 network 10.10.5.192 255.255.255.192
 default-router 10.10.5.193
 dns-server 10.10.4.10
ip dhcp pool SYD-EXEC
 network 10.10.6.0 255.255.255.192
 default-router 10.10.6.1
 dns-server 10.10.4.10
ip dhcp pool CBR-STAFF
 network 10.20.0.0 255.255.255.192
 default-router 10.20.0.1
 dns-server 10.10.4.10
ip dhcp pool CBR-GUEST
 network 10.20.0.64 255.255.255.192
 default-router 10.20.0.65
 dns-server 10.10.4.10
ip dhcp pool CBR-RESTRICTED
 network 10.20.0.128 255.255.255.192
 default-router 10.20.0.129
 dns-server 10.10.4.10
ip dhcp pool CBR-DEVICES
 network 10.20.0.192 255.255.255.224
 default-router 10.20.0.193
 dns-server 10.10.4.10
ip dhcp pool CBR-ITADMIN
 network 10.20.1.0 255.255.255.248
 default-router 10.20.1.1
 dns-server 10.10.4.10
ip dhcp pool CBR-EXEC
 network 10.20.0.240 255.255.255.240
 default-router 10.20.0.241
 dns-server 10.10.4.10
```

NTP

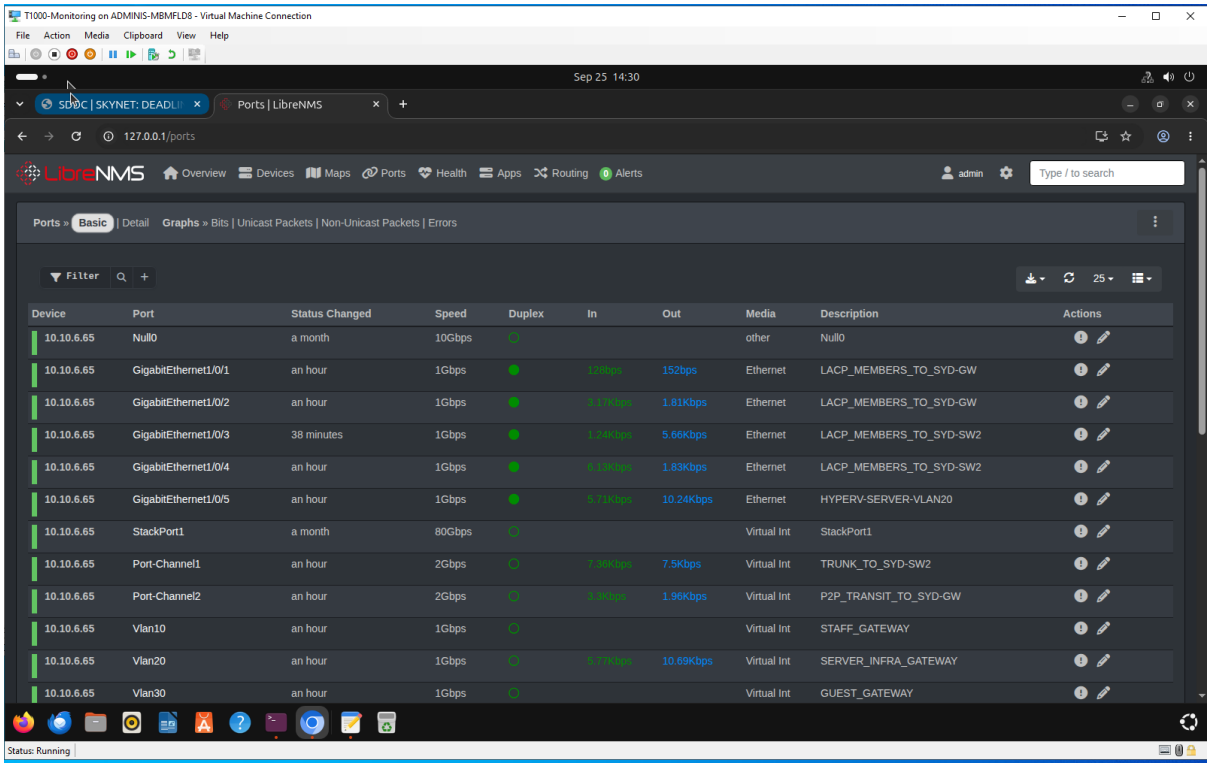


Additional Services

IIS / DNS



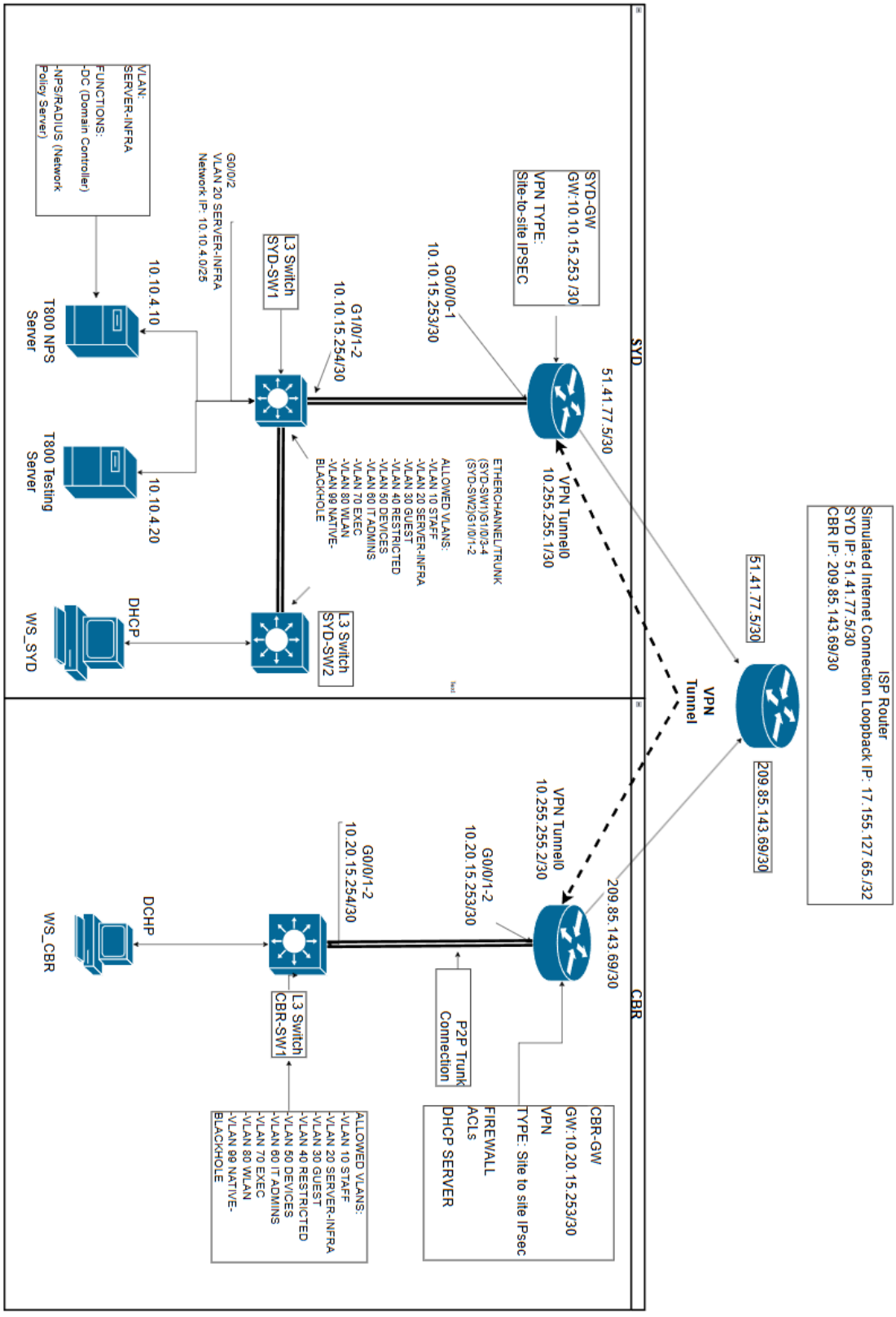
SNMP



References

Cisco Systems, Inc. (2016). *Intelligent WAN (IWAN) Direct Internet Access (DIA) Design Guide*. Cisco Validated Design (CVD) Guides. https://www.cisco.com/c/en/us/td/docs/security/secure-firewall/usecase/b_wan-deployment/dia_usecase/c_overview.html

Network Topology Diagram


ISP Router
Simulated Internet Connection Loopback IP: 17.155.127.65/32
SYD IP: 51.41.77.5/30
CBR IP: 209.85.143.69/30
51.41.77.5/30
209.85.143.69/30
VPN Tunnel
SYD
SYD-GW
GW:10.10.15.253 /30
VPN TYPE:
Site-to-site IPSEC
G0/0/0-1
10.10.15.253/30
51.41.77.5/30
VPN Tunnel0
10.255.255.1/30
G1/0/1-2
10.10.15.254/30
L3 Switch
SYD-SW1
G0/0/2
VLAN 20 SERVER-INFRA
Network IP: 10.10.4.0/25
VLAN:
SERVER-INFRA
FUNCTIONS:
-DC (Domain Controller)
-NPS/RADIUS (Network Policy Server)
10.10.4.10
T800 NPS Server
10.10.4.20
T800 Testing Server
ETHERCHANNEL/TRUNK
(SYD-SW1)G1/0/3-4
(SYD-SW2)G1/0/1-2
ALLOWED VLANS:
-VLAN 10 STAFF
-VLAN 20 SERVER-INFRA
-VLAN 30 GUEST
-VLAN 40 RESTRICTED
-VLAN 50 DEVICES
-VLAN 60 IT ADMINS
-VLAN 70 EXEC
-VLAN 80 WLAN
-VLAN 99 NATIVE-BLACKHOLE
L3 Switch
SYD-SW2
DHCP
WS_SYD
CBR
209.85.143.69/30
VPN Tunnel0
10.255.255.2/30
G0/0/1-2
10.20.15.253/30
CBR-GW
GW:10.20.15.253/30
VPN
TYPE: Site to site IPsec
FIREWALL
ACLs
DHCP SERVER
P2P Trunk Connection
G0/0/1-2
10.20.15.254/30
L3 Switch
CBR-SW1
DCHP
WS_CBR
ALLOWED VLANS:
-VLAN 10 STAFF
-VLAN 20 SERVER-INFRA
-VLAN 30 GUEST
-VLAN 40 RESTRICTED
-VLAN 50 DEVICES
-VLAN 60 IT ADMINS
-VLAN 70 EXEC
-VLAN 80 WLAN
-VLAN 99 NATIVE-BLACKHOLE

ANNEX A - Configuration Files (HQ Site)

Router Configuration

SYD Gateway

```
#######################SYD-GW NETWORKING PROJECT 1 Sydney VPN
Gateway#############
########################### BASIC
SECURITY/SSH/MANAGEMENT#############################

hostname SYD-GW
enable secret Password01
service password-encryption

ip domain name cyberdyne.com
ip ssh version 2

username admin privilege 15 secret Password01

ip access-list standard SYD-MGMT-ACCESS
permit host 10.10.4.10
permit 10.10.5.192 0.0.0.63
permit 10.20.1.0 0.0.0.7
deny any

ip access-list standard SYD-MGMT-ACCESS
27 permit host 10.10.4.20

line vty 0 4
access-class SYD-MGMT-ACCESS in
login local
transport input ssh

line vty 5 14
access-class SYD-MGMT-ACCESS in
login local
transport input ssh

############################BASIC
HARDENING#####################################

no ip source-route
ip options drop

security passwords min-length 8

login block-for 120 attempts 3 within 100
login delay 5
login on-failure log
login on-success log

############################ISP/WAN
CONNECTION##################################

interface GigabitEthernet0/0/2
description WAN_TO_IGN1337
```

```
ip address 51.41.77.6 255.255.255.252
ip nat outside
no shutdown
##############################ROUTING TO
ISP#####################################

ip route 0.0.0.0 0.0.0.0 51.41.77.5

##############################ROUTING TO SYD
LAN#################################

ip route 10.10.0.0 255.255.248.0 10.10.15.253
###NOTE: Temporary static route for Sydney VLAN reachability. Remove ip
route 10.10.0.0 255.255.248.0 10.10.15.253 once OSPF Area 0 is operational
and SYD-GW is successfully learning the Sydney VLAN routes from SYD-SW1.###

##############################LINK TO
L3######################################

interface Port-channel2
description P2P_TRANSIT_TO_SYD-SW1
ip address 10.10.15.254 255.255.255.252
no shutdown

interface GigabitEthernet0/0/0
description LACP_MEMBER_TO_SYD-SW1
no ip address
channel-group 2 mode active
no shutdown

interface GigabitEthernet0/0/1
description LACP_MEMBER_TO_SYD-SW1
no ip address
channel-group 2 mode active
no shutdown



####################################IKE/IPSEC############################
#########

crypto ikev2 proposal SYD-CBR-IKEV2-PROP
encryption aes-cbc-256
integrity sha256
prf sha256
group 14
exit

crypto ikev2 policy SYD-CBR-IKEV2-POL
proposal SYD-CBR-IKEV2-PROP
exit

crypto ikev2 keyring SYD-CBR-IKEV2-KEYRING
peer CBR-GW
address 209.85.143.70 255.255.255.255
pre-shared-key local
825491ec393dba7f48e99d3b5f2cff9ee98612e0d0eab96bc060be5e759eba83
pre-shared-key remote
825491ec393dba7f48e99d3b5f2cff9ee98612e0d0eab96bc060be5e759eba83
```

```
crypto ikev2 profile SYD-CBR-IKEV2-PROFILE
match identity remote address 209.85.143.70 255.255.255.255
identity local address 51.41.77.6
authentication remote pre-share
authentication local pre-share
keyring local SYD-CBR-IKEV2-KEYRING
dpd 10 2 periodic
exit
exit
##^not a typo^##

##do show running-config | section crypto ikev2##

crypto ipsec transform-set SYD-CBR-TRANSFORM esp-aes 256 esp-sha256-hmac
mode transport

crypto ipsec profile SYD-CBR-IPSEC-PROFILE
set transform-set SYD-CBR-TRANSFORM
set pfs group14
set ikev2-profile SYD-CBR-IKEV2-PROFILE

###do show running-config | section crypto ipsec###

################################GRE
TUNNEL###################################

interface Tunnel0
description GRE_OVER_IPSEC_TO_CBR-GW
shutdown
ip address 10.255.255.1 255.255.255.252

tunnel source GigabitEthernet0/0/2
tunnel destination 209.85.143.70
tunnel mode gre ip

ip mtu 1400
ip tcp adjust-mss 1360

tunnel protection ipsec profile SYD-CBR-IPSEC-PROFILE

##do show running-config interface Tunnel0##

no shutdown

##ping 10.255.255.2 source 10.255.255.1##

#####################################OSPF
ROUTING##############################

interface Tunnel0
ip ospf 1 area 0

router ospf 1
router-id 1.1.1.1
passive-interface default
```

```
no passive-interface Tunnel0
log-adjacency-changes



########OSPF SYD-GW TO SYD-SW##########
interface Port-channel2
ip ospf 1 area 0
exit

router ospf 1
no passive-interface Port-channel2

#######################################NAT/PAT#################################
######

interface GigabitEthernet0/0/2
ip nat outside

interface Port-channel2
ip nat inside



ip access-list standard SYD-PAT-SOURCES
permit 10.10.0.0 0.0.3.255
permit 10.10.4.0 0.0.0.127
permit 10.10.4.128 0.0.0.127
permit 10.10.5.0 0.0.0.127
permit 10.10.5.128 0.0.0.63
permit 10.10.5.192 0.0.0.63
permit 10.10.6.0 0.0.0.63
permit 10.10.6.64 0.0.0.7

ip nat inside source list SYD-PAT-SOURCES interface GigabitEthernet0/0/2
overload

##show running-config interface GigabitEthernet0/0/2##
##show running-config interface Port-channel2##

############################NTP#############################################
######

ntp server 10.10.4.10

clock timezone AEST 10 0
clock summer-time AEDT recurring 1 Sun Oct 2:00 1 Sun Apr 3:00 60

#########################################################################
######
end
write memory



#########################################################################
######

#########################################################################
```

```
######

################################################################################
######
```

Sydney Switch 1

```
######################SYD-SW1 NETWORKING PROJECT 1 Sydney L3
Switch##############

en
conf t

hostname SYD-SW1

enable secret Password01

ip domain name cyberdyne.com

username admin privilege 15 secret Password01
crypto key generate rsa modulus 2048

ip ssh version 2

aaa authentication login SSH-LOCAL local
line vty 0 4
login authentication SSH-LOCAL
transport input ssh
access-class SYD-MGMT-ACCESS in

line vty 5 15
login authentication SSH-LOCAL
transport input ssh
access-class SYD-MGMT-ACCESS in

ip access-list standard SYD-MGMT-ACCESS
27 permit host 10.10.4.20

##############################ACCESS
PORTS#######################################

interface GigabitEthernet1/0/8
description ARUBA-AP-GUEST-WIFI
switchport mode access
switchport access vlan 30
spanning-tree portfast
no shut

interface GigabitEthernet1/0/5
description HYPERV-SERVER-VLAN20
switchport mode access
switchport access vlan 20
spanning-tree portfast
```

```
switchport port-security
switchport port-security maximum 4
switchport port-security violation restrict
no shutdown

##############################LAYER 3
ROUTING#####################################

ip routing

##############################LINK TO SYD-
GW##################################

interface range G1/0/1 - 2
description LACP_MEMBERS_TO_SYD-GW
no switchport
channel-group 2 mode active
no shutdown
exit

interface Port-channel2
description P2P_TRANSIT_TO_SYD-GW
no switchport
ip address 10.10.15.253 255.255.255.252
no shutdown
exit

##############################LINK TO SYD-
SW2#################################

interface range GigabitEthernet1/0/3 - 4
description LACP_MEMBERS_TO_SYD-SW2
switchport mode trunk
channel-group 1 mode active
no shutdown

interface Port-channel1
description TRUNK_TO_SYD-SW2
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,99
no shutdown

interface Port-channel1
switchport trunk native vlan 998

##show etherchannel summary##
##show interfaces trunk##

##############################OSPF
ROUTING#########################################

interface Port-channel2
ip ospf 1 area 0
exit

router ospf 1
router-id 3.3.3.3
```

```
passive-interface default
no passive-interface Port-channel2
log-adjacency-changes

interface Vlan10
ip ospf 1 area 0

interface Vlan20
ip ospf 1 area 0

interface Vlan30
ip ospf 1 area 0

interface Vlan40
ip ospf 1 area 0

interface Vlan50
ip ospf 1 area 0

interface Vlan60
ip ospf 1 area 0

interface Vlan70
ip ospf 1 area 0

interface Vlan99
ip ospf 1 area 0

##############################VLAN
CONFIGURATION###################################

vlan 10
name STAFF

vlan 20
name SERVER-INFRA

vlan 30
name GUEST

vlan 40
name RESTRICTED

vlan 50
name DEVICES

vlan 60
name ITADMIN

vlan 70
name EXEC

vlan 99
name MANAGEMENT

vlan 998
```

```
name NATIVE

vlan 999
name BLACKHOLE

##############################SVI / INTER-VLAN
ROUTING##############################

interface Vlan10
description STAFF_GATEWAY
ip address 10.10.0.1 255.255.252.0
no shutdown

interface Vlan20
description SERVER_INFRA_GATEWAY
ip address 10.10.4.1 255.255.255.128
no shutdown

interface Vlan30
description GUEST_GATEWAY
ip address 10.10.4.129 255.255.255.128
no shutdown

interface Vlan40
description RESTRICTED_GATEWAY
ip address 10.10.5.1 255.255.255.128
no shutdown

interface Vlan50
description DEVICES_GATEWAY
ip address 10.10.5.129 255.255.255.192
no shutdown

interface Vlan60
description ITADMIN_GATEWAY
ip address 10.10.5.193 255.255.255.192
no shutdown

interface Vlan70
description EXEC_GATEWAY
ip address 10.10.6.1 255.255.255.192
no shutdown

interface Vlan99
description MANAGEMENT_GATEWAY
ip address 10.10.6.65 255.255.255.248
no shutdown

##show vlan brief##
##show ip interface brief##

##############################ACCESS
PORTS########################################

interface GigabitEthernet1/0/5
description CYBERDYNE_T800_DC_DNS_NPS
switchport mode access
```

```
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface GigabitEthernet1/0/6
description CYBERDYNE_ACCESS_TERMINAL
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

#####TEST INTERFACE#####

interface GigabitEthernet1/0/7
description 8021X_TEST_PORT
switchport access vlan 40
switchport mode access
authentication event fail action authorize vlan 40
authentication port-control auto
dot1x pae authenticator
spanning-tree portfast
spanning-tree bpduguard enable



#####IGNOR FOR NOW#####
interface GigabitEthernet1/0/7
description ACL_TEST_PORT
switchport mode access
switchport access vlan 10
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown



##############################UNUSED
PORTS########################################

interface range GigabitEthernet1/0/8 - 24
description UNUSED
switchport mode access
switchport access vlan 999
shutdown

interface range GigabitEthernet1/1/1 - 4
description UNUSED
switchport mode access
switchport access vlan 999
shutdown

##############################SPANNING
TREE########################################

spanning-tree vlan 10,20,30,40,50,60,70,99,998,999 root primary

##############################DHCP
RELAY#########################################
```

```
interface Vlan10
ip helper-address 10.20.15.254

interface Vlan30
ip helper-address 10.20.15.254

interface Vlan40
ip helper-address 10.20.15.254

interface Vlan50
ip helper-address 10.20.15.254

interface Vlan60
ip helper-address 10.20.15.254

interface Vlan70
ip helper-address 10.20.15.254

################################NTP#############################################
######

ntp server 10.10.4.10

clock timezone AEST 10 0
clock summer-time AEDT recurring 1 Sun Oct 2:00 1 Sun Apr 3:00 60

################################LAYER 3
ACLS#########################################

####RESTRICTED####
ip access-list extended SYD-RESTRICTED-IN
remark Allow DHCP requests to reach central DHCP server
permit udp any eq bootpc any eq bootps
remark Block Restricted VLAN from routing outside its VLAN
deny ip 10.10.5.0 0.0.0.127 any

interface Vlan40
ip access-group SYD-RESTRICTED-IN in

####GUEST####
ip access-list extended SYD-GUEST-IN

remark Allow DHCP
permit udp any eq bootpc any eq bootps

remark Allow DNS to Cyberdyne server
permit udp 10.10.4.128 0.0.0.127 host 10.10.4.10 eq domain
permit tcp 10.10.4.128 0.0.0.127 host 10.10.4.10 eq domain

remark Allow NTP to Cyberdyne server
permit udp 10.10.4.128 0.0.0.127 host 10.10.4.10 eq ntp

remark Block access to all internal networks
deny ip 10.10.4.128 0.0.0.127 10.0.0.0 0.255.255.255
```

```
remark Allow HTTP and HTTPS to Internet
permit tcp 10.10.4.128 0.0.0.127 any eq www
permit tcp 10.10.4.128 0.0.0.127 any eq 443

remark Allow common Internet email services
permit tcp 10.10.4.128 0.0.0.127 any eq smtp
permit tcp 10.10.4.128 0.0.0.127 any eq 465
permit tcp 10.10.4.128 0.0.0.127 any eq 587
permit tcp 10.10.4.128 0.0.0.127 any eq pop3
permit tcp 10.10.4.128 0.0.0.127 any eq 995
permit tcp 10.10.4.128 0.0.0.127 any eq imap
permit tcp 10.10.4.128 0.0.0.127 any eq 993

remark Deny everything else
deny ip any any

interface Vlan30
ip access-group SYD-GUEST-IN in

####STAFF####
ip access-list extended SYD-STAFF-IN

remark Allow DHCP
permit udp any eq bootpc any eq bootps

remark Block SYD Guest
deny ip 10.10.0.0 0.0.3.255 10.10.4.128 0.0.0.127

remark Block SYD Restricted
deny ip 10.10.0.0 0.0.3.255 10.10.5.0 0.0.0.127

remark Block SYD IT Admin
deny ip 10.10.0.0 0.0.3.255 10.10.5.192 0.0.0.63

remark Block SYD Management
deny ip 10.10.0.0 0.0.3.255 10.10.6.64 0.0.0.7

remark Block CBR Guest
deny ip 10.10.0.0 0.0.3.255 10.20.0.64 0.0.0.63

remark Block CBR Restricted
deny ip 10.10.0.0 0.0.3.255 10.20.0.128 0.0.0.63

remark Block CBR IT Admin
deny ip 10.10.0.0 0.0.3.255 10.20.1.0 0.0.0.7

remark Block CBR Management
deny ip 10.10.0.0 0.0.3.255 10.20.1.8 0.0.0.7

remark Allow other internal resources
permit ip 10.10.0.0 0.0.3.255 10.0.0.0 0.255.255.255

remark Allow ICMP for diagnostics
permit icmp 10.10.0.0 0.0.3.255 any

remark Allow HTTP and HTTPS
```

```
permit tcp 10.10.0.0 0.0.3.255 any eq www
permit tcp 10.10.0.0 0.0.3.255 any eq 443

remark Allow FTP
permit tcp 10.10.0.0 0.0.3.255 any eq ftp

remark Allow common email services
permit tcp 10.10.0.0 0.0.3.255 any eq smtp
permit tcp 10.10.0.0 0.0.3.255 any eq 465
permit tcp 10.10.0.0 0.0.3.255 any eq 587
permit tcp 10.10.0.0 0.0.3.255 any eq pop3
permit tcp 10.10.0.0 0.0.3.255 any eq 995
permit tcp 10.10.0.0 0.0.3.255 any eq imap
permit tcp 10.10.0.0 0.0.3.255 any eq 993

remark Deny everything else
deny ip any any



interface Vlan10
ip access-group SYD-STAFF-IN in

####ITADMIN####

ip access-list extended SYD-ITADMIN-IN

remark Allow DHCP
permit udp any eq bootpc any eq bootps

remark Allow full access to internal networks
permit ip 10.10.5.192 0.0.0.63 10.0.0.0 0.255.255.255

remark Allow ICMP for diagnostics
permit icmp 10.10.5.192 0.0.0.63 any

remark Allow HTTP and HTTPS
permit tcp 10.10.5.192 0.0.0.63 any eq www
permit tcp 10.10.5.192 0.0.0.63 any eq 443

remark Allow FTP
permit tcp 10.10.5.192 0.0.0.63 any eq ftp

remark Allow SSH
permit tcp 10.10.5.192 0.0.0.63 any eq 22

remark Allow common email services
permit tcp 10.10.5.192 0.0.0.63 any eq smtp
permit tcp 10.10.5.192 0.0.0.63 any eq 465
permit tcp 10.10.5.192 0.0.0.63 any eq 587
permit tcp 10.10.5.192 0.0.0.63 any eq pop3
permit tcp 10.10.5.192 0.0.0.63 any eq 995
permit tcp 10.10.5.192 0.0.0.63 any eq 143
permit tcp 10.10.5.192 0.0.0.63 any eq 993

remark Deny everything else
deny ip any any
```

```
interface Vlan60
ip access-group SYD-ITADMIN-IN in


####EXEC####

ip access-list extended SYD-EXEC-IN

remark Allow DHCP
permit udp any eq bootpc any eq bootps

remark Allow DNS to Cyberdyne_T800
permit udp 10.10.6.0 0.0.0.63 host 10.10.4.10 eq domain
permit tcp 10.10.6.0 0.0.0.63 host 10.10.4.10 eq domain

remark Allow NTP to Cyberdyne_T800
permit udp 10.10.6.0 0.0.0.63 host 10.10.4.10 eq ntp

remark Allow Sydney server infrastructure
permit ip 10.10.6.0 0.0.0.63 10.10.4.0 0.0.0.127

remark Allow Canberra server infrastructure
permit ip 10.10.6.0 0.0.0.63 10.20.0.224 0.0.0.15

remark Allow Sydney shared devices
permit ip 10.10.6.0 0.0.0.63 10.10.5.128 0.0.0.63

remark Allow Canberra shared devices
permit ip 10.10.6.0 0.0.0.63 10.20.0.192 0.0.0.31

remark Block all other internal networks
deny ip 10.10.6.0 0.0.0.63 10.0.0.0 0.255.255.255

remark Allow ICMP to external destinations
permit icmp 10.10.6.0 0.0.0.63 any

remark Allow HTTP
permit tcp 10.10.6.0 0.0.0.63 any eq www

remark Allow HTTPS
permit tcp 10.10.6.0 0.0.0.63 any eq 443

remark Allow SMTP
permit tcp 10.10.6.0 0.0.0.63 any eq smtp
permit tcp 10.10.6.0 0.0.0.63 any eq 465
permit tcp 10.10.6.0 0.0.0.63 any eq 587

remark Allow POP3
permit tcp 10.10.6.0 0.0.0.63 any eq pop3
permit tcp 10.10.6.0 0.0.0.63 any eq 995

remark Allow IMAP
permit tcp 10.10.6.0 0.0.0.63 any eq 143
permit tcp 10.10.6.0 0.0.0.63 any eq 993
```

```
remark Deny everything else
deny ip any any

interface Vlan70
ip access-group SYD-EXEC-IN in
exit


####VLAN99####

ip access-list standard SYD-MGMT-ACCESS
remark Permit Sydney IT Admin subnet
permit 10.10.5.192 0.0.0.63

remark Permit Canberra IT Admin subnet
permit 10.20.1.0 0.0.0.7

remark Permit CYBERDYNE-T800 testing server
permit host 10.10.4.10

remark Deny all other management sources
deny any


##################################RADIUS
CONFIG####################################

aaa new-model
aaa authentication login SSH-LOCAL local
aaa authentication dot1x default group radius
aaa authorization exec default local
aaa authorization network default group radius
aaa session-id common

ip radius source-interface Vlan99

radius server CYBERDYNE-NPS
address ipv4 10.10.4.10 auth-port 1812 acct-port 1813
key Password01

#####TEST INTERFACE#####

interface GigabitEthernet1/0/7
description 8021X_TEST_PORT
switchport access vlan 40
switchport mode access
authentication event fail action authorize vlan 40
authentication event no-response action authorize vlan 40
authentication port-control auto
dot1x pae authenticator
spanning-tree portfast
spanning-tree bpduguard enable


##########################################################################
######
```

Sydney Switch 2

```
######################SYD-SW2 NETWORKING PROJECT 1 Sydney L2
Switch#############

hostname SYD-SW2
no ip domain lookup

enable secret Password01

ip domain name cyberdyne.com
ip ssh version 2

username admin privilege 15 secret Password01

aaa authentication login SSH-LOCAL local

ip access-list standard SYD-MGMT-ACCESS
permit 10.10.5.192 0.0.0.63
permit 10.20.1.0 0.0.0.7
25 permit host 10.10.4.10
deny any

line vty 0 4
login authentication SSH-LOCAL
transport input ssh
access-class SYD-MGMT-ACCESS in

line vty 5 15
login authentication SSH-LOCAL
transport input ssh
access-class SYD-MGMT-ACCESS in

ip access-list standard SYD-MGMT-ACCESS
27 permit host 10.10.4.20

############################LINK TO SYD-
SW1###################################

interface range GigabitEthernet1/0/1 - 2
description LACP_MEMBERS_TO_SYD-SW1
switchport mode trunk
channel-group 1 mode active
no shutdown



interface Port-channel1
description TRUNK_TO_SYD-SW1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,99
no shutdown

interface Port-channel1
switchport trunk native vlan 998
```

```
##show etherchannel summary##
##show interfaces trunk##

##############################VLAN
CONFIGURATION###################################

vlan 10
name STAFF

vlan 20
name SERVER-INFRA

vlan 30
name GUEST

vlan 40
name RESTRICTED

vlan 50
name DEVICES

vlan 60
name ITADMIN

vlan 70
name EXEC

vlan 99
name MANAGEMENT

vlan 998
name NATIVE

vlan 999
name BLACKHOLE

################################MANAGEMENT#####################################
#####

interface Vlan99
description SWITCH_MANAGEMENT
ip address 10.10.6.66 255.255.255.248
no shutdown
exit

ip default-gateway 10.10.6.65

##show ip interface brief | include Vlan99##

##############################ACCESS
PORTS#########################################

interface GigabitEthernet1/0/3
description 8021X_WS-SYD_PORT
switchport access vlan 40
switchport mode access
```

```
authentication open
authentication event fail action authorize vlan 40
authentication event no-response action authorize vlan 40
authentication port-control auto
dot1x pae authenticator
spanning-tree portfast
spanning-tree bpduguard enable

##############################UNUSED
PORTS#######################################

interface range GigabitEthernet1/0/4 - 24
description UNUSED
switchport mode access
switchport access vlan 999
shutdown

interface range GigabitEthernet1/1/1 - 4
description UNUSED
switchport mode access
switchport access vlan 999
shutdown

##show vlan brief##
##show interfaces trunk##

##############################SPANNING
TREE########################################

spanning-tree vlan 10,20,30,40,50,60,70,99,998,999 root secondary

interface GigabitEthernet1/0/3
spanning-tree portfast
spanning-tree bpduguard enable

################################RADIUS
CONFIG#####################################

aaa new-model
aaa authentication dot1x default group radius
aaa authorization network default group radius
aaa session-id common
aaa authorization exec default local

radius server CYBERDYNE-NPS
address ipv4 10.10.4.10 auth-port 1812 acct-port 1813
key Password01

ip radius source-interface Vlan99

dot1x system-auth-control




##############################NTP#############################################
######
```

```
ntp server 10.10.4.10

clock timezone AEST 10 0
clock summer-time AEDT recurring 1 Sun Oct 2:00 1 Sun Apr 3:00 60
```

ANNEX B - Configuration Files (Remote Site)

Canberra Site

CBR Gateway

```
######################CBR-GW NETWORKING PROJECT 1 Canberra VPN
Gateway#############
########################## BASIC
SECURITY/SSH/MANAGEMENT############################
hostname CBR-GW
service password-encryption
enable secret Password01

no ip domain lookup
ip domain name cyberdyne.com
username admin privilege 15 secret Password01

crypto key generate rsa modulus 2048
ip ssh version 2

ip access-list standard CBR-MGMT-ACCESS
permit 10.10.5.192 0.0.0.63
permit 10.20.1.0 0.0.0.7
deny any
exit

line vty 0 4
access-class CBR-MGMT-ACCESS in
login local
transport input ssh

line vty 5 14
access-class CBR-MGMT-ACCESS in
login local
transport input ssh

ip access-list standard CBR-MGMT-ACCESS
25 permit host 10.10.4.10
27 permit host 10.10.4.20



############################BASIC
HARDENING##################################

no ip source-route
ip options drop

security passwords min-length 8

login block-for 120 attempts 3 within 100
login delay 5
login on-failure log
login on-success log



############################ISP/WAN
```

```
CONNECTION##################################

interface GigabitEthernet0/0/2
description WAN_TO_IGN1337
ip address 209.85.143.70 255.255.255.252
ip nat outside
no shutdown



##############################ROUTING TO
ISP#####################################

ip route 0.0.0.0 0.0.0.0 209.85.143.69



##############################LINK TO
L3######################################

interface Port-channel1
description P2P_TRANSIT_TO_CBR-SW1
ip address 10.20.15.254 255.255.255.252
ip nat inside
no shutdown

interface GigabitEthernet0/0/0
description LACP_MEMBER_TO_CBR-SW1
no ip address
channel-group 1 mode active
no shutdown

interface GigabitEthernet0/0/1
description LACP_MEMBER_TO_CBR-SW1
no ip address
channel-group 1 mode active
no shutdown



#####################################IKE/IPSEC##############################
######

crypto ikev2 proposal CBR-SYD-IKEV2-PROP
encryption aes-cbc-256
integrity sha256
prf sha256
group 14
exit

crypto ikev2 policy CBR-SYD-IKEV2-POL
proposal CBR-SYD-IKEV2-PROP
exit

crypto ikev2 keyring CBR-SYD-IKEV2-KEYRING
peer SYD-GW
address 51.41.77.6 255.255.255.255
pre-shared-key local
825491ec393dba7f48e99d3b5f2cff9ee98612e0d0eab96bc060be5e759eba83
pre-shared-key remote
```

```
825491ec393dba7f48e99d3b5f2cff9ee98612e0d0eab96bc060be5e759eba83
exit
exit
##^not a typo^##

##do show running-config | section crypto ikev2##

crypto ikev2 profile CBR-SYD-IKEV2-PROFILE
match identity remote address 51.41.77.6 255.255.255.255
identity local address 209.85.143.70
authentication remote pre-share
authentication local pre-share
keyring local CBR-SYD-IKEV2-KEYRING
dpd 10 2 periodic
exit

crypto ipsec transform-set CBR-SYD-TRANSFORM esp-aes 256 esp-sha256-hmac
mode transport



crypto ipsec profile CBR-SYD-IPSEC-PROFILE
set transform-set CBR-SYD-TRANSFORM
set pfs group14
set ikev2-profile CBR-SYD-IKEV2-PROFILE

##do show running-config | section crypto ipsec##

################################GRE
TUNNEL#####################################

interface Tunnel0
description GRE_OVER_IPSEC_TO_SYD-GW
shutdown

ip address 10.255.255.2 255.255.255.252

tunnel source GigabitEthernet0/0/2
tunnel destination 51.41.77.6
tunnel mode gre ip

ip mtu 1400
ip tcp adjust-mss 1360

tunnel protection ipsec profile CBR-SYD-IPSEC-PROFILE

no shutdown

##do show ip interface brief | include Tunnel##

##do ping 10.255.255.1 source 10.255.255.2##

##do show crypto ikev2 sa##
##do show crypto ipsec sa##
#####################################OSPF
ROUTING##############################
```

```
interface Tunnel0
ip ospf 1 area 0

router ospf 1
router-id 2.2.2.2
passive-interface default
no passive-interface Port-channel1

##do show ip ospf neighbor##

interface Port-channel1
ip ospf 1 area 0

router ospf 1
router-id 4.4.4.4
passive-interface default
no passive-interface Port-channel1
log-adjacency-changes



########################################NAT/PAT##################################
#####

ip access-list standard CBR-PAT-SOURCES
permit [CBR_NETWORK_1] [WILDCARD_MASK]

ip nat inside source list CBR-PAT-SOURCES interface GigabitEthernet0/0/2
overload



########################################DHCP#####################################
######

###########EXCLUDED ADDRESSES#############
ip dhcp excluded-address 10.10.0.1
ip dhcp excluded-address 10.10.4.129
ip dhcp excluded-address 10.10.5.1
ip dhcp excluded-address 10.10.5.129
ip dhcp excluded-address 10.10.5.193
ip dhcp excluded-address 10.10.6.1

ip dhcp excluded-address 10.20.0.1
ip dhcp excluded-address 10.20.0.65
ip dhcp excluded-address 10.20.0.129
ip dhcp excluded-address 10.20.0.193
ip dhcp excluded-address 10.20.1.1
ip dhcp excluded-address 10.20.0.241

############SYD POOLS###################

ip dhcp pool SYD-STAFF
network 10.10.0.0 255.255.252.0
default-router 10.10.0.1
dns-server 10.10.4.10

ip dhcp pool SYD-GUEST
network 10.10.4.128 255.255.255.128
```

```
default-router 10.10.4.129
dns-server 10.10.4.10

ip dhcp pool SYD-RESTRICTED
network 10.10.5.0 255.255.255.128
default-router 10.10.5.1
dns-server 10.10.4.10

ip dhcp pool SYD-DEVICES
network 10.10.5.128 255.255.255.192
default-router 10.10.5.129
dns-server 10.10.4.10

ip dhcp pool SYD-ITADMIN
network 10.10.5.192 255.255.255.192
default-router 10.10.5.193
dns-server 10.10.4.10

ip dhcp pool SYD-EXEC
network 10.10.6.0 255.255.255.192
default-router 10.10.6.1
dns-server 10.10.4.10

############CBR POOLS###################

ip dhcp pool CBR-STAFF
network 10.20.0.0 255.255.255.192
default-router 10.20.0.1
dns-server 10.10.4.10

ip dhcp pool CBR-GUEST
network 10.20.0.64 255.255.255.192
default-router 10.20.0.65
dns-server 10.10.4.10

ip dhcp pool CBR-RESTRICTED
network 10.20.0.128 255.255.255.192
default-router 10.20.0.129
dns-server 10.10.4.10

ip dhcp pool CBR-DEVICES
network 10.20.0.192 255.255.255.224
default-router 10.20.0.193
dns-server 10.10.4.10

ip dhcp pool CBR-ITADMIN
network 10.20.1.0 255.255.255.248
default-router 10.20.1.1
dns-server 10.10.4.10

ip dhcp pool CBR-EXEC
network 10.20.0.240 255.255.255.240
default-router 10.20.0.241
dns-server 10.10.4.10



!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

```
######

end
write memory



################################################################################
######

################################################################################
######

################################################################################
######
```

CBR Switch

```
######################CBR-SW1 NETWORKING PROJECT 1 CANBERRA L3
Switch##############

en
conf t

hostname CBR-SW1
enable secret Password01



no ip domain lookup
ip domain name cyberdyne.com
username admin privilege 15 secret Password01
ip ssh version 2

ip access-list standard CBR-MGMT-ACCESS
permit 10.10.5.192 0.0.0.63
permit 10.20.1.0 0.0.0.7
deny any

line vty 0 4
access-class CBR-MGMT-ACCESS in
login local
transport input ssh

line vty 5 15
access-class CBR-MGMT-ACCESS in
login local
transport input ssh

ip access-list standard CBR-MGMT-ACCESS
25 permit host 10.10.4.10
27 permit host 10.10.4.20

############################LAYER 3
ROUTING#####################################
```

```
ip routing

##############################LINK TO CBR-
GW####################################

interface range GigabitEthernet1/0/1 - 2
description LACP_MEMBERS_TO_CBR-GW
no switchport
channel-group 1 mode active
no shutdown
exit

interface Port-channel1
description description P2P_TRANSIT_TO_CBR-GW
no switchport
ip address 10.20.15.253 255.255.255.252
no shutdown
exit



##############################OSPF
ROUTING#########################################
interface Port-channel1
ip ospf 1 area 0

router ospf 1
router-id 4.4.4.4
passive-interface default
no passive-interface Port-channel1
log-adjacency-changes

interface Vlan10
ip ospf 1 area 0

interface Vlan20
ip ospf 1 area 0

interface Vlan30
ip ospf 1 area 0

interface Vlan40
ip ospf 1 area 0

interface Vlan50
ip ospf 1 area 0

interface Vlan60
ip ospf 1 area 0

interface Vlan70
ip ospf 1 area 0

interface Vlan99
ip ospf 1 area 0
```

```
##############################DEFAULT
ROUTE#####################################

ip route 0.0.0.0 0.0.0.0 10.20.15.254

##############################VLAN
CONFIGURATION#################################

vlan 10
name STAFF

vlan 20
name SERVER-INFRA

vlan 30
name GUEST

vlan 40
name RESTRICTED

vlan 50
name DEVICES

vlan 60
name ITADMIN

vlan 70
name EXEC

vlan 99
name MANAGEMENT

vlan 998
name NATIVE

vlan 999
name BLACKHOLE




##############################SVI / INTER-VLAN
ROUTING##############################

interface Vlan10
description STAFF_GATEWAY
ip address 10.20.0.1 255.255.255.192
no shutdown

interface Vlan20
description SERVER_INFRA_GATEWAY
ip address 10.20.0.225 255.255.255.240
no shutdown

interface Vlan30
description GUEST_GATEWAY
ip address 10.20.0.65 255.255.255.192
```

```
no shutdown

interface Vlan40
description RESTRICTED_GATEWAY
ip address 10.20.0.129 255.255.255.192
no shutdown

interface Vlan50
description DEVICES_GATEWAY
ip address 10.20.0.193 255.255.255.224
no shutdown

interface Vlan60
description ITADMIN_GATEWAY
ip address 10.20.1.1 255.255.255.248
no shutdown

interface Vlan70
description EXEC_GATEWAY
ip address 10.20.0.241 255.255.255.240
no shutdown

interface Vlan99
description MANAGEMENT_GATEWAY
ip address 10.20.1.9 255.255.255.248
no shutdown

##show vlan brief##
##show ip interface brief##

##############################ACCESS
PORTS########################################

interface GigabitEthernet1/0/3
description WS-CBR
switchport mode access
switchport access vlan 10
spanning-tree portfast
no shutdown

##show vlan brief##
##show ip interface brief | include Vlan10##

##############################UNUSED
PORTS########################################

interface range GigabitEthernet1/0/4 - 24
description UNUSED
switchport mode access
switchport access vlan 999
shutdown

interface range GigabitEthernet1/1/1 - 4
description UNUSED
switchport mode access
switchport access vlan 999
shutdown
```

```
##############################PORT
SECURITY#######################################

interface GigabitEthernet1/0/3
switchport port-security
switchport port-security maximum 3
switchport port-security mac-address sticky
switchport port-security violation restrict

##############################DHCP
RELAY#########################################

interface Vlan10
ip helper-address 10.20.15.254

interface Vlan30
ip helper-address 10.20.15.254

interface Vlan40
ip helper-address 10.20.15.254

interface Vlan50
ip helper-address 10.20.15.254

interface Vlan60
ip helper-address 10.20.15.254

interface Vlan70
ip helper-address 10.20.15.254


##show ip dhcp binding##

##############################LAYER 3
ACLS###########################################
####RESTRICTED####

ip access-list extended CBR-RESTRICTED-IN

remark Allow DHCP requests to central DHCP server
permit udp any eq bootpc any eq bootps

remark Block Restricted VLAN from routing outside its VLAN
deny ip 10.20.0.128 0.0.0.63 any

interface Vlan40
ip access-group CBR-RESTRICTED-IN in

show access-lists CBR-RESTRICTED-IN
show ip interface Vlan40

####GUEST####

ip access-list extended CBR-GUEST-IN
remark Allow DHCP
```

```
permit udp any eq bootpc any eq bootps

remark Allow DNS to Cyberdyne_T800
permit udp 10.20.0.64 0.0.0.63 host 10.10.4.10 eq domain
permit tcp 10.20.0.64 0.0.0.63 host 10.10.4.10 eq domain

remark Allow NTP to Cyberdyne_T800
permit udp 10.20.0.64 0.0.0.63 host 10.10.4.10 eq ntp

remark Block access to internal networks
deny ip 10.20.0.64 0.0.0.63 10.0.0.0 0.255.255.255

remark Allow HTTP
permit tcp 10.20.0.64 0.0.0.63 any eq www

remark Allow HTTPS
permit tcp 10.20.0.64 0.0.0.63 any eq 443

remark Allow SMTP
permit tcp 10.20.0.64 0.0.0.63 any eq smtp
permit tcp 10.20.0.64 0.0.0.63 any eq 465
permit tcp 10.20.0.64 0.0.0.63 any eq 587

remark Allow POP3
permit tcp 10.20.0.64 0.0.0.63 any eq pop3
permit tcp 10.20.0.64 0.0.0.63 any eq 995

remark Allow IMAP
permit tcp 10.20.0.64 0.0.0.63 any eq 143
permit tcp 10.20.0.64 0.0.0.63 any eq 993

remark Deny everything else
deny ip any any

interface Vlan30
ip access-group CBR-GUEST-IN in

do show access-lists CBR-GUEST-IN
do show ip interface Vlan30

####STAFF####

ip access-list extended CBR-STAFF-IN

remark Allow DHCP
permit udp any eq bootpc any eq bootps

remark Block CBR Guest
deny ip 10.20.0.0 0.0.0.63 10.20.0.64 0.0.0.63

remark Block CBR Restricted
deny ip 10.20.0.0 0.0.0.63 10.20.0.128 0.0.0.63

remark Block CBR IT Admin
deny ip 10.20.0.0 0.0.0.63 10.20.1.0 0.0.0.7
```

```
remark Block CBR Management
deny ip 10.20.0.0 0.0.0.63 10.20.1.8 0.0.0.7

remark Block SYD Guest
deny ip 10.20.0.0 0.0.0.63 10.10.4.128 0.0.0.127

remark Block SYD Restricted
deny ip 10.20.0.0 0.0.0.63 10.10.5.0 0.0.0.127

remark Block SYD IT Admin
deny ip 10.20.0.0 0.0.0.63 10.10.5.192 0.0.0.63

remark Block SYD Management
deny ip 10.20.0.0 0.0.0.63 10.10.6.64 0.0.0.7

remark Allow permitted internal resources
permit ip 10.20.0.0 0.0.0.63 10.0.0.0 0.255.255.255

remark Allow ICMP
permit icmp 10.20.0.0 0.0.0.63 any

remark Allow HTTP
permit tcp 10.20.0.0 0.0.0.63 any eq www

remark Allow HTTPS
permit tcp 10.20.0.0 0.0.0.63 any eq 443

remark Allow FTP
permit tcp 10.20.0.0 0.0.0.63 any eq ftp

remark Allow SMTP
permit tcp 10.20.0.0 0.0.0.63 any eq smtp
permit tcp 10.20.0.0 0.0.0.63 any eq 465
permit tcp 10.20.0.0 0.0.0.63 any eq 587

remark Allow POP3
permit tcp 10.20.0.0 0.0.0.63 any eq pop3
permit tcp 10.20.0.0 0.0.0.63 any eq 995

remark Allow IMAP
permit tcp 10.20.0.0 0.0.0.63 any eq 143
permit tcp 10.20.0.0 0.0.0.63 any eq 993
remark Deny everything else
deny ip any any

interface Vlan10
ip access-group CBR-STAFF-IN in

do show access-lists CBR-STAFF-IN
do show ip interface Vlan10

####ITADMIN####

ip access-list extended CBR-ITADMIN-IN

remark Allow DHCP
```

```
permit udp any eq bootpc any eq bootps

remark Allow full access to internal networks
permit ip 10.20.1.0 0.0.0.7 10.0.0.0 0.255.255.255

remark Allow ICMP for diagnostics
permit icmp 10.20.1.0 0.0.0.7 any

remark Allow HTTP
permit tcp 10.20.1.0 0.0.0.7 any eq www

remark Allow HTTPS
permit tcp 10.20.1.0 0.0.0.7 any eq 443

remark Allow FTP
permit tcp 10.20.1.0 0.0.0.7 any eq ftp

remark Allow SSH
permit tcp 10.20.1.0 0.0.0.7 any eq 22

remark Allow SMTP
permit tcp 10.20.1.0 0.0.0.7 any eq smtp
permit tcp 10.20.1.0 0.0.0.7 any eq 465
permit tcp 10.20.1.0 0.0.0.7 any eq 587

remark Allow POP3
permit tcp 10.20.1.0 0.0.0.7 any eq pop3
permit tcp 10.20.1.0 0.0.0.7 any eq 995

remark Allow IMAP
permit tcp 10.20.1.0 0.0.0.7 any eq 143
permit tcp 10.20.1.0 0.0.0.7 any eq 993

remark Deny everything else
deny ip any any

interface Vlan60
ip access-group CBR-ITADMIN-IN in

do show access-lists CBR-ITADMIN-IN
do show ip interface Vlan60

####EXEC####

ip access-list extended CBR-EXEC-IN

remark Allow DHCP
permit udp any eq bootpc any eq bootps

remark Allow DNS to Cyberdyne_T800
permit udp 10.20.0.240 0.0.0.15 host 10.10.4.10 eq domain
permit tcp 10.20.0.240 0.0.0.15 host 10.10.4.10 eq domain

remark Allow NTP to Cyberdyne_T800
permit udp 10.20.0.240 0.0.0.15 host 10.10.4.10 eq ntp
```

```
remark Allow Sydney server infrastructure
permit ip 10.20.0.240 0.0.0.15 10.10.4.0 0.0.0.127

remark Allow Canberra server infrastructure
permit ip 10.20.0.240 0.0.0.15 10.20.0.224 0.0.0.15

remark Allow Sydney shared devices
permit ip 10.20.0.240 0.0.0.15 10.10.5.128 0.0.0.63

remark Allow Canberra shared devices
permit ip 10.20.0.240 0.0.0.15 10.20.0.192 0.0.0.31

remark Block all other internal networks
deny ip 10.20.0.240 0.0.0.15 10.0.0.0 0.255.255.255

remark Allow ICMP to external destinations
permit icmp 10.20.0.240 0.0.0.15 any

remark Allow HTTP
permit tcp 10.20.0.240 0.0.0.15 any eq www

remark Allow HTTPS
permit tcp 10.20.0.240 0.0.0.15 any eq 443

remark Allow SMTP
permit tcp 10.20.0.240 0.0.0.15 any eq smtp
permit tcp 10.20.0.240 0.0.0.15 any eq 465
permit tcp 10.20.0.240 0.0.0.15 any eq 587

remark Allow POP3
permit tcp 10.20.0.240 0.0.0.15 any eq pop3
permit tcp 10.20.0.240 0.0.0.15 any eq 995

remark Allow IMAP
permit tcp 10.20.0.240 0.0.0.15 any eq 143
permit tcp 10.20.0.240 0.0.0.15 any eq 993

remark Deny everything else
deny ip any any

interface Vlan70
ip access-group CBR-EXEC-IN in

do show access-lists CBR-EXEC-IN
do show ip interface Vlan70

####MANAGEMENT ACCESS####

ip access-list standard CBR-MGMT-ACCESS

remark Permit Sydney IT Admin subnet
permit 10.10.5.192 0.0.0.63

remark Permit Canberra IT Admin subnet
permit 10.20.1.0 0.0.0.7
```

```
remark Deny all other management sources
deny any



##############################NTP#############################################
#####

ntp server 10.10.4.10

clock timezone AEST 10 0
clock summer-time AEDT recurring 1 Sun Oct 2:00 1 Sun Apr 3:00 60
#########################################################################
######

end



#########################################################################
######
```

Access Control List Design and Justification

Working reference prepared after physical ACL implementation and testing

1. Design Overview

The Layer 3 ACL design uses inbound filtering on source VLAN SVIs. This means traffic is evaluated as it leaves its originating VLAN, before it is routed toward another VLAN, the remote site, or the Internet.

- Extended ACLs are applied inbound to the source VLAN SVI.

- Rules are ordered from specific exceptions and required services to broader denies.
- An explicit final deny is used where a VLAN is intentionally restricted.
- DHCP is permitted before restrictive rules so clients can still obtain centrally issued leases from CBR-GW.
- DNS and NTP are explicitly permitted where the policy requires access to Cyberdyne_T800 at 10.10.4.10.
- Management access is handled separately using a standard ACL intended for VTY access-class filtering, not as an inbound ACL on VLAN99.
- VLAN20 SERVER-INFRA and VLAN50 DEVICES are intentionally left without inbound SVI ACLs at this stage. Access to them is controlled by source VLAN policies.

Cisco IOS ACLs are stateless. A restrictive inbound ACL on a server or device VLAN can unintentionally block legitimate reply traffic. For this reason, the project primarily controls access from the initiating/source VLAN.

2. VLAN and Subnet Reference

VLAN	Sydney	Canberra	Role
10 STAFF	10.10.0.0/22	10.20.0.0/26	User network
20 SERVER-INFRA	10.10.4.0/25	10.20.0.224/28	Servers and infrastructure
30 GUEST	10.10.4.128/25	10.20.0.64/26	Guest access
40 RESTRICTED	10.10.5.0/25	10.20.0.128/26	Restricted / unauthenticated
50 DEVICES	10.10.5.128/26	10.20.0.192/27	Shared devices / printers / IoT
60 ITADMIN	10.10.5.192/26	10.20.1.0/29	IT administration
70 EXEC	10.10.6.0/26	10.20.0.240/28	Executive users
99 MANAGEMENT	10.10.6.64/29	10.20.1.8/29	Network management

3. High-Level Policy Matrix

Policy	Allowed	Denied	Reason
RESTRICTED	DHCP only for routed traffic	All routed traffic outside its VLAN	Assessment requirement: isolated network
GUEST	DHCP, DNS/NTP to server, web and email Internet services	All other internal resources and unapproved traffic	Internet access without internal network access
STAFF	Permitted internal resources, ICMP, web, email, FTP	Guest, Restricted, ITADMIN and Management at both sites	Assessment-defined staff access policy
ITADMIN	Full internal access, ICMP, web, email, FTP, SSH	Anything not explicitly allowed	Administrative role requires broad network access
EXEC	Server-Infra, Devices, DNS/NTP, ICMP external, web and email	All other internal networks and unapproved traffic	Project-defined least-privilege executive policy
MANAGEMENT	ITADMIN source subnets only for remote device management	All other management sources	Protects device administration plane

4. ACL Placement

ACL	Device	Interface	Direction

SYD-STAFF-IN	SYD-SW1	Vlan10	Inbound
SYD-GUEST-IN	SYD-SW1	Vlan30	Inbound
SYD-RESTRICTED-IN	SYD-SW1	Vlan40	Inbound
SYD-ITADMIN-IN	SYD-SW1	Vlan60	Inbound
SYD-EXEC-IN	SYD-SW1	Vlan70	Inbound
CBR-STAFF-IN	CBR-SW1	Vlan10	Inbound
CBR-GUEST-IN	CBR-SW1	Vlan30	Inbound
CBR-RESTRICTED-IN	CBR-SW1	Vlan40	Inbound
CBR-ITADMIN-IN	CBR-SW1	Vlan60	Inbound
CBR-EXEC-IN	CBR-SW1	Vlan70	Inbound

5. Detailed ACL Design and Justification

5.1 Sydney ACLs

SYD RESTRICTED - VLAN40

Item	Detail
ACL	SYD-RESTRICTED-IN
Source subnet	10.10.5.0/25
Default gateway	10.10.5.1
Placement	SYD-SW1 L3 switch - inbound on source SVI
Allowed	DHCP requests required to obtain a lease.
Denied	All routed IP traffic sourced from the Restricted subnet.
Justification	The assessment requires Restricted users to have no access outside their own VLAN. Same-VLAN Layer 2 communication is unaffected because it is not routed through the SVI ACL.
Testing	Test client obtained DHCP successfully. Routed destinations were blocked and deny counters increased.

Rule logic

permit udp any eq bootpc any eq bootps
deny ip 10.10.5.0 0.0.0.127 any

SYD GUEST - VLAN30

Item	Detail
ACL	SYD-GUEST-IN
Source subnet	10.10.4.128/25
Default gateway	10.10.4.129
Placement	SYD-SW1 L3 switch - inbound on source SVI
Allowed	DHCP; DNS and NTP to 10.10.4.10; HTTP/HTTPS; common email ports.
Denied	All other internal 10.0.0.0/8 resources; all other unapproved traffic.
Justification	The Guest VLAN must reach approved Internet services but must not access internal resources. DNS and NTP are explicit exceptions so guests can use required infrastructure without broader internal access.
Testing	DHCP and DNS succeeded. Attempts to reach Sydney and Canberra internal networks were blocked.

Rule logic

permit udp any eq bootpc any eq bootps
permit udp/tcp 10.10.4.128/25 -> host 10.10.4.10 eq DNS
permit udp 10.10.4.128/25 -> host 10.10.4.10 eq NTP
deny ip 10.10.4.128/25 -> 10.0.0.0/8
permit TCP HTTP/HTTPS and common email services
deny ip any any

SYD STAFF - VLAN10

Item	Detail
ACL	SYD-STAFF-IN
Source subnet	10.10.0.0/22
Default gateway	10.10.0.1
Placement	SYD-SW1 L3 switch - inbound on source SVI
Allowed	Permitted internal resources; ICMP; HTTP/HTTPS; FTP; common email services.
Denied	Guest, Restricted, ITADMIN and Management networks at both sites.
Justification	This directly implements the assessment Staff policy. Specific administrative and isolated networks are denied first, followed by a broader permit for other internal resources such as servers and shared devices.
Testing	Known prohibited networks were unreachable. Permitted server, intersite and simulated Internet tests succeeded.

Rule logic

permit udp any eq bootpc any eq bootps
deny Staff -> SYD/CBR Guest
deny Staff -> SYD/CBR Restricted
deny Staff -> SYD/CBR ITADMIN
deny Staff -> SYD/CBR Management
permit Staff -> remaining 10.0.0.0/8 resources
permit ICMP, HTTP/HTTPS, FTP and common email services
deny ip any any

SYD ITADMIN - VLAN60

Item	Detail
ACL	SYD-ITADMIN-IN
Source subnet	10.10.5.192/26
Default gateway	10.10.5.193
Placement	SYD-SW1 L3 switch - inbound on source SVI
Allowed	Full internal 10.0.0.0/8 access; ICMP; HTTP/HTTPS; FTP; SSH; common email services.
Denied	Traffic not covered by the administrative policy.
Justification	IT administrators need broad reachability for support and infrastructure administration. SSH is explicitly allowed for the IT role, while the final deny limits unrelated external traffic.
Testing	ACL was applied to Vlan60 and verified active. Internal and approved external access were validated.

Rule logic

permit udp any eq bootpc any eq bootps
permit ip 10.10.5.192/26 -> 10.0.0.0/8
permit ICMP
permit TCP HTTP/HTTPS, FTP, SSH and common email services
deny ip any any

SYD EXEC - VLAN70

Item	Detail
ACL	SYD-EXEC-IN
Source subnet	10.10.6.0/26
Default gateway	10.10.6.1
Placement	SYD-SW1 L3 switch - inbound on source SVI
Allowed	DHCP; DNS/NTP to 10.10.4.10; Sydney/Canberra Server-Infra; Sydney/Canberra Devices; external ICMP; web and email.
Denied	All other internal 10.0.0.0/8 networks; unapproved external

	traffic.
Justification	EXEC is a project-defined VLAN. A least-privilege policy was selected: executives can reach business servers and shared devices but cannot initiate connections to Staff, Guest, Restricted, ITADMIN or Management networks.
Testing	Server, shared-device, DNS and simulated Internet tests succeeded. Other internal networks were blocked.

Rule logic

permit DHCP, DNS and NTP
permit EXEC -> SYD/CBR Server-Infra
permit EXEC -> SYD/CBR Devices
deny EXEC -> remaining 10.0.0.0/8
permit external ICMP, HTTP/HTTPS and common email services
deny ip any any

5.2 Canberra ACLs

CBR RESTRICTED - VLAN40

Item	Detail
ACL	CBR-RESTRICTED-IN
Source subnet	10.20.0.128/26
Default gateway	10.20.0.129
Placement	CBR-SW1 L3 switch - inbound on source SVI
Allowed	DHCP requests.
Denied	All routed IP traffic sourced from the Restricted subnet.
Justification	Mirrors the Sydney Restricted policy and satisfies the same isolation requirement.
Testing	Test client obtained DHCP and routed destinations were blocked as expected.

Rule logic

permit udp any eq bootpc any eq bootps
deny ip 10.20.0.128 0.0.0.63 any

CBR GUEST - VLAN30

Item	Detail
ACL	CBR-GUEST-IN
Source subnet	10.20.0.64/26

Default gateway	10.20.0.65
Placement	CBR-SW1 L3 switch - inbound on source SVI
Allowed	DHCP; DNS/NTP to 10.10.4.10 across the VPN; HTTP/HTTPS; common email services.
Denied	All other internal 10.0.0.0/8 resources and unapproved traffic.
Justification	Provides the same Guest security policy at both sites while demonstrating centralised DNS/NTP interoperability across the VPN.
Testing	DHCP and DNS succeeded. Internal Sydney and Canberra destinations were blocked.

Rule logic

permit DHCP
permit DNS/NTP -> host 10.10.4.10
deny CBR Guest -> 10.0.0.0/8
permit HTTP/HTTPS and common email services
deny ip any any

CBR STAFF - VLAN10

Item	Detail
ACL	CBR-STAFF-IN
Source subnet	10.20.0.0/26
Default gateway	10.20.0.1
Placement	CBR-SW1 L3 switch - inbound on source SVI
Allowed	Permitted internal resources; ICMP; HTTP/HTTPS; FTP; common email services.
Denied	Guest, Restricted, ITADMIN and Management networks at both sites.
Justification	Mirrors the Sydney Staff policy so the same user role receives consistent access regardless of site.
Testing	Permitted and prohibited destination tests behaved as expected.

Rule logic

permit DHCP
deny Staff -> CBR/SYD Guest
deny Staff -> CBR/SYD Restricted
deny Staff -> CBR/SYD ITADMIN
deny Staff -> CBR/SYD Management
permit Staff -> remaining 10.0.0.0/8 resources
permit ICMP, HTTP/HTTPS, FTP and common email services
deny ip any any

CBR ITADMIN - VLAN60

Item	Detail
ACL	CBR-ITADMIN-IN
Source subnet	10.20.1.0/29
Default gateway	10.20.1.1
Placement	CBR-SW1 L3 switch - inbound on source SVI
Allowed	Full internal 10.0.0.0/8 access; ICMP; HTTP/HTTPS; FTP; SSH; common email services.
Denied	Traffic outside the defined administrative policy.
Justification	Provides Canberra IT administrators with the same support and management capability as Sydney IT administrators.
Testing	Vlan60 and the test port were verified in VLAN60. WS-CBR received 10.20.1.2/29 and known-active destinations were used for validation.

Rule logic

permit DHCP
permit ip 10.20.1.0/29 -> 10.0.0.0/8
permit ICMP
permit HTTP/HTTPS, FTP, SSH and common email services
deny ip any any

CBR EXEC - VLAN70

Item	Detail
ACL	CBR-EXEC-IN
Source subnet	10.20.0.240/28
Default gateway	10.20.0.241
Placement	CBR-SW1 L3 switch - inbound on source SVI
Allowed	DHCP; DNS/NTP to 10.10.4.10; Sydney/Canberra Server-Infra; Sydney/Canberra Devices; external ICMP; web and email.
Denied	All other internal 10.0.0.0/8 networks and unapproved traffic.
Justification	Mirrors the Sydney least-privilege Executive policy and keeps role-based access consistent between sites.
Testing	Initial testing revealed the ACL object had not yet been attached to Vlan70. After applying it inbound, the SVI showed CBR-EXEC-IN as active and testing continued against the intended policy.

Rule logic

```
permit DHCP, DNS and NTP
permit EXEC -> SYD/CBR Server-Infra
permit EXEC -> SYD/CBR Devices
deny EXEC -> remaining 10.0.0.0/8
```

```
permit external ICMP, HTTP/HTTPS and common email services
deny ip any any
```

6. Management Access ACLs

Management access is intentionally separated from the user VLAN filtering model. A standard ACL is used to identify authorised ITADMIN source networks. This ACL is intended to be applied to the VTY lines with access-class when SSH is enabled.

```
SYD-MGMT-ACCESS / CBR-MGMT-ACCESS
permit 10.10.5.192 0.0.0.63    ! SYD ITADMIN
permit 10.20.1.0 0.0.0.7       ! CBR ITADMIN
deny any
```

Question / Item	Explanation
Why not apply it to Vlan99?	An inbound ACL on Vlan99 would filter traffic originating from the Management VLAN. It would not directly restrict who can open a VTY/SSH session to the device.
Intended application	line vty ... / access-class <MGMT-ACL> in / transport input ssh
Current status	Management ACL design prepared. VTY application, SSH credentials, local users, RSA keys and related presentation credentials are intentionally deferred until presentation day.

7. VLANs Without an Inbound SVI ACL

VLAN	Decision	Justification
VLAN20 SERVER-INFRA	No inbound SVI ACL	Access to servers is restricted from the initiating VLANs. Avoids unnecessary stateless filtering of server reply traffic. The infrastructure role requires broad reachability.
VLAN50 DEVICES	No inbound SVI ACL at this stage	Printers and IoT/shared devices may return traffic using several TCP/UDP protocols. A restrictive inbound ACL could break legitimate sessions. Access to Devices is instead controlled from source VLANs.
VLAN99 MANAGEMENT	No inbound SVI ACL	Remote device management is protected using VTY access-class filtering rather than filtering traffic sourced by VLAN99.

8. Testing and Evidence Method

ACL testing was performed by temporarily assigning a workstation/test port to the VLAN under test, renewing DHCP, checking the received addressing, and then generating both permitted and prohibited traffic.

- Confirm the test port is in the intended access VLAN.
- Use ipconfig /release and ipconfig /renew to obtain the correct DHCP lease.
- Confirm IP address, subnet mask, default gateway and DNS server.
- Test known-active permitted destinations rather than relying on inactive SVI addresses.
- Test prohibited destinations that should be blocked by policy.
- Use nslookup to verify DNS where required.
- Use show ip interface VlanXX to confirm the ACL is actually attached inbound.
- Use show access-lists <ACL-NAME> to inspect rule order and match counters.

Testing lesson recorded during commissioning: a correctly defined ACL has no effect until it is attached to the intended SVI. CBR-EXEC-IN initially existed but Vlan70 showed 'Inbound access list is not set'. Applying 'ip access-group CBR-EXEC-IN in' corrected the issue.
Another testing consideration is that a failed ping does not always prove an ACL deny. Some Canberra VLAN SVIs may be up/down when no active Layer 2 endpoint exists in that VLAN. Known-active endpoints and ACL match counters provide stronger evidence.

9. Assessment Alignment

Requirement	Assessment intent	Project implementation
Restricted VLAN	No access outside its own VLAN	Implemented at both sites with DHCP exception followed by routed deny.
Guest VLAN	No internal resources; web/email Internet	Implemented at both sites with DNS/NTP exceptions and internal 10/8 deny.
Staff VLAN	Internal server resources; Web/Email/FTP Internet; no Guest/Restricted/Administration	Implemented at both sites with explicit protected-network denies followed by permitted internal access.
IT Staff / Infrastructure	Complete internal access plus Web/Email/FTP/SSH Internet	ITADMIN ACLs provide full internal access and approved external services. SERVER-INFRA remains unfiltered inbound to avoid interfering with infrastructure reply traffic.
Basic protocols	ICMP/NTP/DNS where appropriate	Explicit DNS/NTP rules are used for Guest/EXEC; internal permits cover required services for Staff/ITADMIN; ICMP is enabled where justified for troubleshooting.
Management ACL	Prevent unauthorised management access	ITADMIN-only standard management ACLs are designed for VTY access-class application with SSH.

10. Summary

The ACL design combines assessment compliance with a role-based least-privilege model. Mandatory Staff, Guest and Restricted policies are implemented consistently at both sites, ITADMIN retains the access required for administration, and the project-defined EXEC VLAN is intentionally limited to business servers, shared devices and approved Internet services. Management-plane protection is kept separate from inter-VLAN filtering so it can be applied directly to SSH/VTY access.
This document is a working reference for configuration review, troubleshooting, report writing and presentation explanation. Final evidence should use the current device configuration and test outputs captured after the last configuration checkpoint.


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